

A. Eigen/diagonal Decomposition or spectral decomposition Theorem:

In linear algebra, eigendecomposition or sometimes spectral decomposition is the factorization of a matrix into a canonical form, whereby the matrix is represented in terms of its eigenvalues and eigenvectors. Only diagonalizable matrices can be factorized in this way.

Let A be the given Matrix, D is a diagonal matrix with the corresponding eigen values of Matrix A and **P be a matrix of eigenvectors** of a given Matrix A, Then A can be written as an **eigen decomposition** $A = PDP^{-1}$

Example 67: Examine the eigen decomposition for the matrix $A = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}$

$$\text{Let } A = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}$$

$$\therefore A - \lambda I = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} = \begin{pmatrix} 4-\lambda & 1 \\ 1 & 4-\lambda \end{pmatrix}$$

$$|A - \lambda I| = \begin{vmatrix} 4-\lambda & 1 \\ 1 & 4-\lambda \end{vmatrix} = (4-\lambda)(4-\lambda) - 1$$

$$\Rightarrow |A - \lambda I| = 0$$

$$\Rightarrow \begin{vmatrix} 4-\lambda & 1 \\ 1 & 4-\lambda \end{vmatrix} = (4-\lambda)(4-\lambda) - 1 = 0$$

$$\Rightarrow (16 - 4\lambda - 4\lambda + \lambda^2) - 1 = 0$$

$$\Rightarrow \lambda^2 - 8\lambda + 16 - 1 = 0$$

$$\Rightarrow \lambda^2 - 8\lambda + 15 = 0$$

$$\Rightarrow \lambda^2 - 5\lambda - 3\lambda + 15 = 0$$

$$\Rightarrow \lambda(\lambda - 5) - 3(\lambda - 5) = 0$$

$$\Rightarrow (\lambda - 5)(\lambda - 3) = 0$$

$$\Rightarrow \lambda = 5, \lambda = 3$$

$$\therefore D = \begin{pmatrix} 3 & 0 \\ 0 & 5 \end{pmatrix}$$

Now to find out **the eigenvector** of the matrix A,

$$\text{Let a non-zero vector } v = \begin{bmatrix} x \\ y \end{bmatrix}$$

We have, $Av = \lambda v$

$$\Rightarrow \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 4x+y \\ x+4y \end{pmatrix} = 3 \begin{pmatrix} x \\ y \end{pmatrix} \quad [\text{For } \lambda = 3]$$

$$\Rightarrow \begin{pmatrix} 4x+y \\ x+4y \end{pmatrix} = \begin{pmatrix} 3x \\ 3y \end{pmatrix}$$

$$\Rightarrow 4x+y = 3x$$

$$x+4y = 3y$$

$$\Rightarrow x+y = 0$$

$$x+y = 0$$

$$\Rightarrow x+y = 0$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in **particular (2-1) = 1 free variable**, which is y.

Let **y = 1** then **x = -1**

$$\therefore \text{Eigenvector } \mathbf{v} = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \text{ for } \lambda = 3$$

Again for $\lambda = 5$

$$\text{Let a non-zero vector } \mathbf{v} = \begin{bmatrix} x \\ y \end{bmatrix}$$

We have, $\mathbf{Av} = \lambda \mathbf{v}$

$$\Rightarrow \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 5 \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 4x+y \\ x+4y \end{pmatrix} = 5 \begin{pmatrix} x \\ y \end{pmatrix} \quad [\text{For } \lambda = 5]$$

$$\Rightarrow \begin{pmatrix} 4x+y \\ x+4y \end{pmatrix} = \begin{pmatrix} 5x \\ 5y \end{pmatrix}$$

$$\Rightarrow 4x+y = 5x$$

$$x+4y = 5y$$

$$\Rightarrow -x+y = 0$$

$$x-y = 0$$

$$\Rightarrow x-y = 0$$

$$x-y = 0$$

$$\Rightarrow x-y = 0$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in **particular (2-1) = 1 free variable**, which is y.

Let **y = 1**, then **x = 1**

$\therefore$ Eigenvector $\mathbf{v} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ for $\lambda = 5$

P be a matrix of eigenvectors of a given Matrix A.

$$\therefore \mathbf{P} = \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}$$

Now to find out $\mathbf{P}^{-1}$

$$c_{11} = 1, c_{12} = -1, c_{21} = -1, c_{22} = -1$$

$$\mathbf{P}^c = \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}$$

$$\text{Adj } \mathbf{P} = (\mathbf{P}^c)^T = \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}$$

$$|\mathbf{P}| = \begin{vmatrix} -1 & 1 \\ 1 & 1 \end{vmatrix} = -1 - 1 = -2$$

$$\therefore \mathbf{P}^{-1} = \frac{\text{Adj } \mathbf{P}}{|\mathbf{P}|} = -\frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}$$

Then A can be written as an eigen decomposition $\mathbf{A} = \mathbf{PDP}^{-1}$

Justification:

$$\begin{aligned} \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} &= -\frac{1}{2} \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 0 & 5 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix} \\ &= -\frac{1}{2} \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 3 & -3 \\ -5 & -5 \end{pmatrix} \\ &= -\frac{1}{2} \begin{pmatrix} -3-5 & 3-5 \\ 3-5 & -3-5 \end{pmatrix} \\ &= -\frac{1}{2} \begin{pmatrix} -8 & -2 \\ -2 & -8 \end{pmatrix} \\ &= \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} = \mathbf{A} \quad (\text{Proved}) \end{aligned}$$

$$\mathbf{A} = \mathbf{PDP}^{-1} \quad (\text{Proved})$$

Home task: Find eigen decomposition of the matrix $= \begin{pmatrix} 7 & 3 \\ 3 & -1 \end{pmatrix}$ and $= \begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix}$

Q-01: What is the difference between Eigen decomposition and SVD?

Singular Value Decomposition:

In linear algebra, the **singular value decomposition (SVD)** is an important factorization of a rectangular real or complex matrix, with several applications **in signal processing , image processing and compression, gene expression analysis and statistics**. Applications which employ the SVD include computing the pseudo-inverse, matrix approximation, and determining the rank, range and null space of a matrix.

The spectral theorem says that normal matrices, which by definition must be square, can be unitarily diagonalized using a basis of eigenvectors. The SVD can be seen as a generalization of the spectral theorem to arbitrary, not necessarily square, matrices.

The singular value decomposition (SVD) is a generalization of the eigen-decomposition, which can be used to analyze rectangular matrices (the eigen- decomposition is defined only for squared matrices). By analogy with the eigen-decomposition, which decomposes a matrix into two simple matrices, the main idea of the SVD is to decompose a rectangular matrix into three simple matrices: Two orthogonal matrices and one diagonal matrix.

Matrix A is decomposed using SVD to acquire an orthogonal base in the multidimensional feature space:

$$\underbrace{\begin{bmatrix} * & * \\ * & * \\ * & * \end{bmatrix}}_A = \underbrace{\begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \end{bmatrix}}_U \underbrace{\begin{bmatrix} * & \\ & * \end{bmatrix}}_S \underbrace{\begin{bmatrix} * & * \\ * & * \end{bmatrix}}_{V^T}$$

$$\mathbf{A} = \mathbf{U}\mathbf{S}\mathbf{V}^T$$

Here,

$\mathbf{A} \rightarrow$ Given Matrix

$\mathbf{U} \rightarrow$ Matrix of Eigen Vectors of the Matrix $\mathbf{A}\mathbf{A}^T$

$\mathbf{V} \rightarrow$ Matrix of Eigen Vectors of the Matrix $\mathbf{A}^T\mathbf{A}$

$\mathbf{S} \rightarrow$ Roots of Eigen Values of the Matrix $\mathbf{A}\mathbf{A}^T$ or $\mathbf{A}^T\mathbf{A}$

$\mathbf{V}^T \rightarrow$ Transpose of the Matrix $\mathbf{V}$

Example 68: Compute Singular Value Decomposition (SVD) for the given matrix

$$\begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix}$$

Answer: Let, $\mathbf{A} = \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix}$

In order to find $\mathbf{U}$, Now we need to find the eigenvectors for $\mathbf{A}\mathbf{A}^T$

Given, $\mathbf{A} = \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix}$

$$\therefore \mathbf{A}^T = \begin{pmatrix} 2 & 1 \\ -2 & 1 \end{pmatrix}$$

$$\begin{aligned} \text{So, } \mathbf{A}\mathbf{A}^T &= \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ -2 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 4+4 & 2-2 \\ 2-2 & 1+1 \end{pmatrix} \\ &= \begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix} \end{aligned} \quad \text{-----(i)}$$

Now we need to find the eigenvalues of $\mathbf{A}\mathbf{A}^T$

$$\begin{aligned} \mathbf{A}\mathbf{A}^T - \lambda\mathbf{I} &= \begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} \\ &= \begin{pmatrix} 8-\lambda & 0 \\ 0 & 2-\lambda \end{pmatrix} \end{aligned}$$

$$|\mathbf{A}\mathbf{A}^T - \lambda\mathbf{I}| = 0$$

$$\Rightarrow \begin{vmatrix} 8-\lambda & 0 \\ 0 & 2-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (8-\lambda)(2-\lambda) = 0$$

$$\Rightarrow (\lambda-8)(\lambda-2) = 0$$

$$\Rightarrow \lambda = 8, 2$$

Therefore eigenvalues are 8 and 2. We construct matrix D by placing eigenvalues along the main diagonal in decreasing order.

$$D = \begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix}$$

Therefore, taking the square root of matrix D gives,

The matrix of singular values $S = \sqrt{D} = \begin{pmatrix} \sqrt{8} & 0 \\ 0 & \sqrt{2} \end{pmatrix}$

$$S = \begin{pmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix} \text{-----(ii)}$$

Now we need to find the eigenvector of AA^T

We have, $Au_1 = \lambda u_1$

$$[AV = \lambda V]$$

Let $u_1 = \begin{pmatrix} x \\ y \end{pmatrix}$

$$\therefore AA^T u_1 = \lambda u_1$$

$$\begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 8 \begin{pmatrix} x \\ y \end{pmatrix}$$

[For $\lambda = 8$ & from (i) putting the value of AA^T]

$$\Rightarrow \begin{pmatrix} 8x \\ 2y \end{pmatrix} = \begin{pmatrix} 8x \\ 8y \end{pmatrix}$$

$$\Rightarrow 8x = 8x$$

$$2y = 8y$$

$$\Rightarrow 8x - 8x = 0$$

$$2y - 8y = 0$$

$$\Rightarrow 8x = 8x$$

$$-6y = 0$$

$$\Rightarrow 8x = 8x$$

$$y = 0$$

$$\therefore y = 0$$

Let $x = -1$

$$\therefore u_1 = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix} \text{ It is a unit vector or normal vector. Since the length of } u_1$$

$$\text{is } |u_1| = \sqrt{(-1)^2 + (0)^2} = \sqrt{1} = 1$$

Again, Let $\mathbf{u}_2 = \begin{pmatrix} x \\ y \end{pmatrix}$

Here,

$$\therefore \mathbf{A}\mathbf{A}^T \mathbf{u}_2 = \lambda \mathbf{u}_2$$

$$\begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 2 \begin{pmatrix} x \\ y \end{pmatrix} \quad [\text{For } \lambda = 2 \text{ \& from (i) putting the value of } \mathbf{A}\mathbf{A}^T]$$

$$\Rightarrow \begin{pmatrix} 8x \\ 2y \end{pmatrix} = \begin{pmatrix} 2x \\ 2y \end{pmatrix}$$

$$\Rightarrow 8x = 2x$$

$$2y = 2y$$

$$\Rightarrow 8x - 2x = 0$$

$$2y = 2y$$

$$\Rightarrow 6x = 0$$

$$2y = 2y$$

$$\therefore x = 0$$

$$2y = 2y$$

Let $y = 1$

$$\therefore \mathbf{u}_2 = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \text{ It is a unit vector or normal vector. Since the length of } \mathbf{u}_1$$

$$\text{is } |\mathbf{u}_2| = \sqrt{(0)^2 + (1)^2} = \sqrt{1} = 1$$

$$\therefore \mathbf{U} = (\mathbf{u}_1 \quad \mathbf{u}_2) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \quad \text{-----(iii)}$$

Now we need to find the eigenvectors for $\mathbf{A}^T \mathbf{A}$

$$\text{Given, } \mathbf{A} = \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix}$$

$$\therefore \mathbf{A}^T = \begin{pmatrix} 2 & 1 \\ -2 & 1 \end{pmatrix}$$

$$\begin{aligned} \therefore \mathbf{A}^T \mathbf{A} &= \begin{pmatrix} 2 & 1 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix} \quad \text{-----(iv)} \end{aligned}$$

Now we need to find the eigenvalues of $\mathbf{A}^T \mathbf{A}$

$$\begin{aligned}\therefore \mathbf{A}^T \mathbf{A} - \lambda \mathbf{I} &= \begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} \\ &= \begin{pmatrix} 5-\lambda & -3 \\ -3 & 5-\lambda \end{pmatrix}\end{aligned}$$

$$\therefore |\mathbf{A}^T \mathbf{A} - \lambda \mathbf{I}| = 0$$

$$\Rightarrow \begin{vmatrix} 5-\lambda & -3 \\ -3 & 5-\lambda \end{vmatrix} = 0$$

$$(5-\lambda)(5-\lambda) - 9 = 0$$

$$\Rightarrow 25 - 10\lambda + \lambda^2 - 9 = 0$$

$$\Rightarrow \lambda^2 - 10\lambda + 16 = 0$$

$$\Rightarrow (\lambda - 8)(\lambda - 2) = 0$$

$$\Rightarrow \lambda = 8, 2$$

Therefore eigenvalues are 8 and 2. We construct matrix D by placing eigenvalues along the main diagonal in decreasing order.

$$\mathbf{D} = \begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix}$$

Therefore, taking the square root of matrix D gives,

$$\text{The matrix of singular values } \mathbf{S} = \begin{pmatrix} \sqrt{8} & 0 \\ 0 & \sqrt{2} \end{pmatrix}$$

$$\mathbf{S} = \begin{pmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix} \text{------(v)}$$

Now we need find the eigenvectors of $\mathbf{A}^T \mathbf{A}$ which are the columns of V.

$$\text{We have, } \mathbf{A} \mathbf{v}_1 = \lambda \mathbf{v}_1$$

$$[\mathbf{A} \mathbf{V} = \lambda \mathbf{V}]$$

$$\text{Let } \mathbf{v}_1 = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\therefore \mathbf{A}^T \mathbf{A} \mathbf{v}_1 = \lambda \mathbf{v}_1$$

$$\begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 8 \begin{pmatrix} x \\ y \end{pmatrix}$$

[For $\lambda = 8$ & from (iv) putting the value of $\mathbf{A}^T \mathbf{A}$]

$$\begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 8x \\ 8y \end{pmatrix}$$

$$\begin{pmatrix} 5x - 3y \\ -3x + 5y \end{pmatrix} = \begin{pmatrix} 8x \\ 8y \end{pmatrix}$$

$$\Rightarrow 5x - 3y = 8x$$

$$-3x + 5y = 8y$$

$$\Rightarrow -3x - 3y = 0$$

$$-3x - 3y = 0$$

$$\Rightarrow x + y = 0$$

$$x + y = 0$$

$$\Rightarrow \cancel{x} + y = 0$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in particular $(2-1)=1$ free variable, which is y .

Let $y = 1 \therefore x = -1$

$\therefore \mathbf{v}_1 = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$; It is not unit vector or normal vector. Since the length of $\mathbf{v}_1$

is $|\mathbf{v}_1| = \sqrt{(-1)^2 + (1)^2} = \sqrt{2}$; this vector $\mathbf{v}_1$ has to be converted into unit vector by dividing its magnitude or its length.

We have, $\mathbf{v}_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$

$$\therefore \mathbf{v}_1 = \begin{pmatrix} \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

[Dividing by its magnitude or its length $\sqrt{2}$]

$$\mathbf{v}_1 = \begin{pmatrix} \frac{-\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{pmatrix}$$

[Multiplying by $\sqrt{2}$ on numerator and denominator]

Similarly,

We have, $\mathbf{A}\mathbf{v}_2 = \lambda\mathbf{v}_2$

$$[\mathbf{A}\mathbf{V} = \lambda\mathbf{V}]$$

Let $\mathbf{v}_2 = \begin{pmatrix} x \\ y \end{pmatrix}$

$$\therefore \mathbf{A}^T \mathbf{A}\mathbf{v}_2 = \lambda\mathbf{v}_2$$

$$\begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 2 \begin{pmatrix} x \\ y \end{pmatrix}$$

[For $\lambda = 2$ & from (iv) putting the value of $A^T A$]

$$\begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x \\ 2y \end{pmatrix}$$

$$\begin{pmatrix} 5x - 3y \\ -3x + 5y \end{pmatrix} = \begin{pmatrix} 2x \\ 2y \end{pmatrix}$$

$$\Rightarrow 3x - 3y = 0$$

$$-3x + 3y = 0$$

$$\Rightarrow x - y = 0$$

$$x - y = 0$$

$$\Rightarrow x - y = 0$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in particular $(2-1)=1$ free variable, which is y .

Let $y = 1 \therefore x = 1$

$$\therefore \mathbf{v}_2 = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ It is not unit vector or normal vector. Since the length of } \mathbf{v}_2$$

is $|\mathbf{v}_2| = \sqrt{(1)^2 + (1)^2} = \sqrt{2}$; this vector $\mathbf{v}_2$ has to be converted into unit vector by dividing its magnitude or its length.

We have,

$$\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\text{Thus, } \mathbf{v}_2 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

[Dividing by its magnitude or its length $\sqrt{2}$]

$$\Rightarrow \mathbf{v}_2 = \begin{pmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{pmatrix}$$

[Multiplying by $\sqrt{2}$ on numerator and denominator]

Now we need to construct the augmented orthogonal matrix V ,

$$V = (\mathbf{v}_1 \quad \mathbf{v}_2) = \begin{pmatrix} \frac{-\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$$

$$\therefore V^T = \begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} \text{-----(vi)}$$

From (ii), (iii) & (vi), Putting the values of U, S, V^T in the relation $A = USV^T$

Therefore $A = USV^T$

$$A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix} \begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} \text{Answer}$$

Justification Test:

$$A = USV^T$$

$$A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix} \begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} \text{[Putting the value of } U, V^T, S]$$

$$A = USV^T$$

$$A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -2\sqrt{2} \times \frac{\sqrt{2}}{2} + 0 & 2\sqrt{2} \times \frac{\sqrt{2}}{2} + 0 \\ 0 + \sqrt{2} \times \frac{\sqrt{2}}{2} & 0 + \sqrt{2} \times \frac{\sqrt{2}}{2} \end{pmatrix}$$

$$A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -2 & 2 \\ 1 & 1 \end{pmatrix}$$

$$A = \begin{pmatrix} -1 \times (-2) + 0 & (-1) \times 2 + 0 \\ 0 + 1 & 0 + 1 \end{pmatrix}$$

$$A = \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix} \text{[Given Matrix] Proved}$$

Example 69: Compute Singular Value Decomposition (SVD) for the given matrix

$$A = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix}$$

In order to find U, In order to find U, Now we need to find the eigenvectors for AA^T

Given,

$$A = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix}$$

$$\therefore A^T = \begin{bmatrix} 3 & -1 \\ 1 & 3 \\ 1 & 1 \end{bmatrix}$$

$$\text{So, } \mathbf{A}\mathbf{A}^T = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & -1 \\ 1 & 3 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 11 & 1 \\ 1 & 11 \end{bmatrix} \text{-----(i)}$$

Now we need find the eigenvalues of $\mathbf{A}\mathbf{A}^T$

$$|\mathbf{A}\mathbf{A}^T - \lambda \mathbf{I}| = 0$$

$$\left| \begin{pmatrix} 11 & 1 \\ 1 & 11 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right| = 0$$

$$\left| \begin{pmatrix} 11 & 1 \\ 1 & 11 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} \right| = 0$$

Therefore,

$$\begin{vmatrix} 11-\lambda & 1-0 \\ 1-0 & 11-\lambda \end{vmatrix} = 0$$

$$(11-\lambda)(11-\lambda) - 1 = 0$$

$$\Rightarrow 121 - 11\lambda - 11\lambda + \lambda^2 - 1 = 0$$

$$\Rightarrow \lambda^2 - 22\lambda + 121 - 1 = 0$$

$$\Rightarrow \lambda^2 - 22\lambda + 120 = 0$$

$$\Rightarrow \lambda^2 - 10\lambda - 12\lambda + 120 = 0$$

$$\Rightarrow (\lambda - 10)(\lambda - 12) = 0$$

$$\Rightarrow \lambda = 10, 12$$

Therefore eigenvalues are 10 and 12. We construct matrix D by placing eigenvalues along the main diagonal in decreasing order.

$$\mathbf{D} = \begin{pmatrix} 12 & 0 \\ 0 & 10 \end{pmatrix}$$

Therefore, taking the square root of matrix D gives,

$$\text{The matrix of singular values } \mathbf{S} = \sqrt{\mathbf{D}} = \begin{pmatrix} \sqrt{12} & 0 \\ 0 & \sqrt{10} \end{pmatrix} \text{-----(ii)}$$

Now we need to find the eigenvector of $\mathbf{A}\mathbf{A}^T$

$$\text{We have, } \mathbf{A}\mathbf{u}_1 = \lambda \mathbf{u}_1$$

$$[\mathbf{A}\mathbf{V} = \lambda \mathbf{V}]$$

$$\text{Let } \mathbf{u}_1 = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\therefore \mathbf{A}\mathbf{A}^T \mathbf{u}_1 = \lambda \mathbf{u}_1$$

$$\begin{pmatrix} 11 & 1 \\ 1 & 11 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{pmatrix} 11 & 1 \\ 1 & 11 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 12 \begin{pmatrix} x \\ y \end{pmatrix}$$

[For $\lambda = 12$ & from (i) putting the value of $\mathbf{A}\mathbf{A}^T$]

$$\begin{pmatrix} 11 & 1 \\ 1 & 11 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 12x \\ 12y \end{pmatrix}$$

$$\begin{pmatrix} 11x + y \\ x + 11y \end{pmatrix} = \begin{pmatrix} 12x \\ 12y \end{pmatrix}$$

$$\Rightarrow 11x + y = 12x$$

$$x + 11y = 12y$$

$$\Rightarrow -x + y = 0$$

$$x - y = 0$$

$$\Rightarrow x - y = 0$$

$$x - y = 0$$

$$\Rightarrow x - y = 0$$

Let $y = 1 \therefore x = 1$

$\therefore \mathbf{u}_1 = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ It is not unit vector or normal vector. Since the length of $\mathbf{u}_1$

is $|\mathbf{u}_1| = \sqrt{(1)^2 + (1)^2} = \sqrt{2}$; this vector $\mathbf{u}_1$ has to be converted into unit vector by dividing its magnitude or its length.

Since U has an orthonormal basis, $\mathbf{u}_1$ needs to be of length one. We divide $\mathbf{u}_1$ by its magnitude to accomplish this,

$$\text{Thus, } \therefore \mathbf{u}_1 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

[Dividing by its magnitude or its length $\sqrt{2}$]

Again,

We have, $A\mathbf{u}_2 = \lambda\mathbf{u}_2$

$$[AV = \lambda V]$$

$$\therefore AA^T \mathbf{u}_2 = \lambda \mathbf{u}_2$$

$$\text{Let } \mathbf{u}_2 = \begin{pmatrix} x \\ y \end{pmatrix}$$

For $\lambda = 10$

$$\begin{pmatrix} 11 & 1 \\ 1 & 11 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 10 \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{pmatrix} 11 & 1 \\ 1 & 11 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 10 \begin{pmatrix} x \\ y \end{pmatrix}$$

[For $\lambda = 10$ & from (i) putting the value of AA^T]

$$\begin{pmatrix} 11 & 1 \\ 1 & 11 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 10x \\ 10y \end{pmatrix}$$

$$\begin{pmatrix} 11x + y \\ x + 11y \end{pmatrix} = \begin{pmatrix} 10x \\ 10y \end{pmatrix}$$

$$\Rightarrow 11x + y = 10x$$

$$x + 11y = 10y$$

$$\Rightarrow x + y = 0$$

$$x + y = 0$$

$$\Rightarrow x + y = 0$$

$$x + y = 0$$

$$\Rightarrow x + y = 0$$

Let $y = -1 \therefore x = 1$

$\therefore \mathbf{u}_2 = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ It is not unit vector or normal vector. Since the length of $\mathbf{u}_2$

is $|\mathbf{u}_2| = \sqrt{(1)^2 + (-1)^2} = \sqrt{2}$; this vector $\mathbf{u}_2$ has to be converted into unit vector by dividing its magnitude or its length.

Since U has an orthonormal basis, $\mathbf{u}_2$ needs to be of length one. We divide $\mathbf{u}_2$ by its magnitude to accomplish this,

$$\text{Thus, } \therefore \mathbf{u}_2 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{-1}{\sqrt{2}} \end{pmatrix} \text{ [Dividing by its magnitude or its length } \sqrt{2} \text{]}$$

Since $\mathbf{u}_2$ is already of length one, dividing by the magnitude we get the same vector back,

Setting up the augmented matrix U

$$U = (\mathbf{u}_1 \quad \mathbf{u}_2) = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{bmatrix} \text{ -----(iii)}$$

Now we need to find the eigenvectors of $A^T A$ which are the columns of V .

Start with the matrix

$$A = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix}$$

In order to find V , we have to start with $A^T A$ the transpose of A is

$$A^T = \begin{bmatrix} 3 & -1 \\ 1 & 3 \\ 1 & 1 \end{bmatrix}$$

$$\begin{aligned}
\therefore \mathbf{A}^T \mathbf{A} &= \begin{bmatrix} 3 & -1 \\ 1 & 3 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix} \\
&= \begin{pmatrix} 3 \times 3 + 1 & 3 - 3 & 3 - 1 \\ 3 - 3 & 1 + 9 & 1 + 3 \\ 3 - 1 & 1 + 3 & 1 + 1 \end{pmatrix} \\
&= \begin{pmatrix} 10 & 0 & 2 \\ 0 & 10 & 4 \\ 2 & 4 & 2 \end{pmatrix} \quad \text{-----(iv)}
\end{aligned}$$

Now we need to find the eigenvalue of $\mathbf{A}^T \mathbf{A}$

$$\begin{aligned}
|\mathbf{A}^T \mathbf{A} - \lambda \mathbf{I}| &= 0 \\
\left| \begin{pmatrix} 10 & 0 & 2 \\ 0 & 10 & 4 \\ 2 & 4 & 2 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right| &= 0
\end{aligned}$$

$$\left| \begin{pmatrix} 10 & 0 & 2 \\ 0 & 10 & 4 \\ 2 & 4 & 2 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix} \right| = 0$$

Therefore,

$$\left| \begin{pmatrix} 10 - \lambda & 0 & 2 \\ 0 & 10 - \lambda & 4 \\ 2 & 4 & 2 - \lambda \end{pmatrix} \right| = 0$$

$$\begin{vmatrix} 10 - \lambda & 0 & 2 \\ 0 & 10 - \lambda & 4 \\ 2 & 4 & 2 - \lambda \end{vmatrix} = 0$$

$$(10 - \lambda)\{(10 - \lambda)(2 - \lambda) - 16\} + 2\{0 - 2(10 - \lambda)\} = 0$$

$$\Rightarrow (10 - \lambda)(20 - 10\lambda - 2\lambda + \lambda^2 - 16) - 4(10 - \lambda) = 0$$

$$\Rightarrow (10 - \lambda)(20 - 10\lambda - 2\lambda + \lambda^2 - 16 - 4) = 0$$

$$\Rightarrow (10 - \lambda)(\lambda^2 - 12\lambda) = 0$$

$$\Rightarrow \lambda(10 - \lambda)(\lambda - 12) = 0$$

$$\Rightarrow \lambda = 0, 10, 12$$

Therefore eigenvalues are 0, 10 and 12. We construct matrix D by placing eigenvalues along the main diagonal in decreasing order.

$$\mathbf{D} = \begin{pmatrix} 12 & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Therefore, taking the square root of matrix D gives,

The matrix of singular values $S = \sqrt{D} = \begin{pmatrix} \sqrt{12} & 0 & 0 \\ 0 & \sqrt{10} & 0 \\ 0 & 0 & 0 \end{pmatrix}$ -----(v)

Now we need to find the eigenvector of $A^T A$ which, are the columns of V.

We have, $Av_1 = \lambda v_1$

$[AV = \lambda V]$

$\therefore A^T Av_1 = \lambda v_1$

Let $v_1 = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$

$A^T Av_1 = \lambda v_1$

$\begin{pmatrix} 10 & 0 & 2 \\ 0 & 10 & 4 \\ 2 & 4 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 12 \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ [For $\lambda = 12$ & from (iv) putting the value of $A^T A$]

$\begin{pmatrix} 10x + 0 + 2z \\ 0 + 10y + 4z \\ 2x + 4y + 2z \end{pmatrix} = \begin{pmatrix} 12x \\ 12y \\ 12z \end{pmatrix}$

$\Rightarrow 10x + 2z = 12x$

$10y + 4z = 12y$

$2x + 4y + 2z = 12z$

$\Rightarrow 2x + 4y + 2z = 12z$

$10x + 2z = 12x$

$10y + 4z = 12y$

$\Rightarrow 2x + 4y - 10z = 0$

$-2x + 2z = 0$

$-2y + 4z = 0$

$\Rightarrow 2x + 4y - 10z = 0$

$-2x + 0.y + 2z = 0$

$-2y + 4z = 0$

We have,

$L_i \rightarrow -a_{ii}L_1 + a_{ii}L_i$

$i = 2,$

$L_2 \rightarrow -a_{21}L_1 + a_{ii}L_2$

$\rightarrow -(-2)(2x + 4y - 10z) + 2(-2x + 0.y + 2z)$

$\rightarrow 4x + 8y - 20z - 4x + 0 + 4z$

$\rightarrow 8y - 16z$

Thus we obtain a new system,

$$2x + 4y - 10z = 0$$

$$8y - 16z = 0$$

$$-2y + 4z = 0$$

$$\Rightarrow 2x + 4y - 10z = 0$$

$$y - 2z = 0$$

$$y - 2z = 0$$

Since the 1st and 2nd equations are identical. We can disregard one of them

$$\begin{array}{l} 2x + 4y - 10z = 0 \\ y - 2z = 0 \end{array}$$

In echelon form, there are only two equations in three unknowns, then the system has a non-zero solution and in particular $(3-2)=1$ free variable, which is z .

Here free variable is z , Let $z = 1$, $\therefore y = 2$, then $2x + 4 \times 2 - 10 \times 1 = 0$

$$\therefore x = 1$$

$$\therefore v_1 = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$$

$$|v_1| = \sqrt{(1)^2 + (2)^2 + (1)^2} = \sqrt{6}$$

Since V has an orthonormal basis, v_1 needs to be of length one. We divide v_1 by its

magnitude to accomplish this, Thus, $\therefore v_1 = \begin{pmatrix} \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{pmatrix}$

Again,

We have, $Av_2 = \lambda v_2$

$$[AV = \lambda V]$$

$$\therefore A^T Av_2 = \lambda v_2$$

$$\text{Let } v_2 = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$A^T Av_2 = \lambda v_2$$

$$\begin{pmatrix} 10 & 0 & 2 \\ 0 & 10 & 4 \\ 2 & 4 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 10 \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

[For $\lambda = 10$ & from (iv) putting the value of $A^T A$]

$$\begin{pmatrix} 10x + 0 + 2z \\ 0 + 10y + 4z \\ 2x + 4y + 2z \end{pmatrix} = \begin{pmatrix} 10x \\ 10y \\ 10z \end{pmatrix}$$

$$\Rightarrow 10x + 2z = 10x$$

$$10y + 4z = 10y$$

$$2x + 4y + 2z = 10z$$

$$\Rightarrow 2x + 4y + 2z = 10z$$

$$10x + 2z = 10x$$

$$10y + 4z = 10y$$

$$\Rightarrow 2x + 4y - 8z = 0$$

$$2z = 0$$

$$4z = 0$$

$$\Rightarrow 2x + 4y - 8z = 0$$

$$2z = 0$$

$$2z = 0$$

$$\Rightarrow 2x + 4y - 8z = 0$$

$$2z = 0$$

$$\Rightarrow 2x + 4y - 8z = 0$$

$$z = 0$$

$$\therefore z = 0$$

$$\text{Then } \Rightarrow 2x + 4y - 8 \times 0 = 0$$

$$\Rightarrow 2x + 4y = 0$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in particular $(2-1)=1$ free variable, which is y.

Here free variable is y, Let $y = -1$, then $x = 2$

$$\therefore v_2 = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}$$

$$|v_2| = \sqrt{(2)^2 + (-1)^2 + (0)^2} = \sqrt{5}$$

Since V has an orthonormal basis, v_2 needs to be of length one. We divide v_2 by it's

$$\text{magnitude to accomplish this, Thus, } \therefore v_2 = \begin{pmatrix} 2 \\ \sqrt{5} \\ -1 \\ \sqrt{5} \\ 0 \end{pmatrix}$$

Again,

$$\text{We have, } Av_3 = \lambda v_3$$

$$[AV = \lambda V]$$

$$\text{We have, } A^T Av_3 = \lambda v_3$$

$$\text{Let } v_3 = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$A^T Av_3 = \lambda v_3$$

$$\begin{pmatrix} 10 & 0 & 2 \\ 0 & 10 & 4 \\ 2 & 4 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

[For $\lambda = 0$ & from (iv) putting the value of $A^T A$]

$$\begin{pmatrix} 10x + 0 + 2z \\ 0 + 10y + 4z \\ 2x + 4y + 2z \end{pmatrix} = \begin{pmatrix} 0x \\ 0y \\ 0z \end{pmatrix}$$

$$\Rightarrow 10x + 2z = 0$$

$$10y + 4z = 0$$

$$2x + 4y + 2z = 0$$

$$\Rightarrow 2x + 4y + 2z = 0$$

$$10x + 2z = 0$$

$$10y + 4z = 0$$

$$\Rightarrow 2x + 4y + 2z = 0$$

$$10x + 0.y + 2z = 0$$

$$10y + 4z = 0$$

$$L_i \rightarrow -a_{i1}L_1 + a_{i1}L_i$$

$$i = 2,$$

$$L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$\rightarrow -10(2x + 4y + 2z) + 2(10x + 0.y + 2z)$$

$$\rightarrow -20x - 40y - 20z + 20x + 0 + 4z$$

$$\rightarrow -40y - 16z$$

Thus we obtain the following system:

$$\begin{aligned}\Rightarrow 2x + 4y + 2z &= 0 \\ -40y - 16z &= 0 \\ 10y + 4z &= 0\end{aligned}$$

$$\begin{aligned}\Rightarrow 2x + 4y + 2z &= 0 \\ 10y + 4z &= 0 \\ 10y + 4z &= 0\end{aligned}$$

$$\Rightarrow \cancel{2x + 4y + 2z = 0} \\ \cancel{10y + 4z = 0}$$

In echelon form, there are only two equations in three unknowns, then the system has a non-zero solution and in particular $(3-2)=1$ free variable, which is z

Here free variable is z , Let $z = -5$, then $y = 2$, then $x = 1$

$$\therefore \mathbf{v}_3 = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ -5 \end{pmatrix}$$

$$|\mathbf{v}_3| = \sqrt{(1)^2 + (2)^2 + (-5)^2} = \sqrt{30}$$

Since V has an orthonormal basis, $\mathbf{v}_3$ needs to be of length one. We divide $\mathbf{v}_3$ by it's

magnitude to accomplish this, Thus, $\therefore \mathbf{v}_3 = \begin{pmatrix} \frac{1}{\sqrt{30}} \\ \frac{2}{\sqrt{30}} \\ \frac{-5}{\sqrt{30}} \end{pmatrix}$

Now we need to construct the augmented orthogonal matrix V ,

$$V = (\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3)$$

All this to give us

$$V = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{30}} \\ \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{5}} & \frac{2}{\sqrt{30}} \\ \frac{1}{\sqrt{6}} & 0 & \frac{-5}{\sqrt{30}} \end{bmatrix}$$

when we really want its transpose

$$V^T = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{5}} & \frac{-1}{\sqrt{5}} & 0 \\ \frac{1}{\sqrt{30}} & \frac{2}{\sqrt{30}} & \frac{-5}{\sqrt{30}} \end{bmatrix}$$

For S we take the square roots of the non-zero eigenvalues and populate the diagonal with them, putting the largest in s_{11} , the next largest in s_{22} and so on until the smallest value ends up in s_{mn} . The non-zero eigenvalues of U and V are always the same, so that's why it doesn't matter which one we take them from. Because we are doing full SVD, instead of reduced SVD (next section), we have to add a zero column vector to S so that it is of the proper dimensions to allow multiplication between U and V. The diagonal entries in S are the singular values of A, the columns in U are called left singular vectors, and the columns in V are called right singular vectors.

$$S = \begin{bmatrix} \sqrt{12} & 0 & 0 \\ 0 & \sqrt{10} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\Rightarrow S = \begin{bmatrix} \sqrt{12} & 0 & 0 \\ 0 & \sqrt{10} & 0 \end{bmatrix} \text{ [Ignoring third row of the matrix as all elements are zero in the last row]}$$

Now we have all the pieces of the puzzle

$$A = USV^T$$

Putting the value of U, V^T , S in $A = USV^T$

$$A = USV^T$$

$$= \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \sqrt{12} & 0 & 0 \\ 0 & \sqrt{10} & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{5}} & \frac{-1}{\sqrt{5}} & 0 \\ \frac{1}{\sqrt{30}} & \frac{2}{\sqrt{30}} & \frac{-5}{\sqrt{30}} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\sqrt{12}}{\sqrt{2}} & \frac{\sqrt{10}}{\sqrt{2}} & 0 \\ \frac{\sqrt{12}}{\sqrt{2}} & \frac{-\sqrt{10}}{\sqrt{2}} & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{5}} & \frac{-1}{\sqrt{5}} & 0 \\ \frac{1}{\sqrt{30}} & \frac{2}{\sqrt{30}} & \frac{-5}{\sqrt{30}} \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix} \text{ [Proved]}$$

Home Task:

Find the singular value decomposition of the matrix

i. $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$ ii. $A = \begin{bmatrix} 2 & 1 & -2 \end{bmatrix}$

iii. $A = \begin{bmatrix} 2 & 4 \\ 1 & 3 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$

Hints:

$$U = \begin{bmatrix} 0.82 & -0.58 & 0 & 0 \\ 0.58 & 0.82 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$V = \begin{bmatrix} 0.40 & -0.91 \\ 0.91 & 0.40 \end{bmatrix}$$

$$S = \begin{bmatrix} 5.47 & 0 \\ 0 & 0.37 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Singular Value Decomposition:

The Singular Value Decomposition is an important factorization of a real or complex matrix, with several applications in signal processing and statistics. Applications, which employ the SVD, include computing the pseudo-inverse, matrix approximation, and determining the rank, range and null space of a matrix.

The spectral theorem says that normal matrices, which by definition must be square, can be unitarily diagonalized using a basis of Eigen vectors. The SVD can be seen as a generalization of the spectral theorem to arbitrary, not necessarily square, matrices.

Algorithm of Singular Value Decomposition:

To find the Singular Value Decomposition of the matrix:

1. Find the eigenvalues of the matrix $\mathbf{A}^T \mathbf{A}$ and arrange them in descending order
2. Find the number of nonzero eigenvalues of the matrix $\mathbf{A}^T \mathbf{A}$
3. Find the orthogonal eigenvectors of the matrix $\mathbf{A}^T \mathbf{A}$ corresponding to the obtained eigenvalues, and arrange them in the same order to form the column-vectors of the matrix $\mathbf{V}$
4. Form a diagonal matrix $\mathbf{S} = \sum = \sigma$ placing on the leading diagonal of it the square roots $\sigma_i = \sqrt{\lambda_i}$ of $\mathbf{p} = \min\{\mathbf{m}, \mathbf{n}\}$ first eigenvalues of the matrix $\mathbf{A}^T \mathbf{A}$ got in I in descending order
5. Find the first column-vectors of the matrix $\mathbf{U}$, $\mathbf{u}_i = \sigma_i^{-1} \mathbf{A} \mathbf{v}_i (\mathbf{i} = 1 : \mathbf{r})$
6. Add to the matrix $\mathbf{U}$ the rest of $\mathbf{m}-\mathbf{r}$ vectors using the Gram-Schmidt Orthogonalization process

Image Processing with SVD

We want to approximate the $\mathbf{m} \times \mathbf{n}$ matrix $\mathbf{A}$ by using far fewer entries than in the original matrix.

$$\mathbf{A} = \mathbf{U} \mathbf{S} \mathbf{V}^T = \mathbf{S} \mathbf{U} \mathbf{V}^T$$

$$\Rightarrow \mathbf{A} = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \sigma_2 \mathbf{u}_2 \mathbf{v}_2^T + \dots + \sigma_r \mathbf{u}_r \mathbf{v}_r^T + 0 \cdot \mathbf{u}_{r+1} \mathbf{v}_{r+1}^T + \dots$$

Since the singular values are always greater than zero. Adding on the dependent terms where the singular values are equal to zero does not effect the image. The terms at the end of the equation zero out leaving us with:

$$\Rightarrow \mathbf{A} = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \sigma_2 \mathbf{u}_2 \mathbf{v}_2^T + \dots + \sigma_r \mathbf{u}_r \mathbf{v}_r^T$$

Conclusion: We can further approximate the matrix (original image) by leaving off more singular terms of the matrix $\mathbf{A}$. Since the singular values are arranged in descending order, the last terms will have the least effect on the overall image.

Doing this (leaving more singular values) reduces the amount of space required to store the image on a computer.

Justification using above example:

The original image is: $\mathbf{A} = \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix}$

$$\mathbf{S} = \begin{pmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix} \text{ i.e. } \sigma_1 = 2\sqrt{2}, \sigma_2 = \sqrt{2}$$

$$\mathbf{U} = (\mathbf{u}_1 \quad \mathbf{u}_2) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\mathbf{V} = (\mathbf{v}_1 \quad \mathbf{v}_2) = \begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{2}{\sqrt{2}} & \frac{2}{\sqrt{2}} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$$

$$\begin{aligned}
\Rightarrow A &= \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T \\
&= 2\sqrt{2} \begin{pmatrix} -1 \\ 0 \end{pmatrix} \begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} + \sqrt{2} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} \\
&= \begin{pmatrix} -2\sqrt{2} \\ 0 \end{pmatrix} \begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} + \begin{pmatrix} 0 \\ \sqrt{2} \end{pmatrix} \begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} \\
&= \begin{pmatrix} 2 & -2 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} \\
&= \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix} = A \text{ (Original Image can be obtained)}
\end{aligned}$$

We can get approximate image by using less singular values. Say number of singular values are 10 i.e. $\sigma_1, \sigma_2, \dots, \sigma_{10}$.

Then,

$$\Rightarrow A = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \dots + \sigma_{10} u_{10} v_{10}^T$$

If we use 5 singular values then

$$\Rightarrow A = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \dots + \sigma_5 u_5 v_5^T$$

In this result we can get approximate image of the original image (A)

Since large eigenvalues carry significant representation of the matrix, we arrange the values of the Eigen Values from the largest to the smallest and drop/ignore the values towards the last since small eigenvalues are less indicator.

What is the use of SVD?

01. SVD can be used to compute optimal low-rank approximations of arbitrary matrices.
02. Face recognition: represent the face images as eigenfaces and compute distance between the query face image in the principal component space.

B. LU Decomposition

A procedure for decomposing a $N \times N$ matrix A into a product of a lower triangular matrix L and an upper triangular matrix U,

$$A = LU$$

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = \begin{pmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{pmatrix} \begin{pmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{pmatrix}$$

LU decomposition is implemented in *Mathematica* as LU Decomposition[m].

Example 70

Solve the linear system of equations using LU decomposition

$$u_1 + 2u_2 + 3u_3 = 5$$

$$2u_1 - 4u_2 + 6u_3 = 18$$

$$3u_1 - 9u_2 - 3u_3 = 6$$

Solution:

Given the linear system of equations

$$u_1 + 2u_2 + 3u_3 = 5$$

$$2u_1 - 4u_2 + 6u_3 = 18$$

$$3u_1 - 9u_2 - 3u_3 = 6$$

This can be written as

$$\mathbf{A}\mathbf{u} = \mathbf{b} \quad \text{-----(i)}$$

Where,

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & -4 & 6 \\ 3 & -9 & -3 \end{pmatrix} \quad \mathbf{u} = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 5 \\ 18 \\ 6 \end{pmatrix}$$

Again, According to LU decomposition,

$$\mathbf{A} = \mathbf{L}\mathbf{U} \quad \text{-----(ii)}$$

Where,

$$\mathbf{L} = \begin{pmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{pmatrix} \quad \text{and} \quad \mathbf{U} = \begin{pmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{pmatrix}$$

From (ii), $\mathbf{A} = \mathbf{L}\mathbf{U}$

$$\Rightarrow \mathbf{L}\mathbf{U} = \mathbf{A}$$

$$\begin{pmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{pmatrix} \begin{pmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & -4 & 6 \\ 3 & -9 & -3 \end{pmatrix}$$

$$\begin{pmatrix} l_{11} + 0 + 0 & l_{11}u_{12} + 0 + 0 & l_{11}u_{13} + 0 + 0 \\ l_{21} + 0 + 0 & l_{21}u_{12} + l_{22} + 0 & l_{21}u_{13} + l_{22}u_{23} + 0 \\ l_{31} + 0 + 0 & l_{31}u_{12} + l_{32} + 0 & l_{31}u_{13} + l_{32}u_{23} + l_{33} \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & -4 & 6 \\ 3 & -9 & -3 \end{pmatrix}$$

Now,

We can write

$$l_{11} = 1 \quad \text{-----(iii)}$$

$$l_{11}u_{12} = 2 \quad \text{-----(iv)}$$

$$l_{11}u_{13} = 3 \quad \text{-----(v)}$$

$$l_{21} = 2 \quad \text{-----(vi)}$$

$$l_{21}u_{12} + l_{22} = -4 \quad \text{-----(vii)}$$

$$l_{21}u_{13} + l_{22}u_{23} = 6 \quad \text{-----(viii)}$$

$$l_{31} = 3 \quad \text{----- (ix)}$$

$$l_{31}u_{12} + l_{32} = -9 \quad \text{----- (x)}$$

$$l_{31}u_{13} + l_{32}u_{23} + l_{33} = -3 \quad \text{----- (xi)}$$

From (iii) & (iv)

$$l_{11} = 1$$

$$l_{11}u_{12} = 2$$

$$1.u_{12} = 2$$

$$u_{12} = 2 \quad \text{----- (xii)}$$

From (iii) & (v)

$$l_{11}u_{13} = 3$$

$$1.u_{13} = 3$$

$$u_{13} = 3 \quad \text{----- (xiii)}$$

From (vii),

$$l_{21}u_{12} + l_{22} = -4$$

$$2.2 + l_{22} = -4 \quad [u_{12} = 2 ; l_{21} = 2]$$

$$4 + l_{22} = -4$$

$$\therefore l_{22} = -8 \quad \text{----- (xiv)}$$

From (viii),

$$l_{21}u_{13} + l_{22}u_{23} = 6$$

$$2.3 - 8u_{23} = 6 \quad [l_{21} = 2; u_{13} = 3; \therefore l_{22} = -8]$$

$$-8u_{23} = 6 - 6$$

$$u_{23} = 0 \quad \text{-----(xv)}$$

From (x),

$$l_{31}u_{12} + l_{32} = -9$$

$$3.2 + l_{32} = -9 \quad [l_{31} = 3; u_{12} = 2]$$

$$l_{32} = -9 - 6$$

$$l_{32} = -15 \quad \text{-----(xvi)}$$

From (xi)

$$l_{31}u_{13} + l_{32}u_{23} + l_{33} = -3$$

$$3.3 - 15.0 + l_{33} = -3 \quad [l_{31} = 3; u_{13} = 3; l_{32} = -15; u_{23} = 0]$$

$$l_{33} = -3 - 9$$

$$l_{33} = -12 \quad \text{-----(xvii)}$$

$$\therefore L = \begin{pmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{pmatrix}$$

$$\Rightarrow L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & -8 & 0 \\ 3 & -15 & -12 \end{pmatrix}$$

And

$$U = \begin{pmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{pmatrix}$$

$$U = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

From (i),

$$\mathbf{A}\mathbf{u} = \mathbf{b}$$

$$\mathbf{L}\mathbf{U}\mathbf{u} = \mathbf{b} \quad [\text{from (ii), } \mathbf{A} = \mathbf{LU}]$$

$$\mathbf{L}\mathbf{U}\mathbf{u} = \mathbf{b} \quad \text{-----(xviii)}$$

Let,

$$\mathbf{U}\mathbf{u} = \mathbf{v} \quad \text{-----(xix)}$$

Where,

$$\text{Where, } \mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$$

From (xviii),

$$\mathbf{L}\mathbf{U}\mathbf{u} = \mathbf{b}$$

$$\mathbf{L}\mathbf{v} = \mathbf{b}$$

$$\Rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 2 & -8 & 0 \\ 3 & -15 & -12 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 5 \\ 18 \\ 6 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 1 \times v_1 + 0 \times v_2 + 0 \times v_3 \\ 2 \times v_1 - 8 \times v_2 + 0 \times v_3 \\ 3 \times v_1 - 15 \times v_2 - 12 \times v_3 \end{pmatrix} = \begin{pmatrix} 5 \\ 18 \\ 6 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} v_1 \\ 2v_1 - 8v_2 \\ 3v_1 - 15v_2 - 12v_3 \end{pmatrix} = \begin{pmatrix} 5 \\ 18 \\ 6 \end{pmatrix}$$

$$v_1 = 5 \quad \text{-----(xx)}$$

$$2v_1 - 8v_2 = 18 \quad \text{-----(xxi)}$$

$$3v_1 - 15v_2 - 12v_3 = 6 \quad \text{-----(xxii)}$$

From (xxi),

$$2v_1 - 8v_2 = 18$$

$$2 \times 5 - 8v_2 = 18 \quad [v_1 = 5]$$

$$10 - 8v_2 = 18$$

$$-8v_2 = 18 - 10$$

$$-8v_2 = 8$$

$$\therefore v_2 = -1 \quad \text{-----(xxiii)}$$

From (xxii),

$$3v_1 - 15v_2 - 12v_3 = 6$$

$$3.5 - 15(-1) - 12v_3 = 6 \quad [v_1 = 5; v_2 = -1]$$

$$15 + 15 - 12v_3 = 6$$

$$30 - 12v_3 = 6$$

$$-12v_3 = 6 - 30$$

$$-12v_3 = -24$$

$$\therefore v_3 = 2 \quad \text{-----(xxiv)}$$

From (xix),

$$Uu = v$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} = \begin{pmatrix} 5 \\ -1 \\ 2 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 1 \times u_1 + 2 \times u_2 + 3 \times u_3 \\ 0 \times u_1 + 1 \times u_2 + 0 \times u_3 \\ 0 \times u_1 + 0 \times u_2 + 1 \times u_3 \end{pmatrix} = \begin{pmatrix} 5 \\ -1 \\ 2 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} u_1 + 2u_2 + 3u_3 \\ u_2 \\ u_3 \end{pmatrix} = \begin{pmatrix} 5 \\ -1 \\ 2 \end{pmatrix}$$

$$u_1 + 2u_2 + 3u_3 = 5 \quad \text{-----(xxv)}$$

$$u_2 = -1 \quad \text{-----(xxvi)}$$

$$u_3 = 2 \quad \text{-----(xxvii)}$$

From (xxv),

$$u_1 + 2u_2 + 3u_3 = 5$$

$$u_1 + 2(-1) + 3.2 = 5$$

$$u_1 - 2 + 6 = 5$$

$$u_1 = 5 - 4$$

$$u_1 = 1$$

$$u_1 = 1, u_2 = -1, u_3 = 2 \text{ Answer}$$

Example 71: Given the following system of linear equations, determine the value of each of the variables using the LU decomposition method.

$$6x_1 - 2x_2 = 14$$

$$9x_1 - x_2 + x_3 = 21$$

$$3x_1 - 7x_2 + 5x_3 = 9$$

Vector Analysis

THE DOT AND CROSS PRODUCT

Q # 13: Find a unit vector perpendicular to the vectors $\vec{a} = 3\hat{i} + \hat{j}$ and $\vec{b} = -\hat{i} + 2\hat{j} + 2\hat{k}$

Answer: A vector perpendicular to $\vec{a}$ and $\vec{b}$ is $\vec{a} \times \vec{b}$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 1 & 0 \\ -1 & 2 & 2 \end{vmatrix} = \hat{i}(2 - 0) - \hat{j}(6 - 0) + \hat{k}(6 + 1) = 2\hat{i} - 6\hat{j} + 7\hat{k}$$

A unit vector, perpendicular to $\vec{a}$ and $\vec{b}$, in this direction is obtained by simply dividing $\vec{a} \times \vec{b}$ by its magnitude. Thus

$$\hat{e} = \frac{2\hat{i} - 6\hat{j} + 7\hat{k}}{\sqrt{2^2 + (-6)^2 + 7^2}} = \frac{2\hat{i} - 6\hat{j} + 7\hat{k}}{\sqrt{89}} \text{ is the required vector.}$$

$$\therefore \text{Magnitude of Unit vector of } \hat{e} \\ = |\hat{e}| = \sqrt{\left(\frac{2}{\sqrt{89}}\right)^2 + \left(\frac{-6}{\sqrt{89}}\right)^2 + \left(\frac{7}{\sqrt{89}}\right)^2} = \sqrt{\frac{4}{89} + \frac{36}{89} + \frac{49}{89}} = \sqrt{\frac{89}{89}} = 1$$

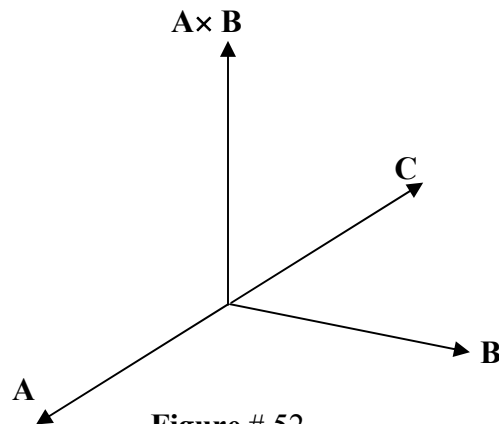
Home Task

Find a unit vector parallel to the resultant of vectors, $\vec{A} = 2\hat{i} + 4\hat{j} - 5\hat{k}$ and $\vec{B} = \hat{i} + 2\hat{j} + 3\hat{k}$

$$\hat{e} = \frac{\vec{A} + \vec{B}}{|\vec{A} + \vec{B}|}$$

Q#14: Show that $\vec{A} = \hat{i} + 2\hat{j} - 3\hat{k}$, $\vec{B} = 2\hat{i} - \hat{j} + 2\hat{k}$ and $\vec{C} = 3\hat{i} + \hat{j} - \hat{k}$ are coplanar.

Answer:



If $\vec{A}$ is a third vector perpendicular to $(\vec{B} \times \vec{C})$, then $\vec{A}$, $\vec{B}$ and $\vec{C}$ are coplanar and $\vec{A} \cdot (\vec{B} \times \vec{C}) = 0$

Therefore, three vectors $\vec{A}$, $\vec{B}$, $\vec{C}$ are coplanar if $\vec{A} \cdot (\vec{B} \times \vec{C}) = 0$

$$\vec{B} \times \vec{C} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -1 & 2 \\ 3 & 1 & -1 \end{vmatrix} = \hat{i}(1-2) - \hat{j}(-2-6) + \hat{k}(2+3) = -\hat{i} + 8\hat{j} + 5\hat{k}$$

$$\vec{A} \cdot (\vec{B} \times \vec{C}) = (\hat{i} + 2\hat{j} - 3\hat{k}) \cdot (-\hat{i} + 8\hat{j} + 5\hat{k}) = -1 + 16 - 15 = 0$$

Therefore $\vec{A}$, $\vec{B}$, $\vec{C}$ are coplanar.

Q#15: On which condition of λ the vectors $\vec{i} + 2\vec{j} + 3\vec{k}$, $\lambda\vec{i} + 4\vec{j} + 7\vec{k}$, $-3\vec{i} - 2\vec{j} - 5\vec{k}$ are collinear

Solution

We know that if the points (x_1, y_1, z_1) , (x_2, y_2, z_2) and (x_3, y_3, z_3) be collinear then

$$\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} 1 & 2 & 3 \\ \lambda & 4 & 7 \\ -3 & -2 & -5 \end{vmatrix} = 0$$

$$\Rightarrow 1[4 \times (-5) - 7 \times (-2)] - 2[\lambda(-5) - (-3) \times 7] + 3[\lambda(-2) - 4(-3)] = 0$$

$$\Rightarrow 1[-20 + 14] - 2[-5\lambda + 21] + 3[-2\lambda + 12] = 0$$

$$\Rightarrow -6 + 10\lambda - 42 - 6\lambda + 36 = 0$$

$$\Rightarrow 4\lambda - 48 + 36 = 0$$

$$\Rightarrow 4\lambda - 12 = 0$$

$$\Rightarrow 4\lambda = 12$$

$$\Rightarrow \lambda = 3$$

Q-17: If $\vec{a} \cdot \vec{b} = \sqrt{3}$ and $\vec{a} \times \vec{b} = \hat{i} + 2\hat{j} + 2\hat{k}$, find the angle between $\vec{a}$ and $\vec{b}$

Answer: We have, $\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$

Given, $\vec{a} \cdot \vec{b} = \sqrt{3}$

$$\therefore |\vec{a}| |\vec{b}| \cos \theta = \sqrt{3} \quad \text{-----(i)}$$

Again, we have, $\vec{a} \times \vec{b} = |\vec{a}| |\vec{b}| \sin \theta \hat{n}$

Given, $\vec{a} \times \vec{b} = \hat{i} + 2\hat{j} + 2\hat{k}$

$$|\vec{a}| |\vec{b}| \sin \theta \hat{n} = \hat{i} + 2\hat{j} + 2\hat{k}$$

$$\hat{n} |\vec{a}| |\vec{b}| \sin \theta = \hat{i} + 2\hat{j} + 2\hat{k} \quad \text{-----(ii)}$$

We have, $\vec{a} \times \vec{b} = |\vec{a}| |\vec{b}| \sin \theta \hat{n}$

Again, we can write, $\hat{n} = \frac{\vec{a} \times \vec{b}}{|\vec{a}| |\vec{b}| \sin \theta} = \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|} \quad \text{-----(iii)}$

Given, $\vec{a} \times \vec{b} = \hat{i} + 2\hat{j} + 2\hat{k}$

$$\therefore \left| \vec{a} \times \vec{b} \right| = \sqrt{1^2 + 2^2 + 2^2} = \sqrt{9} = 3$$

From (iii),

$$\hat{\eta} = \frac{\vec{a} \times \vec{b}}{\left| \vec{a} \times \vec{b} \right|}$$

$$\hat{\eta} = \frac{\hat{i} + 2\hat{j} + 2\hat{k}}{3}$$

Putting the value of $\hat{\eta}$ in (ii), we get,

$$\hat{\eta} \left| \vec{a} \right| \left| \vec{b} \right| \sin \theta = \hat{i} + 2\hat{j} + 2\hat{k}$$

$$\frac{\hat{i} + 2\hat{j} + 2\hat{k}}{3} \times \left| \vec{a} \right| \left| \vec{b} \right| \sin \theta = \hat{i} + 2\hat{j} + 2\hat{k}$$

$$\frac{1}{3} \times \left| \vec{a} \right| \left| \vec{b} \right| \sin \theta = 1$$

$$\left| \vec{a} \right| \left| \vec{b} \right| \sin \theta = 3 \quad \text{-----(iv)}$$

(iv) $\div$ (i)

$$\frac{\left| \vec{a} \right| \left| \vec{b} \right| \sin \theta}{\left| \vec{a} \right| \left| \vec{b} \right| \cos \theta} = \frac{3}{\sqrt{3}}$$

$$\frac{\sin \theta}{\cos \theta} = \sqrt{3}$$

$$\tan \theta = \sqrt{3}$$

$$\tan \theta = \tan 60^\circ$$

$$\therefore \theta = 60^\circ \text{ Answer}$$

Q-18: Find all vectors $\vec{v}$ such that $(\hat{i} + 2\hat{j} + \hat{k}) \times \vec{v} = 3\hat{i} + \hat{j} - 5\hat{k}$

Answer:

Given,

$$(\hat{i} + 2\hat{j} + \hat{k}) \times \vec{v} = 3\hat{i} + \hat{j} - 5\hat{k} \text{-----(i)}$$

$$\text{Let } \vec{v} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$(\hat{i} + 2\hat{j} + \hat{k}) \times \vec{v} = (\hat{i} + 2\hat{j} + \hat{k}) \times x\hat{i} + y\hat{j} + z\hat{k} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 1 \\ x & y & z \end{vmatrix}$$

$$\therefore (\hat{i} + 2\hat{j} + \hat{k}) \times \vec{v} = \hat{i}(2z - y) - \hat{j}(z - x) + \hat{k}(y - 2x) \text{-----(ii)}$$

Given,

$$(\hat{i} + 2\hat{j} + \hat{k}) \times \vec{v} = 3\hat{i} + \hat{j} - 5\hat{k}$$

From (ii), We can write,

$$\therefore (\hat{i} + 2\hat{j} + \hat{k}) \times \vec{v} = \hat{i}(2z - y) - \hat{j}(z - x) + \hat{k}(y - 2x) = 3\hat{i} + \hat{j} - 5\hat{k}$$

$$\Rightarrow \hat{i}(2z - y) - \hat{j}(z - x) + \hat{k}(y - 2x) = 3\hat{i} + \hat{j} - 5\hat{k}$$

Equating the coefficient of $\hat{i}$, $\hat{j}$, and $\hat{k}$ on both sides

$$\therefore 2z - y = 3$$

$$-(z - x) = 1$$

$$y - 2x = -5$$

That is,

$$\therefore 2z - y = 3$$

$$\Rightarrow x - z = 1$$

$$y - 2x = -5$$

$$x - z = 1$$

$$\Rightarrow y - 2x = -5$$

$$2z - y = 3$$

$$x + 0.y - z = 1$$

$$\Rightarrow -2x + y + 0.z = -5 \text{-----(iii)}$$

$$0.x - y + 2z = 3$$

We have,

$$\mathbf{L}_i \rightarrow -\mathbf{a}_{i1}\mathbf{L}_1 + \mathbf{a}_{i1}\mathbf{L}_i$$

Here, $a_{11} = 1$, $a_{12} = 0$, $a_{13} = -1$, $a_{21} = -2$, $a_{22} = 1$, $a_{23} = 0$, $a_{31} = 0$, $a_{32} = -1$, $a_{33} = 2$

1st time:

$$\mathbf{i} = 2, \mathbf{L}_2 \rightarrow -\mathbf{a}_{21}\mathbf{L}_1 + \mathbf{a}_{11}\mathbf{L}_2$$

$$= -(-2)(x + 0.y - z - 1) + 1(-2x + y + 0.z + 5)$$

$$= 2x - 2z - 2 - 2x + y + 5$$

$$= y - 2z + 3$$

$$\begin{aligned} \mathbf{i} = 3, \mathbf{L}_3 &\rightarrow -\mathbf{a}_{31}\mathbf{L}_1 + \mathbf{a}_{11}\mathbf{L}_3 \\ &= -0(x + 0.y - z - 1) + 1(0.x - y + 2z - 3) \\ &= -y + 2z - 3 \end{aligned}$$

Thus we obtain the following new system

$$\begin{aligned} x + 0.y - z &= 1 \\ y - 2z &= -3 \\ -y + 2z &= 3 \end{aligned}$$

2nd time:

$$\begin{aligned} x + 0.y - z &= 1 \\ y - 2z &= -3 \text{-----} -\mathbf{L}_1 \\ -y + 2z &= 3 \text{-----} -\mathbf{L}_2 \end{aligned}$$

We have,

$$\mathbf{L}_i \rightarrow -\mathbf{a}_{i1}\mathbf{L}_1 + \mathbf{a}_{11}\mathbf{L}_i$$

Here, $a_{11} = 1, a_{12} = -2, a_{21} = -1, a_{22} = 2$

$$\begin{aligned} \mathbf{i} = 2, \mathbf{L}_2 &\rightarrow -\mathbf{a}_{21}\mathbf{L}_1 + \mathbf{a}_{11}\mathbf{L}_2 \\ &= -(-1)(y - 2z + 3) + 1(-y + 2z - 3) \\ &= y - 2z + 3 - y + 2z - 3 \\ &= 0 \end{aligned}$$

Thus we obtain the following new system

$$\begin{aligned} x + 0.y - z &= 1 \\ y - 2z &= -3 \\ 0 &= 0 \end{aligned}$$

Thus we obtain the following new system

$$\begin{aligned} x + 0.y - z &= 1 \\ y - 2z &= -3 \end{aligned}$$

In echelon form, there are only two equations in three unknowns, then the system has a non-zero solution and in particular $(3-2)=1$ free variable, which is z . We obtain more than one solution of the system; hence we will get infinite number of vectors $\vec{v}$

$$\begin{array}{cc} \begin{array}{c} \uparrow a_{11} \\ 1 \end{array} & \begin{array}{c} \uparrow a_{12} \\ -2 \end{array} \\ y - 2z &= -3 \\ \begin{array}{c} \uparrow a_{21} \\ -1 \end{array} & \begin{array}{c} \uparrow a_{22} \\ 2 \end{array} \\ -y + 2z &= 3 \end{array}$$

Chapter 1: Differentiation of Vectors

In this chapter, the basic concept of Vector Differentiation will be discussed.
In many practical problems, we often deal with vectors that change with time, e.g. Velocity, acceleration, etc.

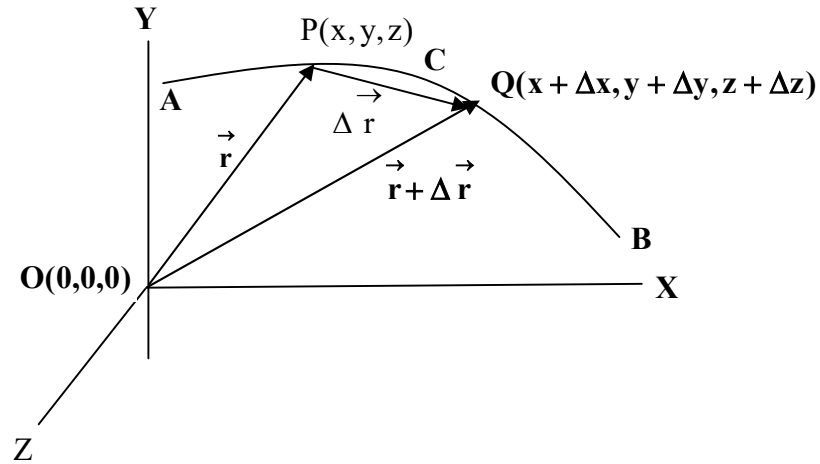


Figure # 57

We consider a position vector $\vec{OP} = \vec{r}$, which is drawn from O to P then $\vec{OP}$ moves from P to Q. Then $\Delta \vec{r}$ is small increment from P to Q. So, $\vec{OQ} = \vec{r} + \Delta \vec{r}$ is a new vector drawn from O to Q. Assume that $\vec{r}$ is a vector function of x, y, z and depends on a scalar variable t . Here, $\vec{PQ}$ represents the displacement vector of the object during the interval of time Δt

From figure,

$$\vec{OP} = \vec{r} = x\hat{i} + y\hat{j} + z\hat{k} \text{-----(i)}$$

$$\therefore \Delta \vec{r} = \Delta x\hat{i} + \Delta y\hat{j} + \Delta z\hat{k} \text{-----(ii)}$$

$$\vec{OQ} = (x + \Delta x)\hat{i} + (y + \Delta y)\hat{j} + (z + \Delta z)\hat{k}$$

$$\vec{OQ} = (x\hat{i} + y\hat{j} + z\hat{k}) + (\Delta x\hat{i} + \Delta y\hat{j} + \Delta z\hat{k})$$

$$\vec{OQ} = \vec{r} + \Delta \vec{r} \text{-----(iii)}$$

$$\vec{a} = -\omega^2 \vec{r} \text{-----(i)}$$

$$\vec{a} \propto \vec{r}$$

$$\left| \vec{a} \right| \propto \left| \vec{r} \right| \text{-----(ii)}$$

From equation (1), the acceleration is opposite to the direction of $\vec{r}$. i.e. it is directed toward the origin, Its magnitude is proportional to $\left| \vec{r} \right|$ which is the distance from the origin.

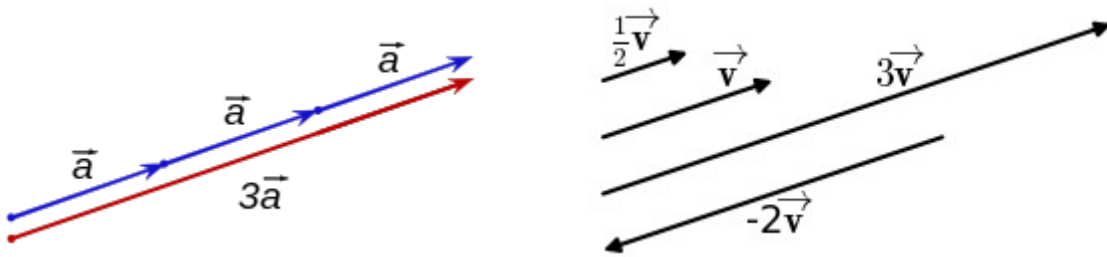


Figure # 60

c) Here $\vec{r} = \cos \omega t \hat{i} + \sin \omega t \hat{j}$ and $\vec{V} = -\omega \sin \omega t \hat{i} + \omega \cos \omega t \hat{j}$

$$\begin{aligned} \vec{r} \times \vec{V} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos \omega t & \sin \omega t & 0 \\ -\omega \sin \omega t & \omega \cos \omega t & 0 \end{vmatrix} \\ &= \hat{i}(\sin \omega t \times 0 - 0 \times \omega \cos \omega t) - \hat{j}(\cos \omega t \times 0 - (-\omega \sin \omega t) \times 0) + \hat{k}(\cos \omega t \times \omega \cos \omega t - (-\omega \sin \omega t) \times \sin \omega t) \\ &= \hat{i} \times 0 - \hat{j} \times 0 + \hat{k}(\omega \cos^2 \omega t + \omega \sin^2 \omega t) \\ &= \hat{k} \omega (\cos^2 \omega t + \sin^2 \omega t) \\ &= \hat{k} \omega.1 \quad [\because \cos^2 \omega t + \sin^2 \omega t = 1] \\ &= \hat{k} \omega \end{aligned}$$

Q#22: A particle moves along a circular path in such a way that its x- and y-coordinates at time t are $x = 2 \cos t$, $y = 2 \sin t$

a) Find the instantaneous velocity and speed of the particle at time t.

- b) Sketch the path of the particle and show the position and velocity vectors at time $t = \frac{\pi}{4}$ with the velocity vector drawn so that its initial point is at the top of the position vector
- c) Show that at each instant the acceleration vector is perpendicular to the velocity vector

Answer:

a)

Let the position vector at any time t is:

$$\vec{OP} = \vec{r}(t) = x\hat{i} + y\hat{j} \quad [\text{Page no 48, Figure no 57, Equation no (i)}]$$

At time t , the position vector is:

$$\vec{OP} = \vec{r}(t) = 2\cos t\hat{i} + 2\sin t\hat{j} \quad [\text{Given, } x = 2\cos t, y = 2\sin t] \quad \text{-----(i)}$$

So the instantaneous velocity is:

$$\vec{V}(t) = \frac{d\vec{r}}{dt} = -2\sin t\hat{i} + 2\cos t\hat{j} \quad \text{-----(ii)}$$

So the instantaneous speed is:

$$|\vec{V}(t)| = \sqrt{(-2\sin t)^2 + (2\cos t)^2} = \sqrt{4\sin^2 t + 4\cos^2 t} = 2$$

Answer (b)

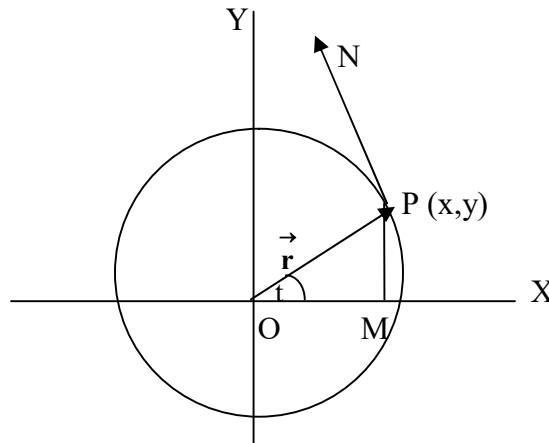


Figure #: 61

$$\vec{OP} = \vec{r}(t) = 2\cos t\hat{i} + 2\sin t\hat{j}$$

Here, $OM = x = 2\cos t$ and $PM = y = 2\sin t$

$$\angle POM = t$$

$$\frac{PM}{OP} = \sin t$$

$$\frac{2 \sin t}{OP} = \sin t \quad [\text{Given } PM = y = 2 \sin t]$$

$$OP \sin t = 2 \sin t$$

$$OP = 2 \text{ -----(iii)}$$

Similarly,

$$\frac{OM}{OP} = \cos t$$

$$\frac{2 \cos t}{OP} = \cos t \quad [OM = x = 2 \cos t]$$

$$OP \cos t = 2 \cos t$$

$$OP = 2 \text{ -----(iv)}$$

Hence the radius of the circle is $OP = 2$.

At time $t = \frac{\pi}{4}$, the position and velocity vector of the particles are:

$$\vec{OP} = \vec{r}(t) = 2 \cos t \hat{i} + 2 \sin t \hat{j} \quad [\text{From (i)}]$$

$$\vec{r}\left(\frac{\pi}{4}\right) = 2 \cos \frac{\pi}{4} \hat{i} + 2 \sin \frac{\pi}{4} \hat{j}$$

$$\vec{r}\left(\frac{\pi}{4}\right) = 2 \frac{1}{\sqrt{2}} \hat{i} + 2 \frac{1}{\sqrt{2}} \hat{j}$$

$$\vec{OP} = \vec{r}\left(\frac{\pi}{4}\right) = \sqrt{2} \hat{i} + \sqrt{2} \hat{j} \quad \text{When } t = \frac{\pi}{4}.$$

From (ii),

$$\vec{V}(t) = \frac{d\vec{r}}{dt} = -2 \sin t \hat{i} + 2 \cos t \hat{j}$$

$$\vec{V}\left(\frac{\pi}{4}\right) = -2 \sin \frac{\pi}{4} \hat{i} + 2 \cos \frac{\pi}{4} \hat{j}$$

$$\vec{PN} = \vec{V}\left(\frac{\pi}{4}\right) = -2 \frac{1}{\sqrt{2}} \hat{i} + 2 \frac{1}{\sqrt{2}} \hat{j} = \vec{V}\left(\frac{\pi}{4}\right) = -\sqrt{2} \hat{i} + \sqrt{2} \hat{j}$$

Answer (c):

We have,

$$\vec{V}(t) = \frac{d\vec{r}}{dt} = -2 \sin t \hat{i} + 2 \cos t \hat{j} \quad [\text{From (ii)}]$$

At time t , the acceleration vector is:

$$\therefore \vec{a}(t) = \frac{d\vec{v}}{dt} = -2 \cos t \hat{i} - 2 \sin t \hat{j} \text{ -----(v)}$$

Test: From (ii) & (v),

$$\vec{V}(t) \cdot \vec{a}(t) = (-2 \sin t \hat{i} + 2 \cos t \hat{j}) \cdot (-2 \cos t \hat{i} - 2 \sin t \hat{j}) = 4 \sin t \cos t - 4 \sin t \cos t = 0$$

Since the dot product of Velocity Vector (ii) and acceleration Vector (v) is Zero, Hence acceleration vector is perpendicular to the velocity vector.

(Proved)

Q#23: A particle moves through 3-space in such a way that its velocity is $\vec{v}(t) = \hat{i} + t \hat{j} + t^2 \hat{k}$. Find the co-ordinates of the particle at time $t = 1$ given that the particle is at the point $(-1, 2, 4)$ at time $t = 0$

Answer: We have,

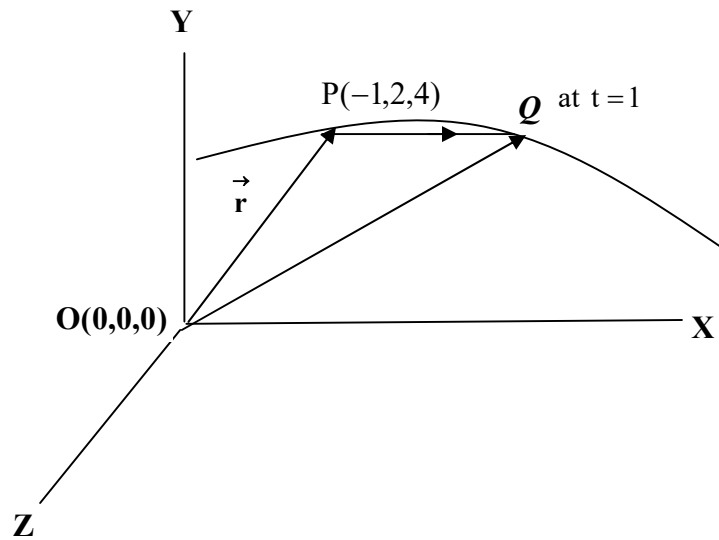


Figure 62

$$\vec{v}(t) = \frac{d\vec{r}}{dt} \text{ Where } \vec{r} \text{ is a position vector.}$$

$$\text{Given, } \vec{v}(t) = \hat{i} + t \hat{j} + t^2 \hat{k}$$

$$\therefore \vec{v}(t) = \frac{d\vec{r}(t)}{dt} = \hat{i} + t \hat{j} + t^2 \hat{k}$$

$$\frac{d\vec{r}(t)}{dt} = \hat{i} + t \hat{j} + t^2 \hat{k}$$

$$d\vec{r}(t) = (\hat{i} + t \hat{j} + t^2 \hat{k})dt \text{ -----(i)}$$

Integrate (i) both sides, we get,

$$\int d\vec{r}(t) = \int (\hat{i} + t \hat{j} + t^2 \hat{k})dt$$

$$\int d\vec{r}(t) = \int \hat{i} dt + \int t \hat{j} dt + \int t^2 \hat{k} dt$$

$$\vec{r}(t) = t \hat{i} + \frac{t^2}{2} \hat{j} + \frac{t^3}{3} \hat{k} + C \quad \text{-----(ii)}$$

Where C is a vector constant of integration. Since the coordinates of the particle at time $t = 0$ are $(-1, 2, 4)$, the position vector at time $t = 0$ is

We have the position vector

$$\vec{r}(t) = x \hat{i} + y \hat{j} + z \hat{k} \quad \text{[Page no 48, Figure no 57, Equation no (i)]}$$

$$\vec{r}(0) = (-1) \hat{i} + 2 \hat{j} + 4 \hat{k} \quad \text{[at time } t = 0, \text{ the position vector is at } (-1, 2, 4) \text{]}$$

$$\vec{r}(0) = (-1) \hat{i} + 2 \hat{j} + 4 \hat{k} \quad \text{-----(iii)}$$

Again, putting $t = 0$ in (ii), we get,

$$\vec{r}(t) = t \hat{i} + \frac{t^2}{2} \hat{j} + \frac{t^3}{3} \hat{k} + C$$

$$\vec{r}(0) = 0 \hat{i} + \frac{0^2}{2} \hat{j} + \frac{0^3}{3} \hat{k} + C$$

$$\vec{r}(0) = 0 + 0 + 0 + C$$

$$\vec{r}(0) = C \quad \text{-----(iv)}$$

Comparing (iii) and (iv), we get,

$$\vec{r}(0) = (-1) \hat{i} + 2 \hat{j} + 4 \hat{k} = C$$

$$C = -\hat{i} + 2 \hat{j} + 4 \hat{k} \quad \text{-----(v)}$$

Putting the value of C in (ii), we get,

$$\vec{r}(t) = t \hat{i} + \frac{t^2}{2} \hat{j} + \frac{t^3}{3} \hat{k} + C$$

$$\vec{r}(t) = t \hat{i} + \frac{t^2}{2} \hat{j} + \frac{t^3}{3} \hat{k} - \hat{i} + 2 \hat{j} + 4 \hat{k}$$

$$\vec{r}(t) = (t-1) \hat{i} + \left(\frac{t^2}{2} + 2\right) \hat{j} + \left(\frac{t^3}{3} + 4\right) \hat{k} \quad \text{-----(vi)}$$

Thus, at time $t = 1$, the position vector of the particle is

From (vi),

$$\vec{r}(t) = (t-1) \hat{i} + \left(\frac{t^2}{2} + 2\right) \hat{j} + \left(\frac{t^3}{3} + 4\right) \hat{k}$$

$$\vec{r}(1) = (1-1) \hat{i} + \left(\frac{1^2}{2} + 2\right) \hat{j} + \left(\frac{1^3}{3} + 4\right) \hat{k}$$

$$\vec{r}(1) = 0 \hat{i} + \frac{5}{2} \hat{j} + \frac{13}{3} \hat{k}.$$

So, the coordinates of the particle at time $t = 1$ is $(0, \frac{5}{2}, \frac{13}{3})$ the Answer

Chapter 2: Gradient, Divergence & Curl

Gradient:

- Measures the rate of change in a scalar field; the gradient of a scalar field is a vector field. The derivative/differentiation/rate of change of a scalar field result **in a vector field** called the gradient.
- Computes the gradient of a scalar function. That is, it finds the Gradient, the slope, how fast you change, in any given direction.
- A gradient is applied to a scalar quantity that is a function of a 3D vector field: position. The gradient measures the direction in which the scalar quantity changes the most, as well as the rate of change with respect to position. A common example of this is height as a function of latitude and longitude, often applied to mountain ranges. A measure of the slope, and direction of the slope, is often called the gradient.

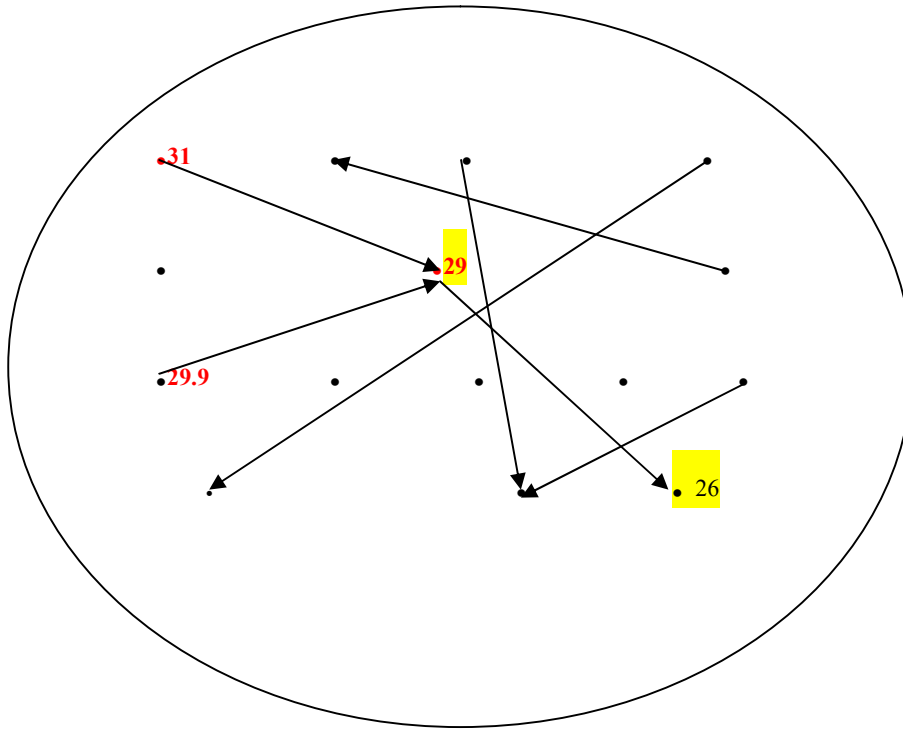


Figure # 64

Mathematical Expression of Gradient, divergence, curl of a Vector Field

Vector differential operator: $\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}$ and is denoted by the symbol $\vec{\nabla}$ (pronounced 'del' or sometimes 'Nabla')

That is $\vec{\nabla} = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}$ -----(i)

Beware! $\vec{\nabla}$ Cannot exist alone: it is an operator and must operate on a stated scalar function $\phi(x, y, z)$.

If F is a vector, $\vec{\nabla} F$ has no meaning

Grad (gradient of a scalar function)

If a scalar function $\phi(x, y, z)$ is continuously differentiable with respect to its variables x, y, z , throughout the region, then the *gradient* of ϕ , Written $\text{grad } \phi$, is defined as the vector

grad ϕ = $\vec{\nabla} \phi = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \phi = \frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k}$ -----(ii) *where ϕ is a function of x, y, z*

of x, y, z

Note that, while ϕ is a scalar function, $\text{grad } \phi$ is a vector function

Divergence of a vector field: If we form the scalar (dot) product of $\vec{\nabla}$ with a vector function $\vec{A}(x, y, z) = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$ we get a scalar result called the divergence of $\vec{A}$:

$$\text{div } \vec{A} \equiv \vec{\nabla} \cdot \vec{A} \equiv \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (A_x \hat{i} + A_y \hat{j} + A_z \hat{k})$$

$$\text{div } \vec{A} \equiv \vec{\nabla} \cdot \vec{A} \equiv \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \text{ -----(iii)}$$

Curl of a vector field: The curl of a vector field $\vec{A}(x, y, z) = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$ is defined by

$$\vec{\nabla} \times \vec{A} = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \times (A_x \hat{i} + A_y \hat{j} + A_z \hat{k}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} \text{ -----(iv)}$$

Q# 25: If $\phi(x, y, z) = 3x^2y - y^3z^2$, find $\vec{\nabla} \Phi$ (or grad Φ) at the point (1, -2, -1).

$$\begin{aligned}
 \text{Answer: } \vec{\nabla} \Phi &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) (3x^2y - y^3z^2) \\
 &= \hat{i} \frac{\partial}{\partial x} (3x^2y - y^3z^2) + \hat{j} \frac{\partial}{\partial y} (3x^2y - y^3z^2) + \hat{k} \frac{\partial}{\partial z} (3x^2y - y^3z^2) \\
 &= \hat{i} (6xy - 0) + \hat{j} (3x^2 - 3y^2z^2) + \hat{k} (0 - 2y^3z) \\
 &= 6xy \hat{i} + (3x^2 - 3y^2z^2) \hat{j} - 2y^3z \hat{k} \\
 &= 6(1)(-2) \hat{i} + \{3(1)^2 - 3(-2)^2(-1)^2\} \hat{j} - 2(-2)^3(-1) \hat{k} \\
 &= -12 \hat{i} - 9 \hat{j} - 16 \hat{k} \text{ (Answer).}
 \end{aligned}$$

Q# 26: Find $\vec{\nabla} \phi$ if (a) $\phi = \ln |\vec{r}|$ (b) $\phi = \frac{1}{|\vec{r}|}$

Let $\vec{r} = x \hat{i} + y \hat{j} + z \hat{k}$ [Page no 48, Figure no 57, Equation no (i)]

$$(a) \text{ Then } |\vec{r}| = \sqrt{x^2 + y^2 + z^2} \text{ and } |\vec{r}|^2 = x^2 + y^2 + z^2$$

$$\begin{aligned}
 \text{Then } \phi &= \ln |\vec{r}| = \ln \sqrt{x^2 + y^2 + z^2} = \ln (x^2 + y^2 + z^2)^{1/2} = \frac{1}{2} \ln(x^2 + y^2 + z^2) \\
 \therefore \vec{\nabla} \phi &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \phi = \left(\frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k} \right) = \hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y} + \hat{k} \frac{\partial \phi}{\partial z} \\
 &= \frac{1}{2} \hat{i} \frac{\partial}{\partial x} \ln(x^2 + y^2 + z^2) + \frac{1}{2} \hat{j} \frac{\partial}{\partial y} \ln(x^2 + y^2 + z^2) + \frac{1}{2} \hat{k} \frac{\partial}{\partial z} \ln(x^2 + y^2 + z^2) \\
 &= \frac{1}{2} \hat{i} \left(\frac{1}{x^2 + y^2 + z^2} \right) \left\{ \frac{\partial}{\partial x} (x^2 + y^2 + z^2) \right\} + \frac{1}{2} \hat{j} \left(\frac{1}{x^2 + y^2 + z^2} \right) \left\{ \frac{\partial}{\partial y} (x^2 + y^2 + z^2) \right\} + \frac{1}{2} \\
 &\quad \hat{k} \left(\frac{1}{x^2 + y^2 + z^2} \right) \left\{ \frac{\partial}{\partial z} (x^2 + y^2 + z^2) \right\} \\
 &= \frac{1}{2} \hat{i} \left(\frac{1}{x^2 + y^2 + z^2} \right) (2x + 0 + 0) + \frac{1}{2} \hat{j} \left(\frac{1}{x^2 + y^2 + z^2} \right) (0 + 2y + 0) + \frac{1}{2} \\
 &\quad \hat{k} \left(\frac{1}{x^2 + y^2 + z^2} \right) (0 + 0 + 2z) \\
 &= \frac{1}{2} \hat{i} \left(\frac{2x}{x^2 + y^2 + z^2} \right) + \frac{1}{2} \hat{j} \left(\frac{2y}{x^2 + y^2 + z^2} \right) + \frac{1}{2} \hat{k} \left(\frac{2z}{x^2 + y^2 + z^2} \right)
 \end{aligned}$$

$$= \frac{1}{2} \times 2 \left\{ \frac{x\hat{i} + y\hat{j} + z\hat{k}}{x^2 + y^2 + z^2} \right\} = \left\{ \frac{x\hat{i} + y\hat{j} + z\hat{k}}{x^2 + y^2 + z^2} \right\} = \frac{\vec{r}}{|\vec{r}|^2} \text{ Answer}$$

(b) Given, $\phi = \frac{1}{|\vec{r}|} = \frac{1}{\sqrt{x^2 + y^2 + z^2}} = (x^2 + y^2 + z^2)^{-1/2}$

$$\begin{aligned} \vec{\nabla} \phi &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \phi = \left(\frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k} \right) = \hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y} + \hat{k} \frac{\partial \phi}{\partial z} \\ &= \hat{i} \frac{\partial}{\partial x} (x^2 + y^2 + z^2)^{-1/2} + \hat{j} \frac{\partial}{\partial y} (x^2 + y^2 + z^2)^{-1/2} + \hat{k} \frac{\partial}{\partial z} (x^2 + y^2 + z^2)^{-1/2} \\ &= \hat{i} \left(-\frac{1}{2} \right) (x^2 + y^2 + z^2)^{-1/2-1} \frac{\partial}{\partial x} (x^2 + y^2 + z^2) + \hat{j} \left(-\frac{1}{2} \right) \\ &\quad (x^2 + y^2 + z^2)^{-1/2-1} \frac{\partial}{\partial y} (x^2 + y^2 + z^2) + \hat{k} \left(-\frac{1}{2} \right) (x^2 + y^2 + z^2)^{-1/2-1} \\ &\quad \frac{\partial}{\partial z} (x^2 + y^2 + z^2) \\ &= \hat{i} \left(-\frac{1}{2} \right) (x^2 + y^2 + z^2)^{-3/2} (2x + 0 + 0) + \hat{j} \left(-\frac{1}{2} \right) (x^2 + y^2 + z^2)^{-3/2} \\ &\quad (0 + 2y + 0) + \hat{k} \left(-\frac{1}{2} \right) (x^2 + y^2 + z^2)^{-3/2} \frac{\partial}{\partial z} (0 + 0 + 2z) \\ &= \hat{i} (-x) (x^2 + y^2 + z^2)^{-3/2} + \hat{j} (-y) (x^2 + y^2 + z^2)^{-3/2} + \hat{k} (-z) (x^2 + y^2 + z^2)^{-3/2} \\ &= \frac{-x\hat{i}}{(x^2 + y^2 + z^2)^{3/2}} + \frac{-y\hat{j}}{(x^2 + y^2 + z^2)^{3/2}} + \frac{-z\hat{k}}{(x^2 + y^2 + z^2)^{3/2}} \\ &= -\frac{(x\hat{i} + y\hat{j} + z\hat{k})}{(x^2 + y^2 + z^2)^{3/2}} = -\frac{(x\hat{i} + y\hat{j} + z\hat{k})}{(x^2 + y^2 + z^2)^1 \cdot (x^2 + y^2 + z^2)^{1/2}} \\ &= -\frac{\vec{r}}{|\vec{r}|^2 \cdot |\vec{r}|} = -\frac{\vec{r}}{|\vec{r}|^3} \text{ Answer} \end{aligned}$$

Q#27: Find the level curve of $f(x, y) = -x^2 + y^2$ passing through (2, 3). Draw Graph the gradient at the point (2, 3)

Answer: Given, $f(x, y) = -x^2 + y^2$

$$f(2, 3) = -2^2 + 3^2 = -4 + 9 = 5$$

Hence the level curve is the hyperbola, i.e.

$$f(x, y) = -x^2 + y^2 = 5$$

i.e. $-x^2 + y^2 = 5$



i.e. $x^2 - y^2 = -5$

$\Rightarrow \frac{x^2}{-5} - \frac{y^2}{-5} = 1$ [This is the equation of a hyperbola, i. e. $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$] -----(i)

From (i),

$\Rightarrow \frac{x^2}{-5} - \frac{y^2}{-5} = 1 \quad \Rightarrow x^2 - y^2 = -5 \quad \Rightarrow y^2 = 5 + x^2 \quad \Rightarrow y = \pm\sqrt{5+x^2}$

$\Rightarrow y = \pm\sqrt{5+x^2}$ -----(ii)

x	0	-1	-2	-3	1	2	3	-4	4	
$y = \pm\sqrt{5+x^2}$	$\pm\sqrt{5}$	$\pm\sqrt{6}$	± 3	$\pm\sqrt{14}$	$\pm\sqrt{6}$	± 3	$\pm\sqrt{14}$	$\pm\sqrt{19}$	$\pm\sqrt{19}$	
	$= \pm 2.23$	± 2.44	± 3	± 3.74	± 2.44	± 3	± 3.74	± 4.35	± 4.35	

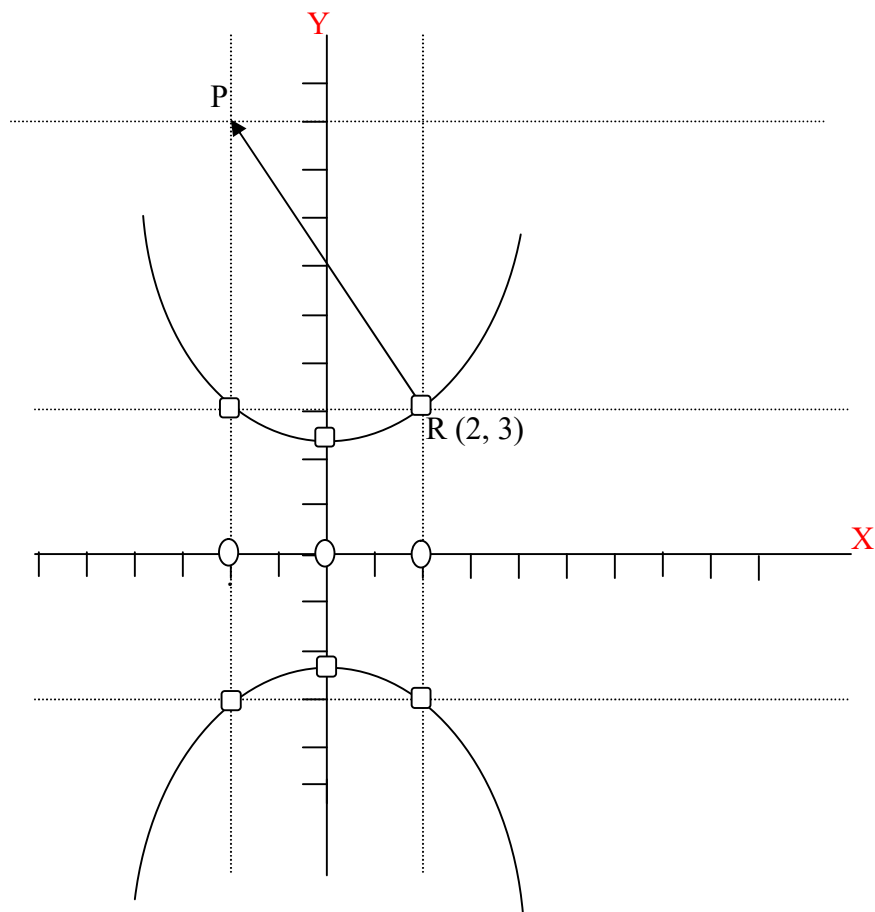


Figure # 69

Given, $f(x, y) = -x^2 + y^2$

Now, Gradient of the function, i.e.

$$\vec{\nabla} f(x, y) = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) (-x^2 + y^2)$$

$$\vec{\nabla} f(x, y) = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) (-x^2 + y^2)$$

$$\vec{\nabla} f(x, y) = \hat{i} \frac{\partial}{\partial x} (-x^2 + y^2) + \hat{j} \frac{\partial}{\partial y} (-x^2 + y^2) + \hat{k} \frac{\partial}{\partial z} (-x^2 + y^2)$$

$$\vec{\nabla} f(x, y) = \hat{i} \frac{\partial}{\partial x} (-x^2) + \hat{i} \frac{\partial}{\partial x} (y^2) + \hat{j} \frac{\partial}{\partial y} (-x^2) + \hat{j} \frac{\partial}{\partial y} (y^2) + \hat{k} \frac{\partial}{\partial z} (-x^2) + \hat{k} \frac{\partial}{\partial z} (y^2)$$

$$\vec{\nabla} f(x, y) = \hat{i}(-2x) + \hat{i} \times 0 + \hat{j} \times 0 + \hat{j}(2y) + \hat{k} \times 0 + \hat{k} \times 0$$

$$\vec{\nabla} f(x, y) = -2x \hat{i} + 2y \hat{j}$$

$$\vec{\nabla} f(2, 3) = -2 \times 2 \hat{i} + 2 \times 3 \hat{j}$$

$$\vec{\nabla} f(2, 3) = -4 \hat{i} + 6 \hat{j}$$

Hence the gradient Vector is $\vec{RP} = \vec{\nabla} f(2, 3) = -4 \hat{i} + 6 \hat{j}$ the Answer

Q# 28: Sketch the level curve for the function $f(x, y) = x^2 + y^2$ through the point (3, 4) and draw the gradient vector at this point.

✗ VOID

Answer: Given, the function $f(x, y) = x^2 + y^2$ through the point (3, 4),

$$f(3, 4) = 3^2 + 4^2$$

$$f(3, 4) = 9 + 16 = 25$$

Since $f(3, 4) = 25$, the level curve through the point (3, 4) has the equation

$f(x, y) = x^2 + y^2 = 25$, which is the circle. That is $x^2 + y^2 = 25$ whose centre (0, 0) and radius 5.

Now,

$$f(x, y) = x^2 + y^2$$

Now, Gradient of the function,

$$\vec{\nabla} f(x, y) = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) (x^2 + y^2)$$

$$\vec{\nabla} f(x, y) = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) (x^2 + y^2)$$

$$\vec{\nabla} f(x, y) = \hat{i} \frac{\partial}{\partial x} (x^2 + y^2) + \hat{j} \frac{\partial}{\partial y} (x^2 + y^2) + \hat{k} \frac{\partial}{\partial z} (x^2 + y^2)$$

$$\vec{\nabla} f(x, y) = \hat{i} \frac{\partial}{\partial x} (x^2) + \hat{i} \frac{\partial}{\partial x} (y^2) + \hat{j} \frac{\partial}{\partial y} (x^2) + \hat{j} \frac{\partial}{\partial y} (y^2) + \hat{k} \frac{\partial}{\partial z} (x^2) + \hat{k} \frac{\partial}{\partial z} (y^2)$$

$$\vec{\nabla} f(x,y) = \hat{i}(2x) + \hat{i} \times 0 + \hat{j} \times 0 + \hat{j}(2y) + \hat{k} \times 0 + \hat{k} \times 0$$

$$\vec{\nabla} f(x,y) = 2x \hat{i} + 2y \hat{j} \quad \text{-----(i)}$$

The gradient vector at (3,4) is

$$\therefore \vec{\nabla} f(3,4) = 2 \times 3 \hat{i} + 2 \times 4 \hat{j}$$

$$\vec{\nabla} f(3,4) = 6\hat{i} + 8\hat{j} \quad \text{-----(ii)}$$

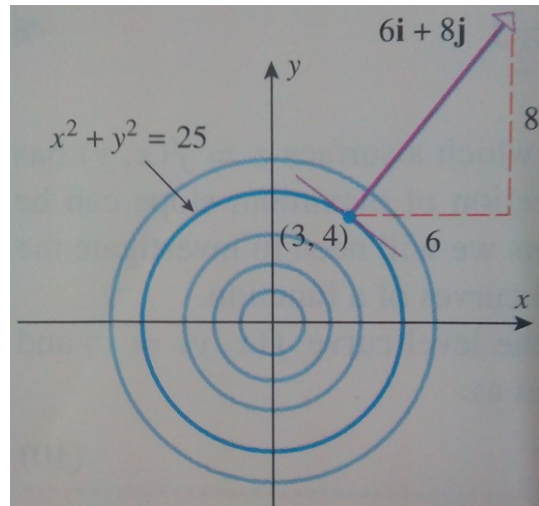


Figure # 70

Hence the gradient vector is perpendicular to the circle at (3,4) .

Q# 32: Find a unit normal to the surface $x^2y + 2xz = 4$ at the point $(2, -2, 3)$

Answer: Given, $\phi(x, y, z) = x^2y + 2xz = 4$

$$\begin{aligned}\therefore \vec{\nabla} \phi &= \left(\frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k} \right) = \left(\frac{\partial}{\partial x} (x^2y + 2xz) \hat{i} + \frac{\partial}{\partial y} (x^2y + 2xz) \hat{j} + \frac{\partial}{\partial z} (x^2y + 2xz) \hat{k} \right) \\ &= \hat{i} \frac{\partial}{\partial x} (x^2y + 2xz) + \hat{j} \frac{\partial}{\partial y} (x^2y + 2xz) + \hat{k} \frac{\partial}{\partial z} (x^2y + 2xz) \\ &= (2xy + 2z) \hat{i} + x^2 \hat{j} + 2x \hat{k} \\ &= (2 \times 2 \times (-2) + 2 \times 3) \hat{i} + 2^2 \hat{j} + 2 \times 2 \hat{k} \text{ at the point } (2, -2, 3) \\ &= -2 \hat{i} + 4 \hat{j} + 4 \hat{k}\end{aligned}$$

$$\text{Then a unit normal to the surface} = \frac{-2 \hat{i} + 4 \hat{j} + 4 \hat{k}}{\sqrt{(-2)^2 + (4)^2 + (4)^2}} = -\frac{1}{3} \hat{i} + \frac{2}{3} \hat{j} + \frac{2}{3} \hat{k} \text{ Answer}$$

Q# 33:

Find the level surface of $F(x, y, z) = x^2 + y^2 + z^2$ passing through (1,1,1). Graph the gradient at the point.

Answer: Given, $F(x, y, z) = x^2 + y^2 + z^2$

$$\therefore F(1,1,1) = 1^2 + 1^2 + 1^2 = 3$$

$$\text{Hence } F(x, y, z) = x^2 + y^2 + z^2 = 3$$

Because $F(1,1,1) = 3$, the level surface passing through (1,1,1) is the sphere $x^2 + y^2 + z^2 = 3$.

The gradient of the function is

$$F(x, y, z) = x^2 + y^2 + z^2$$

$$F(x, y, z) = x^2 + y^2 + z^2$$

$$\therefore \nabla F(x, y, z) = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) F$$

$$\therefore \nabla F(x, y, z) = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) (x^2 + y^2 + z^2)$$

$$\therefore \nabla F(x, y, z) =$$

$$\hat{i} \frac{\partial}{\partial x} (x^2 + y^2 + z^2) + \hat{j} \frac{\partial}{\partial y} (x^2 + y^2 + z^2) + \hat{k} \frac{\partial}{\partial z} (x^2 + y^2 + z^2)$$

$$\therefore \nabla F(x, y, z) = \hat{i} \frac{\partial}{\partial x} (2x + 0 + 0) + \hat{j} \frac{\partial}{\partial y} (0 + 2y + 0) + \hat{k} \frac{\partial}{\partial z} (0 + 0 + 2z)$$

$$\therefore \nabla F(x, y, z) = \hat{i}(2x) + \hat{j}(2y) + \hat{k}(2z)$$

$$\therefore \nabla F(x, y, z) = 2x \hat{i} + 2y \hat{j} + 2z \hat{k}$$

And so, at the given point

$$\therefore \nabla F(1,1,1) = 2.1 \hat{i} + 2.1 \hat{j} + 2.1 \hat{k}$$

$$\therefore \nabla F(1,1,1) = 2 \hat{i} + 2 \hat{j} + 2 \hat{k}$$

The level surface and $\nabla F(1,1,1)$ are illustrated in figure no 74

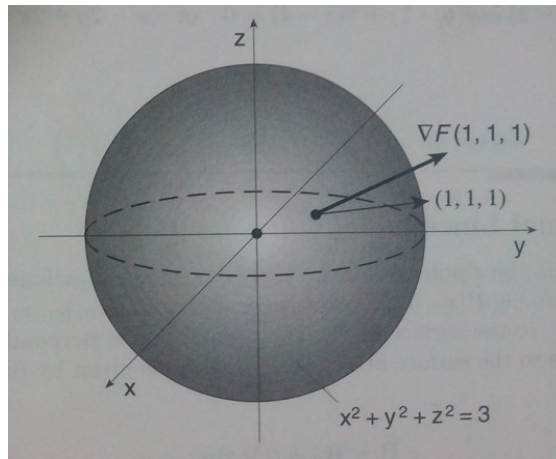


Figure # 74

Q# 35: Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at the point $(2, -1, 2)$

Answer:

Given, $z = x^2 + y^2 - 3$

$$\Rightarrow x^2 + y^2 - z = 3$$

Let, $\phi_1(x, y, z) = x^2 + y^2 + z^2 = 9$ and $\phi_2(x, y, z) = x^2 + y^2 - z = 3$

$$\begin{aligned} \therefore \nabla \phi_1 &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \phi_1 = \left(\frac{\partial \phi_1}{\partial x} \hat{i} + \frac{\partial \phi_1}{\partial y} \hat{j} + \frac{\partial \phi_1}{\partial z} \hat{k} \right) = \hat{i} \frac{\partial}{\partial x} \phi_1 + \hat{j} \frac{\partial}{\partial y} \phi_1 + \hat{k} \frac{\partial}{\partial z} \phi_1 \\ &= \hat{i} \frac{\partial}{\partial x} (x^2 + y^2 + z^2) + \hat{j} \frac{\partial}{\partial y} (x^2 + y^2 + z^2) + \hat{k} \frac{\partial}{\partial z} (x^2 + y^2 + z^2) \\ &= 2x \hat{i} + 2y \hat{j} + 2z \hat{k} \\ &= 4\hat{i} - 2\hat{j} + 4\hat{k} \text{ at the point } (2, -1, 2) \end{aligned}$$

A normal to $x^2 + y^2 + z^2 = 9$ at $(2, -1, 2)$ is $\nabla\phi_1 = 4\hat{i} - 2\hat{j} + 4\hat{k}$

Again,

$$\phi_2(x, y, z) = x^2 + y^2 - z = 3$$

$$\begin{aligned}\therefore \nabla\phi_2 &= \left(\frac{\delta}{\delta x}\hat{i} + \frac{\delta}{\delta y}\hat{j} + \frac{\delta}{\delta z}\hat{k}\right)\phi_2 = \left(\frac{\delta\phi_2}{\delta x}\hat{i} + \frac{\delta\phi_2}{\delta y}\hat{j} + \frac{\delta\phi_2}{\delta z}\hat{k}\right) = \hat{i}\frac{\delta}{\delta x}\phi_2 + \hat{j}\frac{\delta}{\delta y}\phi_2 + \hat{k}\frac{\delta}{\delta z}\phi_2 \\ &= \hat{i}\frac{\delta}{\delta x}(x^2 + y^2 - z) + \hat{j}\frac{\delta}{\delta y}(x^2 + y^2 - z) + \hat{k}\frac{\delta}{\delta z}(x^2 + y^2 - z) \\ &= 2x\hat{i} + 2y\hat{j} - \hat{k} \\ &= 4\hat{i} - 2\hat{j} - \hat{k} \text{ at the point } (2, -1, 2)\end{aligned}$$

A normal to $x^2 + y^2 - z = 3$ at $(2, -1, 2)$ is $\nabla\phi_2 = 4\hat{i} - 2\hat{j} - \hat{k}$

Let ϕ be the angle between the normals to the surfaces at the point $(2, -1, 2)$

Then we have,

$$\nabla\phi_1 \cdot \nabla\phi_2 = |\nabla\phi_1| |\nabla\phi_2| \cos \phi, \quad \text{[from equation no (i), page no 78]}$$

$$\Rightarrow (4\hat{i} - 2\hat{j} + 4\hat{k}) \cdot (4\hat{i} - 2\hat{j} - \hat{k}) = \left|4\hat{i} - 2\hat{j} + 4\hat{k}\right| \left|4\hat{i} - 2\hat{j} - \hat{k}\right| \cos \phi$$

$$\Rightarrow 16 + 4 - 4 = \sqrt{(4)^2 + (-2)^2 + (4)^2} \sqrt{(4)^2 + (-2)^2 + (-1)^2} \cos \phi$$

$$\Rightarrow 16 = 6\sqrt{21} \cos \phi$$

$$\Rightarrow \cos \phi = \frac{16}{6\sqrt{21}}$$

$$\therefore \phi = \cos^{-1}\left(\frac{16}{6\sqrt{21}}\right) \text{ Answer } [\because \phi = \theta]$$

Directional derivatives

I. M M M M M M M M M M

II.

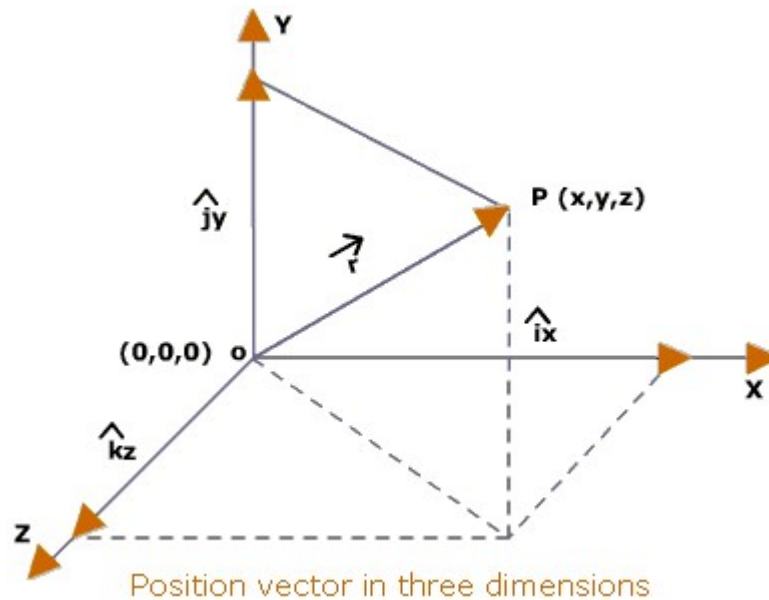


Figure # 76

Let $\vec{OP}$ is a position vector $\vec{r}$ where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ [Page no 48, Figure no 57, Equation no (i)]

and $d\vec{r}$ is a small displacement corresponding to changes dx, dy, dz in x, y, z respectively, then

$$d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k} \dots\dots\dots(i)$$

If $\phi(x, y, z)$ is a scalar function at P, then the *gradient* of ϕ

$$\text{grad } \phi = \vec{\nabla} \phi = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \phi \dots\dots\dots(ii)$$

$$\begin{aligned} \text{Then } \text{grad } \phi \cdot d\vec{r} &= \vec{\nabla} \phi \cdot d\vec{r} = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \phi \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k}) \\ &= \left(\frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k} \right) \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k}) \\ &= \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy + \frac{\partial \phi}{\partial z} dz \\ &= d\phi \\ &= \text{The total differential } d\phi \text{ of } \phi \end{aligned}$$

$$\text{grad } \phi \cdot d\vec{r} = d\phi$$

$$d\phi = \text{grad } \phi \cdot d\vec{r}$$

$$d\phi = \vec{\nabla} \phi \cdot d\vec{r} \text{-----(iii)}$$

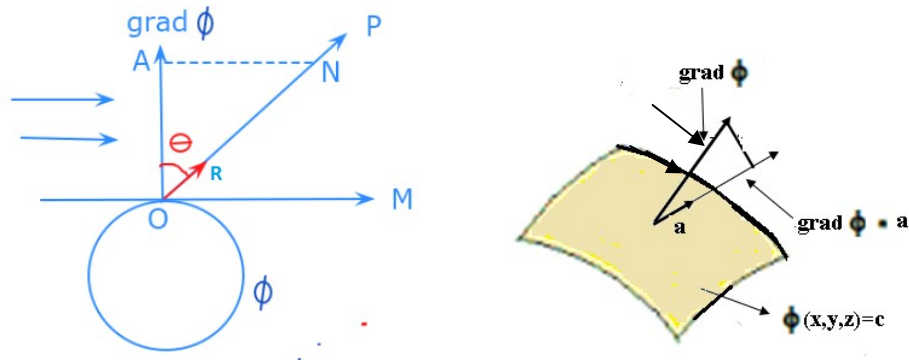


Figure # 77

We have just established that

$$d\phi = dr \cdot \text{grad } \phi$$

If ds is the small element of arc between $P(r)$ and $Q(r+dr)$ then $ds = \left| \frac{d\vec{r}}{ds} \right|$

$$\frac{d\vec{r}}{ds} = \frac{d\vec{r}}{\left| \frac{d\vec{r}}{ds} \right|}$$

and $\frac{d\vec{r}}{ds}$ is thus a unit vector in the direction of $d\vec{r}$.

$$\therefore \frac{d\phi}{ds} = \frac{d\vec{r}}{ds} \cdot \text{Grad } \phi$$

If we denote this unit vector by $\hat{a}$, i.e. $\frac{d\vec{r}}{ds} = \hat{a}$, the result becomes

$$\frac{d\phi}{ds} = \hat{a} \cdot \text{Grad } \phi$$

$\frac{d\phi}{ds}$ is the projection of $\text{grad } \phi$ on the unit vector $\hat{a}$ is called the directional derivative of ϕ in the direction of $\hat{a}$. It gives the rate of change of ϕ with distance measured in the

direction of $\hat{a}$ and $\frac{d\phi}{ds} = \hat{a} \cdot \text{grad } \phi$. Grad ϕ will be a maximum when $\hat{a}$ and grad ϕ have the same direction, since then, $\hat{a} \cdot \text{grad } \phi = |\hat{a}| |\text{grad } \phi| \cos \theta$ and θ will be zero

Thus direction of grad ϕ gives the direction in which the maximum rate of change of ϕ occurs.

Q# 37: Find the directional derivative of the function $\phi = x^2z + 2xy^2 + yz^2$ at the point of (1, 2, -1) in the direction of the vector $\vec{A} = 2\hat{i} + 3\hat{j} + \hat{k}$.

We start off with $\phi = x^2z + 2xy^2 + yz^2$

$$\therefore \nabla \phi = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \phi$$

$$\therefore \nabla \phi = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) (x^2z + 2xy^2 + yz^2)$$

$$\therefore \nabla \phi =$$

$$\hat{i} \frac{\partial}{\partial x} (x^2z + 2xy^2 + yz^2) + \hat{j} \frac{\partial}{\partial y} (x^2z + 2xy^2 + yz^2) + \hat{k} \frac{\partial}{\partial z} (x^2z + 2xy^2 + yz^2)$$

$$\therefore \nabla \phi = \hat{i}(2xz + 2.1.y^2 + 0) + \hat{j}(0 + 2x.2y + 1.z^2) + \hat{k}(x^2.1 + 0 + y.2z)$$

$$\therefore \nabla \phi = \hat{i}(2xz + 2y^2) + \hat{j}(4xy + z^2) + \hat{k}(x^2 + 2yz)$$

At the point (1, 2, -1)

$$\nabla \phi = \hat{i}(2xz + 2y^2) + \hat{j}(4xy + z^2) + \hat{k}(x^2 + 2yz)$$

$$\therefore \nabla \phi = \hat{i}[2.1.(-1) + 2(2^2)] + \hat{j}[4.1.2 + (-1)^2] + \hat{k}[(1^2 + 2.2.(-1))]$$

$$\therefore \nabla \phi = \hat{i}[-2 + 8] + \hat{j}[8 + 1] + \hat{k}[(1 - 4)]$$

$$\therefore \nabla \phi = \hat{i}[6] + \hat{j}[9] + \hat{k}(-3)$$

$$\therefore \nabla \phi = 6\hat{i} + 9\hat{j} - 3\hat{k}$$

Next we have to find out the unit vector $\hat{a}$ where $\vec{A} = 2\hat{i} + 3\hat{j} - 4\hat{k}$

$$\therefore \hat{a} = \frac{\vec{A}}{|\vec{A}|}$$

$$\therefore \hat{a} = \frac{2\hat{i} + 3\hat{j} - 4\hat{k}}{\sqrt{2^2 + 3^2 + (-4)^2}}$$

$$\therefore \hat{a} = \frac{2\hat{i} + 3\hat{j} - 4\hat{k}}{|4 + 9 + 16|}$$

$$\therefore \hat{a} = \frac{2\hat{i} + 3\hat{j} - 4\hat{k}}{|29|}$$

$$\text{Hence } \frac{d\phi}{ds} = \hat{a} \cdot \vec{\nabla} \phi$$

$$\frac{d\phi}{ds} = \frac{2\hat{i} + 3\hat{j} - 4\hat{k}}{|29|} \cdot (6\hat{i} + 9\hat{j} - 3\hat{k})$$

$$\frac{d\phi}{ds} = \frac{1}{|29|} (2\hat{i} + 3\hat{j} - 4\hat{k}) \cdot (6\hat{i} + 9\hat{j} - 3\hat{k})$$

$$\frac{d\phi}{ds} = \frac{1}{|29|} (12 + 27 + 12)$$

$$\frac{d\phi}{ds} = \frac{51}{|29|} \text{ Answer}$$

Q#38: Find the directional derivative of the function $\phi = x^2y + y^2z + z^2x$ at the point of $(1, -1, 2)$ in the direction of the vector $\vec{A} = 4\hat{i} + 2\hat{j} - 5\hat{k}$.

We have $\phi = x^2y + y^2z + z^2x$

$$\therefore \nabla \phi = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \phi$$

$$\therefore \nabla \phi = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) (x^2y + y^2z + z^2x)$$

$$\therefore \nabla \phi = \hat{i} \frac{\partial}{\partial x} (x^2y + y^2z + z^2x) + \hat{j} \frac{\partial}{\partial y} (x^2y + y^2z + z^2x) + \hat{k} \frac{\partial}{\partial z} (x^2y + y^2z + z^2x)$$

$$\therefore \nabla \phi = \hat{i}(2xy + 0 + z^2 \cdot 1) + \hat{j}(x^2 \cdot 1 + 2yz + 0) + \hat{k}(0 + y^2 \cdot 1 + 2zx)$$

$$\therefore \nabla \phi = \hat{i}(2xy + z^2) + \hat{j}(x^2 + 2yz) + \hat{k}(y^2 + 2zx)$$

At the point $(1, -1, 2)$

$$\therefore \nabla \phi = \hat{i}(2xy + z^2) + \hat{j}(x^2 + 2yz) + \hat{k}(y^2 + 2zx)$$

$$\therefore \nabla \phi = \hat{i}[2 \cdot 1 \cdot (-1) + 2^2] + \hat{j}[1^2 + 2(-1) \cdot 2] + \hat{k}[(-1)^2 + 2 \cdot 2 \cdot 1]$$

$$\therefore \nabla \phi = \hat{i}[-2 + 4] + \hat{j}[1 - 4] + \hat{k}[1 + 4]$$

$$\therefore \nabla \phi = 2\hat{i} - 3\hat{j} + 5\hat{k}$$

Next we have to find out the unit vector $\hat{a}$ where $\vec{A} = 4\hat{i} + 2\hat{j} - 5\hat{k}$

$$\therefore \hat{a} = \frac{\vec{A}}{|\vec{A}|}$$

$$\therefore \hat{a} = \frac{4\hat{i} + 2\hat{j} - 5\hat{k}}{\sqrt{4^2 + 2^2 + (-5)^2}}$$

$$\therefore \hat{a} = \frac{4\hat{i} + 2\hat{j} - 5\hat{k}}{\sqrt{45}}$$

$$\text{Hence } \frac{d\phi}{ds} = \hat{a} \cdot \vec{\nabla} \phi$$

$$\frac{d\phi}{ds} = \frac{(4\hat{i} + 2\hat{j} - 5\hat{k})}{\sqrt{45}} \cdot (2\hat{i} - 3\hat{j} + 5\hat{k})$$

$$\frac{d\phi}{ds} = \frac{1}{\sqrt{45}} (4\hat{i} + 2\hat{j} - 5\hat{k}) \cdot (2\hat{i} - 3\hat{j} + 5\hat{k})$$

$$\frac{d\phi}{ds} = \frac{1}{\sqrt{45}} (8 - 6 - 25)$$

$$\frac{d\phi}{ds} = \frac{1}{\sqrt{45}} (-23) \text{ Answer}$$

Q#39: Find the directional derivative of the function $\phi = (x, y, z) = x^2 - y^2 + 2z^2$ at the point of (1,2,3) in the direction of the vector $\vec{A} = 4\hat{i} - 2\hat{j} + \hat{k}$.

Answer: Let, $\phi(x, y, z) = x^2 - y^2 + 2z^2$

$$\frac{\partial \phi}{\partial x} = \frac{\partial}{\partial x} (x^2 - y^2 + 2z^2)$$

$$\frac{\partial \phi}{\partial x} = (2x - 0 + 0)$$

$$\frac{\partial \phi}{\partial x} = 2x$$

$$\phi = (x, y, z) = x^2 - y^2 + 2z^2$$

$$\frac{\partial \phi}{\partial y} = \frac{\partial}{\partial y} (x^2 - y^2 + 2z^2)$$

$$\frac{\partial \phi}{\partial y} = (0 - 2y + 0)$$

$$\frac{\partial \phi}{\partial y} = -2y$$

$$\phi(x, y, z) = x^2 - y^2 + 2z^2$$

Q# 41: If $\vec{V}(x,y,z) = xz\hat{i} + xyz\hat{j} - y^2\hat{k}$ Find divergence of $\vec{V}$ that is $\text{div } \vec{V}$

Answer: $\text{div } \vec{V} = \nabla \cdot \vec{V} = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (xz\hat{i} + xyz\hat{j} - y^2\hat{k})$

$$= \frac{\partial}{\partial x}(xz) + \frac{\partial}{\partial y}(xyz) - \frac{\partial}{\partial z}(y^2) \quad [\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{k} \cdot \hat{i} = 0 \text{ etc.}]$$

$$= z + xz \text{ Answer}$$

Q# 42: Let $\vec{V}$ be a constant vector field. Show that $\text{div } \vec{V} = 0$

Answer: Let, $\vec{V} = a\hat{i} + b\hat{j} + c\hat{k}$, where a, b, c are constants, Then

$$\text{div } \vec{V} = \nabla \cdot \vec{V} = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (a\hat{i} + b\hat{j} + c\hat{k})$$

$$= \frac{\partial}{\partial x}(a) + \frac{\partial}{\partial y}(b) + \frac{\partial}{\partial z}(c) \quad [\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{k} \cdot \hat{i} = 0 \text{ etc.}]$$

$$= 0$$

Q# 43: what is solenoidal?

Answer: If $\vec{A}$ is solenoidal then $\nabla \cdot \vec{A} = 0$

Home Task: Determine the constant a so that the vector

$\vec{F} = (x + 2y)\hat{i} + (y - 2z)\hat{j} + (x + az)\hat{k}$ is solenoidal

Q# 44: Show that the vector field $\vec{v} = \frac{-x\hat{i} - y\hat{j}}{\sqrt{x^2 + y^2}}$ is a “sink field”. Plot and give a physical interpretation. [Here $\hat{V}$ is a unit vector; Since it is divided by its length]

Answer: given,

$$\vec{v} = \frac{-x\hat{i} - y\hat{j}}{\sqrt{x^2 + y^2}}$$

$$\vec{v} = \frac{-x\hat{i} - y\hat{j}}{\sqrt{(-x)^2 + (-y)^2}}$$

$$\vec{v} = \frac{-x\hat{i} - y\hat{j}}{\sqrt{x^2 + y^2}}$$

$$\vec{v} = \frac{-x}{\sqrt{x^2 + y^2}}\hat{i} + \frac{-y}{\sqrt{x^2 + y^2}}\hat{j}$$

$$\begin{aligned}
\therefore \operatorname{div} \vec{v} &\equiv \vec{\nabla} \cdot \vec{v} \equiv \left(\frac{\delta}{\delta x} \hat{i} + \frac{\delta}{\delta y} \hat{j} + \frac{\delta}{\delta z} \hat{k} \right) \cdot \left(\frac{-x}{\sqrt{x^2 + y^2}} \hat{i} + \frac{-y}{\sqrt{x^2 + y^2}} \hat{j} + 0 \hat{k} \right) \\
\operatorname{div} \vec{v} &\equiv \vec{\nabla} \cdot \vec{v} \equiv \frac{\delta}{\delta x} \left(\frac{-x}{\sqrt{x^2 + y^2}} \right) + \frac{\delta}{\delta y} \left(\frac{-y}{\sqrt{x^2 + y^2}} \right) \\
\operatorname{div} \vec{v} &\equiv \vec{\nabla} \cdot \vec{v} \equiv \frac{\delta}{\delta x} \left(\frac{-x}{\sqrt{x^2 + y^2}} \right) + \frac{\delta}{\delta y} \left(\frac{-y}{\sqrt{x^2 + y^2}} \right) \\
\operatorname{div} \vec{v} &\equiv \vec{\nabla} \cdot \vec{v} \equiv \frac{(\sqrt{x^2 + y^2}) \frac{\delta}{\delta x} (-x) - (-x) \frac{\delta}{\delta x} (\sqrt{x^2 + y^2})}{(\sqrt{x^2 + y^2})^2} + \frac{(\sqrt{x^2 + y^2}) \frac{\delta}{\delta y} (-y) - (-y) \frac{\delta}{\delta y} (\sqrt{x^2 + y^2})}{(\sqrt{x^2 + y^2})^2} \\
\operatorname{div} \vec{v} &\equiv \vec{\nabla} \cdot \vec{v} \equiv \frac{(\sqrt{x^2 + y^2})(-1) + x \frac{\delta}{\delta x} (\sqrt{x^2 + y^2})}{(\sqrt{x^2 + y^2})^2} + \frac{(\sqrt{x^2 + y^2})(-1) + y \frac{\delta}{\delta y} (\sqrt{x^2 + y^2})}{(\sqrt{x^2 + y^2})^2} \\
\operatorname{div} \vec{v} &\equiv \vec{\nabla} \cdot \vec{v} \equiv \frac{-(\sqrt{x^2 + y^2}) + x \frac{\delta}{\delta x} (\sqrt{x^2 + y^2})}{(\sqrt{x^2 + y^2})^2} + \frac{-(\sqrt{x^2 + y^2}) + y \frac{\delta}{\delta y} (\sqrt{x^2 + y^2})}{(\sqrt{x^2 + y^2})^2} \\
\operatorname{div} \vec{v} &\equiv \vec{\nabla} \cdot \vec{v} \equiv \frac{-(\sqrt{x^2 + y^2}) + x \frac{\delta}{\delta x} (x^2 + y^2)^{1/2}}{(\sqrt{x^2 + y^2})^2} + \frac{-(\sqrt{x^2 + y^2}) + y \frac{\delta}{\delta y} (x^2 + y^2)^{1/2}}{(\sqrt{x^2 + y^2})^2} \\
\operatorname{div} \vec{v} &\equiv \vec{\nabla} \cdot \vec{v} \equiv \frac{-(\sqrt{x^2 + y^2}) + x \frac{1}{2} (x^2 + y^2)^{\frac{1}{2}-1} \cdot \frac{\delta}{\delta x} (x^2 + y^2)}{(\sqrt{x^2 + y^2})^2} + \frac{-(\sqrt{x^2 + y^2}) + y \frac{1}{2} (x^2 + y^2)^{\frac{1}{2}-1} \cdot \frac{\delta}{\delta y} (x^2 + y^2)}{(\sqrt{x^2 + y^2})^2} \\
\operatorname{div} \vec{v} &\equiv \vec{\nabla} \cdot \vec{v} \equiv \frac{-(\sqrt{x^2 + y^2}) + x \frac{1}{2} (x^2 + y^2)^{-\frac{1}{2}} \cdot (2x)}{(\sqrt{x^2 + y^2})^2} + \frac{-(\sqrt{x^2 + y^2}) + y \frac{1}{2} (x^2 + y^2)^{-\frac{1}{2}} \cdot (2y)}{(\sqrt{x^2 + y^2})^2} \\
\operatorname{div} \vec{v} &\equiv \vec{\nabla} \cdot \vec{v} \equiv \frac{-(\sqrt{x^2 + y^2}) + x^2 (x^2 + y^2)^{-\frac{1}{2}}}{(\sqrt{x^2 + y^2})^2} + \frac{-(\sqrt{x^2 + y^2}) + y^2 (x^2 + y^2)^{-\frac{1}{2}}}{(\sqrt{x^2 + y^2})^2} \\
\operatorname{div} \vec{v} &\equiv \vec{\nabla} \cdot \vec{v} \equiv \frac{-(\sqrt{x^2 + y^2}) + \frac{x^2}{(x^2 + y^2)^{\frac{1}{2}}}}{x^2 + y^2} + \frac{-(\sqrt{x^2 + y^2}) + \frac{y^2}{(x^2 + y^2)^{\frac{1}{2}}}}{x^2 + y^2} \\
\operatorname{div} \vec{v} &\equiv \vec{\nabla} \cdot \vec{v} \equiv \frac{-(\sqrt{x^2 + y^2}) + \frac{x^2}{\sqrt{x^2 + y^2}}}{x^2 + y^2} + \frac{-(\sqrt{x^2 + y^2}) + \frac{y^2}{\sqrt{x^2 + y^2}}}{x^2 + y^2}
\end{aligned}$$

$$\begin{aligned}\operatorname{div} \vec{v} &\equiv \vec{\nabla} \cdot \vec{v} \equiv \frac{-x^2 - y^2 + x^2}{\sqrt{x^2 + y^2}} + \frac{-x^2 - y^2 + y^2}{\sqrt{x^2 + y^2}} \\ \operatorname{div} \vec{v} &\equiv \vec{\nabla} \cdot \vec{v} \equiv \frac{-y^2}{\sqrt{x^2 + y^2}} + \frac{-x^2}{\sqrt{x^2 + y^2}} \\ \operatorname{div} \vec{v} &\equiv \vec{\nabla} \cdot \vec{v} \equiv \frac{-y^2}{(x^2 + y^2)(\sqrt{x^2 + y^2})} + \frac{-x^2}{(x^2 + y^2)(\sqrt{x^2 + y^2})} \\ \operatorname{div} \vec{v} &\equiv \vec{\nabla} \cdot \vec{v} \equiv \frac{-y^2 - x^2}{(x^2 + y^2)(\sqrt{x^2 + y^2})} \\ \operatorname{div} \vec{v} &\equiv \vec{\nabla} \cdot \vec{v} \equiv \frac{-(x^2 + y^2)}{(x^2 + y^2)(\sqrt{x^2 + y^2})} \\ \operatorname{div} \vec{v} &\equiv \vec{\nabla} \cdot \vec{v} \equiv \frac{-1}{\sqrt{x^2 + y^2}} < 0\end{aligned}$$

So, $\vec{v}$ a “sink field”

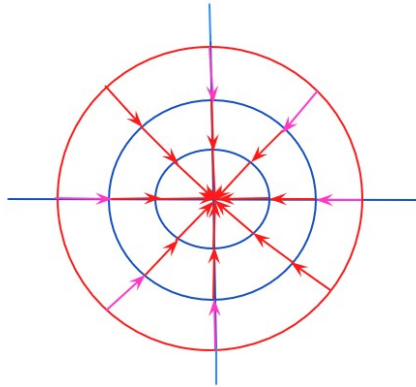


Figure # 78

Q# 45: Let $\vec{V}$ be a constant vector field. Show that $\operatorname{curl} \vec{V} = 0$

Answer Let, $\vec{V} = a\hat{i} + b\hat{j} + c\hat{k}$, where a, b, c are constants, Then

$$\operatorname{curl} \vec{V} = \nabla \times \vec{V} = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \times (a\hat{i} + b\hat{j} + c\hat{k})$$

$$\begin{aligned}
&= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ a & b & c \end{vmatrix} \\
&= \hat{i} \left(\frac{\partial c}{\partial y} - \frac{\partial b}{\partial z} \right) - \hat{j} \left(\frac{\partial c}{\partial x} - \frac{\partial a}{\partial z} \right) + \hat{k} \left(\frac{\partial b}{\partial x} - \frac{\partial a}{\partial y} \right) \\
&= \hat{i}(0-0) - \hat{j}(0-0) + \hat{k}(0-0) \\
&= 0
\end{aligned}$$

Q# 46: If $\vec{v}(x,y,z) = xz\hat{i} + xyz\hat{j} - y^2\hat{k}$ Find $\text{curl } \vec{V}$

Answer: The curl of a vector field $\vec{v}(x,y,z) = xz\hat{i} + xyz\hat{j} - y^2\hat{k}$ is defined by

$$\begin{aligned}
\nabla \times \vec{v} &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \times (xz\hat{i} + xyz\hat{j} - y^2\hat{k}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xz & xyz & -y^2 \end{vmatrix} \\
&= \hat{i} \left[\frac{\partial}{\partial y}(-y^2) - \frac{\partial}{\partial z}(xyz) \right] - \hat{j} \left[\frac{\partial}{\partial x}(-y^2) - \frac{\partial}{\partial z}(xz) \right] + \hat{k} \left[\frac{\partial}{\partial x}(xyz) - \frac{\partial}{\partial y}(xz) \right] \\
&= \hat{i}[-2y - xy] - \hat{j}[0 - x] + \hat{k}[yz - 0] \\
&= -[2y + xy]\hat{i} + \hat{j}x + \hat{k}yz \\
&= -[2y + xy]\hat{i} + x\hat{j} + yz\hat{k}
\end{aligned}$$

Q# 47: What is irrotational Field or conservative vector field?

A vector field $\vec{V}$ for which the curl vanishes, that is: $\vec{V} \times \vec{V} = 0$

Q# 48: Determine $\vec{F}$ is a conservative vector field or not where

$$\vec{F} = x^2y\hat{i} + xyz\hat{j} - x^2y^2\hat{k}$$

Answer:

So all that we need to do is compute the curl and see if we get the zero vector or not.

The curl of a vector field $\vec{F} = x^2y\hat{i} + xyz\hat{j} - x^2y^2\hat{k}$ is defined by

$$\begin{aligned}
\nabla \times \vec{F} &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \times (x^2y \hat{i} + xyz \hat{j} - x^2y^2 \hat{k}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2y & xyz & -x^2y^2 \end{vmatrix} \\
&= \hat{i} \left[\frac{\partial}{\partial y} (-x^2y^2) - \frac{\partial}{\partial z} (xyz) \right] - \hat{j} \left[\frac{\partial}{\partial x} (-x^2y^2) - \frac{\partial}{\partial z} (x^2y) \right] \\
&\quad + \hat{k} \left[\frac{\partial}{\partial x} (xyz) - \frac{\partial}{\partial y} (x^2y) \right] \\
&= \hat{i} [-2x^2y - xy] - \hat{j} [-2xy^2 - 0] + \hat{k} [yz - x^2] \\
&\neq 0
\end{aligned}$$

So, the curl isn't the zero vectors and so this vector field is not conservative.

Home Task

Find the values of constants a, b and c so that,

$\vec{F} = (x + 2y + az) \hat{i} + (bx - 3y - z) \hat{j} + (4x + cy + 2z) \hat{k}$ is irrotational.

Q# 49: If $\vec{A} = (xz^3 \hat{i} - 2x^2yz \hat{j} + 2yz^4 \hat{k})$, find $\nabla \times \vec{A}$ (or curl A) at the point (1, -1, 1).

Answer:

$$\begin{aligned}
\nabla \times \vec{A} &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \times (xz^3 \hat{i} - 2x^2yz \hat{j} + 2yz^4 \hat{k}) \\
&= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xz^3 & -2x^2yz & 2yz^4 \end{vmatrix} \\
&= \hat{i} \left[\frac{\partial}{\partial y} (2yz^4) - \frac{\partial}{\partial z} (-2x^2yz) \right] - \hat{j} \left[\frac{\partial}{\partial x} (2yz^4) - \frac{\partial}{\partial z} (xz^3) \right] + \hat{k} \left[\frac{\partial}{\partial x} (-2x^2yz) - \frac{\partial}{\partial y} (xz^3) \right] \\
&= \hat{i} [2z^4 + 2x^2y] - \hat{j} [0 - 3xz^2] + \hat{k} [-4xyz - 0] \\
&= \hat{i} [2 \cdot 1^4 + 2 \cdot 1^2 \cdot (-1)] - \hat{j} [0 - 3 \cdot 1 \cdot 1^2] + \hat{k} [-4 \cdot 1 \cdot (-1) \cdot 1 - 0] \\
&= \hat{i} [2 - 2] - \hat{j} [0 - 3] + \hat{k} [4] \\
&= 3 \hat{j} + 4 \hat{k}
\end{aligned}$$

$$\begin{aligned}
&= -\left(\frac{\delta^2}{\delta x^2} + \frac{\delta^2}{\delta y^2} + \frac{\delta^2}{\delta z^2}\right)(A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) + \left(\hat{i} \frac{\delta}{\delta x} + \hat{j} \frac{\delta}{\delta y} + \hat{k} \frac{\delta}{\delta z}\right)\left(\frac{\delta A_1}{\delta x} + \frac{\delta A_2}{\delta y} + \frac{\delta A_3}{\delta z}\right) \\
&= -\nabla^2 (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) + \nabla \left(\frac{\delta A_1}{\delta x} + \frac{\delta A_2}{\delta y} + \frac{\delta A_3}{\delta z}\right) \\
&= -\nabla^2 \vec{A} + \nabla \left[\left(\hat{i} \frac{\delta}{\delta x} + \hat{j} \frac{\delta}{\delta y} + \hat{k} \frac{\delta}{\delta z}\right) \cdot (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k})\right] \\
&= -\nabla^2 \vec{A} + \nabla(\nabla \cdot \vec{A}) \text{ Proved}
\end{aligned}$$

Answer b)

$$\text{Let, } \vec{A} = A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}$$

$$\phi \vec{A} = \phi(A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k})$$

$$\phi \vec{A} = \phi A_1 \hat{i} + \phi A_2 \hat{j} + \phi A_3 \hat{k}$$

$$\text{L.H.S. } \vec{\nabla} \times (\phi \vec{A}) = \left(\hat{i} \frac{\delta}{\delta x} + \hat{j} \frac{\delta}{\delta y} + \hat{k} \frac{\delta}{\delta z}\right) \times (\phi A_1 \hat{i} + \phi A_2 \hat{j} + \phi A_3 \hat{k})$$

$$\begin{aligned}
&= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\delta}{\delta x} & \frac{\delta}{\delta y} & \frac{\delta}{\delta z} \\ \phi A_1 & \phi A_2 & \phi A_3 \end{vmatrix} \\
&= \hat{i} \left[\frac{\delta}{\delta y} (\phi A_3) - \frac{\delta}{\delta z} (\phi A_2) \right] - \hat{j} \left[\frac{\delta}{\delta x} (\phi A_3) - \frac{\delta}{\delta z} (\phi A_1) \right] + \hat{k} \left[\frac{\delta}{\delta x} (\phi A_2) - \frac{\delta}{\delta y} (\phi A_1) \right] \\
&= \hat{i} \left[\phi \frac{\delta}{\delta y} (A_3) + A_3 \frac{\delta \phi}{\delta y} - \phi \frac{\delta}{\delta z} (A_2) - A_2 \frac{\delta \phi}{\delta z} \right] - \hat{j} \left[\phi \frac{\delta}{\delta x} (A_3) + A_3 \frac{\delta \phi}{\delta x} - \phi \frac{\delta}{\delta z} (A_1) - A_1 \frac{\delta \phi}{\delta z} \right] \\
&\quad + \hat{k} \left[\phi \frac{\delta}{\delta x} (A_2) + A_2 \frac{\delta \phi}{\delta x} - \phi \frac{\delta}{\delta y} (A_1) - A_1 \frac{\delta \phi}{\delta y} \right] \\
&= \phi \left[\hat{i} \left\{ \frac{\delta}{\delta y} (A_3) - \frac{\delta}{\delta z} (A_2) \right\} - \hat{j} \left\{ \frac{\delta}{\delta x} (A_3) - \frac{\delta}{\delta z} (A_1) \right\} + \hat{k} \left\{ \frac{\delta}{\delta x} (A_2) - \frac{\delta}{\delta y} (A_1) \right\} \right] \\
&\quad + \hat{i} \left[A_3 \frac{\delta \phi}{\delta y} - A_2 \frac{\delta \phi}{\delta z} \right] - \hat{j} \left[A_3 \frac{\delta \phi}{\delta x} - A_1 \frac{\delta \phi}{\delta z} \right] + \hat{k} \left[A_2 \frac{\delta \phi}{\delta x} - A_1 \frac{\delta \phi}{\delta y} \right]
\end{aligned}$$

Q# 55: Show that $\vec{r} \cdot \vec{r} = r^2$

Also, $\vec{r} \cdot \vec{r} = (\hat{x}\hat{i} + \hat{y}\hat{j} + \hat{z}\hat{k}) \cdot (\hat{x}\hat{i} + \hat{y}\hat{j} + \hat{z}\hat{k})$

$$\begin{aligned}\vec{r} \cdot \vec{r} &= x^2 + y^2 + z^2 [\because \hat{i} \cdot \hat{i} = 1; \hat{j} \cdot \hat{j} = 1; \hat{k} \cdot \hat{k} = 1] \\ &= r^2 [\because r^2 = x^2 + y^2 + z^2]\end{aligned}$$

Similarly,

$$\vec{A} \cdot \vec{A} = A^2$$

$$\vec{\nabla} \cdot \vec{\nabla} = \nabla^2$$

Q# 56: Show that, $\vec{\nabla} = \frac{\vec{r}}{r} \frac{\partial}{\partial r}$

Proof: L.H.S = $\vec{\nabla}$

$$\begin{aligned}&= \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \\ &= \hat{i} \frac{\partial}{\partial x} \frac{\partial r}{\partial r} + \hat{j} \frac{\partial}{\partial y} \frac{\partial r}{\partial r} + \hat{k} \frac{\partial}{\partial z} \frac{\partial r}{\partial r} \\ &= \hat{i} \frac{\partial r}{\partial x} \frac{\partial}{\partial r} + \hat{j} \frac{\partial r}{\partial y} \frac{\partial}{\partial r} + \hat{k} \frac{\partial r}{\partial z} \frac{\partial}{\partial r} \\ &= \left(\hat{i} \frac{\partial r}{\partial x} + \hat{j} \frac{\partial r}{\partial y} + \hat{k} \frac{\partial r}{\partial z} \right) \frac{\partial}{\partial r} \\ &= \left(\hat{i} \frac{x}{r} + \hat{j} \frac{y}{r} + \hat{k} \frac{z}{r} \right) \frac{\partial}{\partial r} [\because \frac{\partial r}{\partial x} = \frac{x}{r}; \frac{\partial r}{\partial y} = \frac{y}{r}; \frac{\partial r}{\partial z} = \frac{z}{r}] \\ &= \frac{\hat{x}\hat{i} + \hat{y}\hat{j} + \hat{z}\hat{k}}{r} \frac{\partial}{\partial r} \\ &= \frac{\vec{r}}{r} \frac{\partial}{\partial r}\end{aligned}$$

Q# 57: Show that $\vec{\nabla} \cdot (\phi \vec{A}) = \phi (\vec{\nabla} \cdot \vec{A}) + (\vec{\nabla} \phi) \cdot \vec{A}$

Solution

Let $\vec{A} = A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}$ is a vector and ϕ is a function of a variable or variables

$$\begin{aligned}\text{L.H.S } \vec{\nabla} \cdot (\phi \vec{A}) &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (\phi \vec{A}) \\ &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot \phi (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) \\ &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (\phi A_1 \hat{i} + \phi A_2 \hat{j} + \phi A_3 \hat{k})\end{aligned}$$

$$\begin{aligned}
&= \frac{\delta}{\delta x}(\phi A_1) + \frac{\delta}{\delta y}(\phi A_2) + \frac{\delta}{\delta z}(\phi A_3) \quad [\because \hat{i} \cdot \hat{i} = 1; \hat{j} \cdot \hat{j} = 1; \hat{k} \cdot \hat{k} = 1] \\
&= \phi \frac{\delta}{\delta x}(A_1) + A_1 \frac{\delta}{\delta x}(\phi) + \phi \frac{\delta}{\delta y}(A_2) + A_2 \frac{\delta}{\delta y}(\phi) + \phi \frac{\delta}{\delta z}(A_3) + A_3 \frac{\delta}{\delta z}(\phi) \\
&\quad [\because \frac{d}{dx}(uv) = u \frac{d}{dx} v + v \frac{d}{dx} u] \\
&= A_1 \frac{\delta}{\delta x}(\phi) + A_2 \frac{\delta}{\delta y}(\phi) + A_3 \frac{\delta}{\delta z}(\phi) + \phi \frac{\delta}{\delta x}(A_1) + \phi \frac{\delta}{\delta y}(A_2) + \phi \frac{\delta}{\delta z}(A_3) \\
&= A_1 \frac{\delta}{\delta x}(\phi) + A_2 \frac{\delta}{\delta y}(\phi) + A_3 \frac{\delta}{\delta z}(\phi) + \phi \left\{ \frac{\delta}{\delta x}(A_1) + \frac{\delta}{\delta y}(A_2) + \frac{\delta}{\delta z}(A_3) \right\} \\
&= A_1 \frac{\delta \phi}{\delta x} + A_2 \frac{\delta \phi}{\delta y} + A_3 \frac{\delta \phi}{\delta z} + \phi \left\{ \frac{\delta}{\delta x}(A_1) + \frac{\delta}{\delta y}(A_2) + \frac{\delta}{\delta z}(A_3) \right\} \\
&= \frac{\delta \phi}{\delta x} A_1 + \frac{\delta \phi}{\delta y} A_2 + \frac{\delta \phi}{\delta z} A_3 + \phi \left\{ \frac{\delta}{\delta x}(A_1) + \frac{\delta}{\delta y}(A_2) + \frac{\delta}{\delta z}(A_3) \right\} \\
&= \phi \left\{ \frac{\delta}{\delta x}(A_1) + \frac{\delta}{\delta y}(A_2) + \frac{\delta}{\delta z}(A_3) \right\} + \frac{\delta \phi}{\delta x} A_1 + \frac{\delta \phi}{\delta y} A_2 + \frac{\delta \phi}{\delta z} A_3 \\
&= \phi \left(\frac{\delta}{\delta x} \hat{i} + \frac{\delta}{\delta y} \hat{j} + \frac{\delta}{\delta z} \hat{k} \right) \cdot (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) + \left(\frac{\delta \phi}{\delta x} \hat{i} + \frac{\delta \phi}{\delta y} \hat{j} + \frac{\delta \phi}{\delta z} \hat{k} \right) \cdot (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) \\
&\quad [\because \hat{i} \cdot \hat{i} = 1; \hat{j} \cdot \hat{j} = 1; \hat{k} \cdot \hat{k} = 1] \\
&= \phi \left(\frac{\delta}{\delta x} \hat{i} + \frac{\delta}{\delta y} \hat{j} + \frac{\delta}{\delta z} \hat{k} \right) \cdot (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) + \left(\frac{\delta}{\delta x} \hat{i} + \frac{\delta}{\delta y} \hat{j} + \frac{\delta}{\delta z} \hat{k} \right) \phi \cdot (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) \\
&= \phi(\vec{\nabla} \cdot \vec{A}) + (\vec{\nabla} \phi) \cdot \vec{A} \quad [\because \vec{\nabla} = \frac{\delta}{\delta x} \hat{i} + \frac{\delta}{\delta y} \hat{j} + \frac{\delta}{\delta z} \hat{k}] \\
\therefore \vec{\nabla} \cdot (\phi \vec{A}) &= \phi(\vec{\nabla} \cdot \vec{A}) + (\vec{\nabla} \phi) \cdot \vec{A}
\end{aligned}$$

Q# 58: Show that $\nabla^2(\ln r) = \frac{1}{r^2}$

$$\text{L.H.S} = \nabla^2(\ln r)$$

$$\begin{aligned}
&= \vec{\nabla} \cdot \vec{\nabla}(\ln r) \quad [\because \vec{\nabla} \cdot \vec{\nabla} = \nabla^2] \\
&= \vec{\nabla} \cdot \left[\frac{\vec{r}}{r} \frac{\partial}{\partial r} \ln r \right] \quad [\because \vec{\nabla} = \frac{\vec{r}}{r} \frac{\partial}{\partial r}] \\
&= \vec{\nabla} \cdot \left[\frac{\vec{r}}{r} \frac{\partial}{\partial r} (\ln r) \right] \\
&= \vec{\nabla} \cdot \left[\frac{\vec{r}}{r} \frac{1}{r} \right] \quad [\because \frac{\partial}{\partial r}(\ln r) = \frac{1}{r}]
\end{aligned}$$

$$\begin{aligned}
&= \vec{\nabla} \cdot \left[\frac{\vec{r}}{r^2} \right] \\
&= \vec{\nabla} \cdot \left[\frac{1}{r^2} \vec{r} \right] \\
&= \frac{1}{r^2} [\vec{\nabla} \cdot \vec{r}] + \left[\vec{\nabla} \left(\frac{1}{r^2} \right) \right] \cdot \vec{r} & [\because \vec{\nabla} \cdot (\phi \vec{A}) = \phi (\vec{\nabla} \cdot \vec{A}) + (\vec{\nabla} \phi) \cdot \vec{A}] \\
&= \frac{3}{r^2} + \left[\frac{\vec{r}}{r} \frac{\partial}{\partial r} \left(\frac{1}{r^2} \right) \right] \cdot \vec{r} & [\because \vec{\nabla} \cdot \vec{r} = 3 \text{ \& } \because \vec{\nabla} = \frac{\vec{r}}{r} \frac{\partial}{\partial r}] \\
&= \frac{3}{r^2} + \left[\frac{\vec{r}}{r} \frac{\partial}{\partial r} (r^{-2}) \right] \cdot \vec{r} \\
&= \frac{3}{r^2} + \frac{\vec{r}}{r} (-2r^{-2-1}) \cdot \vec{r} \\
&= \frac{3}{r^2} + \frac{\vec{r}}{r} \left(-\frac{2}{r^3} \right) \cdot \vec{r} \\
&= \frac{3}{r^2} - \frac{2}{r^4} \left(\vec{r} \cdot \vec{r} \right) \\
&= \frac{3}{r^2} - \frac{2}{r^4} \times r^2 & [\because \vec{r} \cdot \vec{r} = r^2] \\
&= \frac{3}{r^2} - \frac{2}{r^2} \\
&= \frac{1}{r^2} .
\end{aligned}$$

Q# 59: Find the directional derivative of $\frac{1}{r}$ in the direction of $\vec{r}$

Answer: Let $\phi = \frac{1}{r}$

Therefore, the directional derivative of $\frac{1}{r}$ in the direction of $\vec{r} = \vec{\nabla} \phi \cdot \hat{r}$

We can write, $\hat{r} = \frac{\vec{r}}{|\vec{r}|} = \frac{\vec{r}}{r}$

Here, $\phi = \frac{1}{r}$

$$\Rightarrow \vec{\nabla} \phi = \vec{\nabla} \frac{1}{r}$$

$$\Rightarrow \vec{\nabla} \phi = \frac{\vec{r}}{r} \frac{\partial}{\partial r} \frac{1}{r} \quad [\because \vec{\nabla} = \frac{\vec{r}}{r} \frac{\partial}{\partial r}]$$

$$\begin{aligned}
\Rightarrow \vec{\nabla} \phi &= \frac{\vec{r}}{r} \frac{\partial}{\partial r} (r^{-1}) \\
\Rightarrow \vec{\nabla} \phi &= \frac{\vec{r}}{r} (-1)(r^{-1-1}) \\
\Rightarrow \vec{\nabla} \phi &= \frac{\vec{r}}{r} (-1)(r^{-2}) \\
\Rightarrow \vec{\nabla} \phi &= -\frac{\vec{r}}{r} \frac{1}{r^2} \\
\Rightarrow \vec{\nabla} \phi &= -\frac{\vec{r}}{r^3}
\end{aligned}$$

Therefore, the directional derivative of $\frac{1}{r}$ in the direction of $\vec{r}$

$$\begin{aligned}
&= \vec{\nabla} \phi \cdot \hat{r} \\
&= -\frac{\vec{r}}{r^3} \cdot \hat{r} \\
&= -\frac{\vec{r}}{r^3} \cdot \frac{\vec{r}}{r} \\
&= -\frac{1}{r^4} (\vec{r} \cdot \vec{r}) \\
&= -\frac{1}{r^4} r^2 \\
&= -\frac{1}{r^2}
\end{aligned}
\quad
\begin{aligned}
&[\vec{\nabla} \phi = -\frac{\vec{r}}{r^3}] \\
&[\hat{r} = \frac{\vec{r}}{|\vec{r}|} = \frac{\vec{r}}{r}] \\
&[\because \vec{r} \cdot \vec{r} = r^2]
\end{aligned}$$

Q# 60: Show that $\nabla^2 \mathbf{r}^n = n(n+1) \mathbf{r}^{n-2}$

$$\begin{aligned}
\text{L.H.S} &= \nabla^2 \mathbf{r}^n = \vec{\nabla} \cdot \vec{\nabla} (\mathbf{r}^n) \\
&= \vec{\nabla} \cdot \left[\frac{\vec{r}}{r} \frac{\partial}{\partial r} \mathbf{r}^n \right] \\
&= \vec{\nabla} \cdot \left[\frac{\vec{r}}{r} n \mathbf{r}^{n-1} \right] \\
&= \vec{\nabla} \cdot \left[n \vec{r} r^{n-1} r^{-1} \right] \\
&= \vec{\nabla} \cdot \left[n \vec{r} r^{n-1-1} \right] \\
&= \vec{\nabla} \cdot \left[n \vec{r} r^{n-2} \right] \\
&= n \left[\vec{\nabla} \cdot (\vec{r} r^{n-2}) \right] \\
&= n \left[\vec{\nabla} \cdot (r^{n-2} \vec{r}) \right] \\
&= n \left[r^{n-2} (\vec{\nabla} \cdot \vec{r}) + \vec{\nabla} (r^{n-2}) \cdot \vec{r} \right]
\end{aligned}
\quad
\begin{aligned}
&[\because \vec{\nabla} \cdot \vec{\nabla} = \nabla^2] \\
&[\because \vec{\nabla} = \frac{\vec{r}}{r} \frac{\partial}{\partial r}] \\
&[\because \vec{\nabla} \cdot (\phi \vec{A}) = \phi (\vec{\nabla} \cdot \vec{A}) + (\vec{\nabla} \phi) \cdot \vec{A}]
\end{aligned}$$

$$\begin{aligned}
&= n \left[3r^{n-2} + \frac{\vec{r}}{r} \frac{\partial}{\partial r} (r^{n-2}) \cdot \vec{r} \right] \quad [\because \vec{\nabla} = \frac{\vec{r}}{r} \frac{\partial}{\partial r}] \text{ \& } [\vec{\nabla} \cdot \vec{r} = 3] \\
&= n \left[3r^{n-2} + \frac{1}{r} \frac{\partial}{\partial r} (r^{n-2}) (\vec{r} \cdot \vec{r}) \right] \\
&= n \left[3r^{n-2} + \frac{1}{r} (n-2)(r^{n-2-1}) (\vec{r} \cdot \vec{r}) \right] \\
&= n \left[3r^{n-2} + \frac{1}{r} (n-2) r^{n-3} r^2 \right] \quad [\because \vec{r} \cdot \vec{r} = r^2] \\
&= n \left[3r^{n-2} + (n-2) r^{n-3} r \right] \\
&= n \left[3r^{n-2} + (n-2) r^{n-3+1} \right] \\
&= n \left[3r^{n-2} + (n-2) r^{n-2} \right] \\
&= n[3 + n - 2] r^{n-2} \\
&= n(n+1) r^{n-2}
\end{aligned}$$

Q# 61: Show that $\vec{\nabla} \cdot (r^3 \vec{r}) = 6r^3$

$$\text{L.H.S} = \vec{\nabla} \cdot (r^3 \vec{r})$$

$$\begin{aligned}
&= r^3 (\vec{\nabla} \cdot \vec{r}) + \vec{\nabla} (r^3) \cdot \vec{r} \quad [\because \vec{\nabla} \cdot (\phi \vec{A}) = \phi (\vec{\nabla} \cdot \vec{A}) + (\vec{\nabla} \phi) \cdot \vec{A}] \\
&= 3r^3 + \left[\frac{\vec{r}}{r} \frac{\partial}{\partial r} r^3 \right] \cdot \vec{r} \quad [\because \vec{\nabla} \cdot \vec{r} = 3 \text{ \& } \vec{\nabla} = \frac{\vec{r}}{r} \frac{\partial}{\partial r}] \\
&= 3r^3 + \left[\frac{\vec{r}}{r} 3r^{3-1} \right] \cdot \vec{r} \\
&= 3r^3 + \frac{1}{r} 3r^2 (\vec{r} \cdot \vec{r}) \\
&= 3r^3 + 3r(r^2) \quad [\because \vec{r} \cdot \vec{r} = r^2] \\
&= 3r^3 + 3r^3 \\
&= 6r^3
\end{aligned}$$

Q# 62: Show that $\vec{\nabla} \cdot \left[r \vec{\nabla} \left(\frac{1}{r^3} \right) \right] = \frac{3}{r^4}$.

$$\begin{aligned}
\text{L.H.S} &= \vec{\nabla} \cdot \left[r \vec{\nabla} \left(\frac{1}{r^3} \right) \right] \\
&= \vec{\nabla} \cdot \left[r \left\{ \frac{\vec{r}}{r} \frac{\partial}{\partial r} \left(\frac{1}{r^3} \right) \right\} \right] \quad [\because \vec{\nabla} = \frac{\vec{r}}{r} \frac{\partial}{\partial r}] \\
&= \vec{\nabla} \cdot \left[r \left\{ \frac{\vec{r}}{r} \frac{\partial}{\partial r} r^{-3} \right\} \right]
\end{aligned}$$

Chapter 3: Vector Integration

Categories of curves

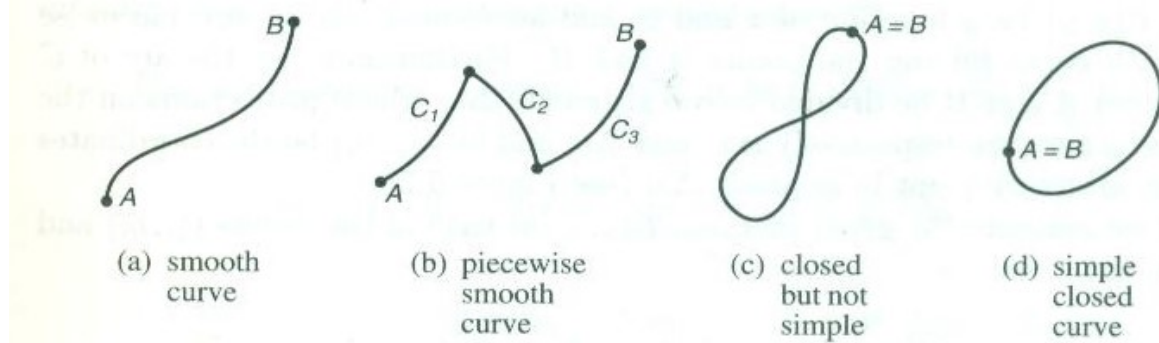


Figure # 79

1. C is smooth curve
2. C is piecewise smooth if it consists of a finite number of smooth curves $C_1, C_2, \dots, C_n$ joined end to end i.e. $C = C_1 \cup C_2 \cup \dots \cup C_n$
3. C is a closed curve if $A = B$
4. C is a simple closed curve if $A = B$ and the curve does not cross itself

First, we approximate the curve C by a polygonal path - a path made up of straight-line segments - as shown in figures below.

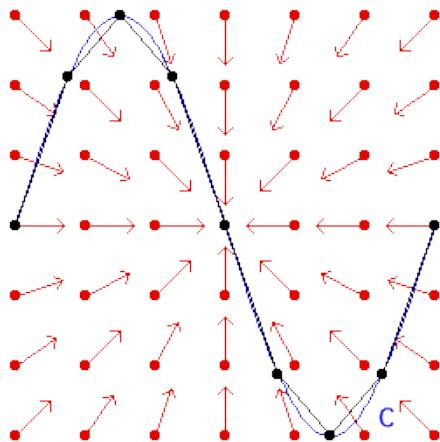


Figure # 80

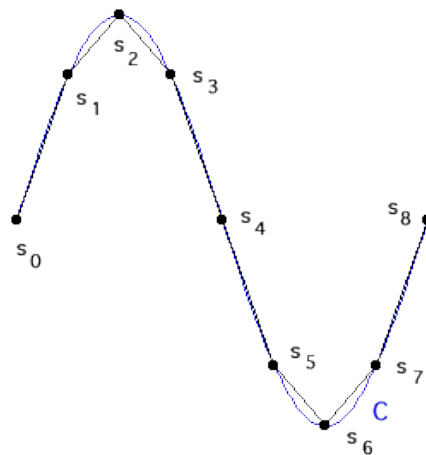


Figure # 81

Path integral may refer to: Line integral, Suppose a force F acting at each point on a smooth curve C .

Line Integral

Any Integral which is evaluated along the curve is called Line Integral, and it is denoted by $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F}$ is a vector point function, $\vec{r}$ is a position vector and C is the curve

Theorem: the work performed by a vector field on a particle moving along a parametric curve C is obtained by integrating the scalar tangential component of force along C.

$$W = \oint_C \vec{F} \cdot d\vec{r}$$

Q # 74: Establish the path Integral $\int_C \vec{F} \cdot d\vec{S} = \lim_{n \rightarrow \infty} \sum_{i=1}^n \vec{F}(\vec{S}_{i-1}) \cdot (\vec{S}_i - \vec{S}_{i-1})$

Answer:

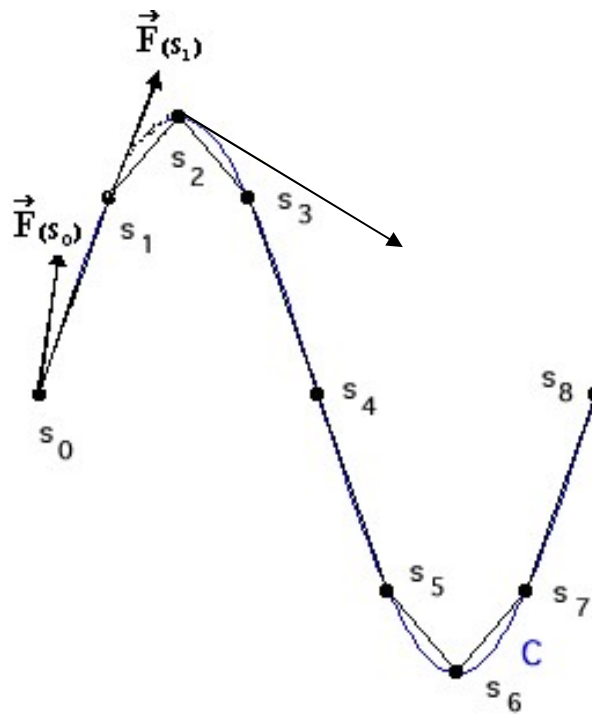


Figure # 82

We choose points $S_0, S_1, S_2, \dots, S_n$ along the path C and then connect these points as shown in the figure above.

and the amount of total work done on the whole path by :

$$W_1 = \vec{F}(S_0) \cdot (\vec{S}_1 - \vec{S}_0) = \vec{F}(S_{1-1}) \cdot (\vec{S}_1 - \vec{S}_{1-1})$$

$$W_2 = \vec{F}(S_1) \cdot (\vec{S}_2 - \vec{S}_1) = \vec{F}(S_{2-1}) \cdot (\vec{S}_2 - \vec{S}_{2-1})$$

$$W_3 = \vec{F}(S_2) \cdot (\vec{S}_3 - \vec{S}_2) = \vec{F}(S_{3-1}) \cdot (\vec{S}_3 - \vec{S}_{3-1})$$

$$\mathbf{W}_n = \vec{\mathbf{F}}_{(s_{n-1})} \cdot (\vec{\mathbf{S}}_n - \vec{\mathbf{S}}_{n-1})$$

$$\text{Total Work: } \mathbf{W}_1 + \mathbf{W}_2 + \mathbf{W}_3 + \text{-----} + \mathbf{W}_n = \sum_{i=1}^n \mathbf{W}_i = \sum_{i=1}^n \vec{\mathbf{F}}_{(\vec{\mathbf{S}}_{i-1})} \cdot (\vec{\mathbf{S}}_i - \vec{\mathbf{S}}_{i-1})$$

Then we estimate the Total work done on the **i**-th segment of the path by

$$\sum_{i=1}^n \mathbf{W}_i = \vec{\mathbf{F}}_{(s_{i-1})} \cdot (\vec{\mathbf{S}}_i - \vec{\mathbf{S}}_{i-1})$$

By using a large number of small segments we can obtain a very good estimate for the amount of work done. The exact amount of work done is obtained by taking the limit of these estimates. This limit is called the **line integral** of the vector field **F** over the path **C** and the amount of work done on the whole path by:

$$\text{Total Work done} = \int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}} = \lim_{n \rightarrow \infty} \sum_{i=1}^n \vec{\mathbf{F}}_{(\vec{\mathbf{S}}_{i-1})} \cdot (\vec{\mathbf{S}}_i - \vec{\mathbf{S}}_{i-1}) \quad \text{Answer}$$

$$[\text{Where } ds = s_1 - s_0 = s_2 - s_1 = s_3 - s_2 = \text{.....} = s_n - s_{n-1}]$$

Mathematical Expression to find out work done along a curve

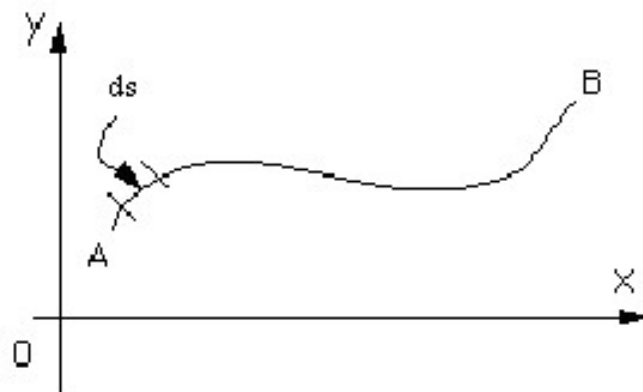


Figure # 83

Then the work done by that force in moving the particle small distance **ds** is given by:
Work Done = F × ds

Now if we want to know the total work done in moving the particle all the way from **A** to **B**, we need to add up all the small contributions, each of the form **F × ds**.

However **F** may have different strengths at different positions, i.e. **F** is a function of position, so what we need to add up are lots of contributions like **F(s) × ds**.

It should be familiar to you that when we add up lots of small things like that we do it by **integration**. The integral in this case is then: $\int_{AB} \vec{F} ds$

Notice that instead of upper and lower limits we just have AB written at the bottom of the integral sign; to show we're integrating **along the line from A to B**.

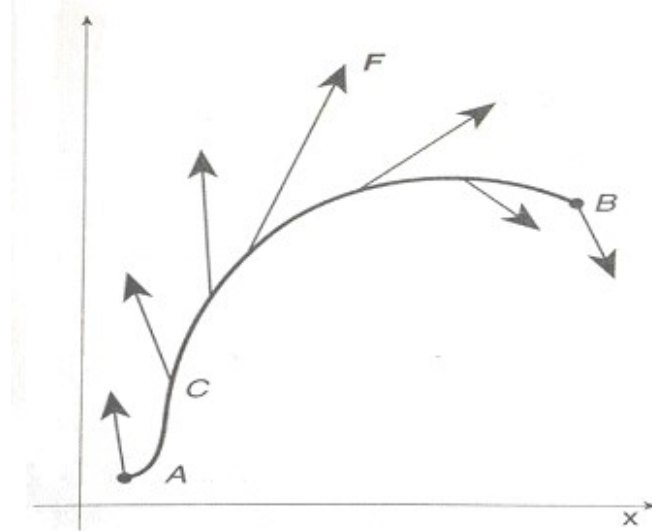


Figure # 84

A force $\vec{F}$ acting at each point on a smooth curve C.

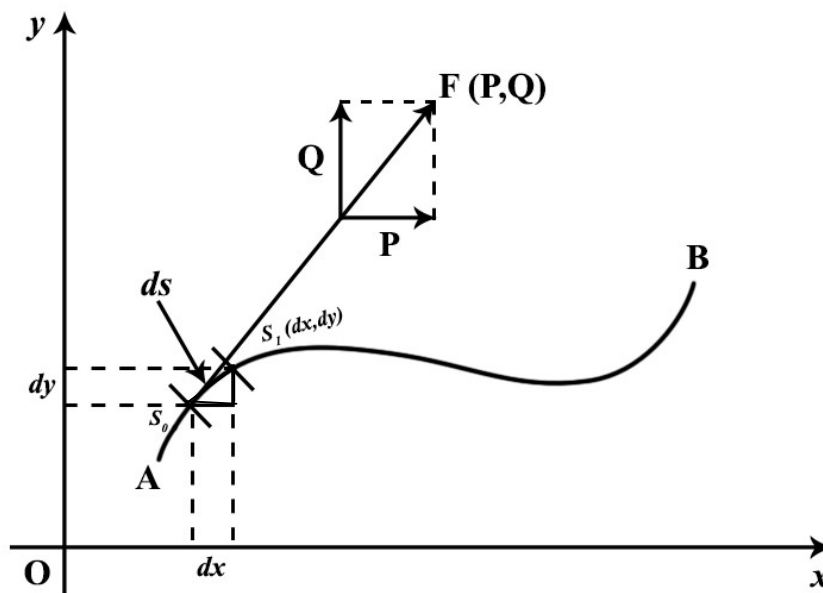


Figure # 85

Here I have divided the force $\vec{F}$ into a component in the x-direction, called P, and a component in the y-direction, called Q. The work done, $\vec{F} \times d\vec{s}$, can therefore be written as $Pdx + Qdy$.

So we can rewrite the integral above: $\int_{AB} \vec{F} d\vec{s} = \int_{AB} P dx + Q dy$

Because,

The work done by $\vec{F}$ in moving the particle from the tail to the head $d\vec{S}$ is approximately:

$$\vec{F} \cdot d\vec{S} \text{ -----(i)}$$

If $\vec{F}(x,y) = P(x,y)\hat{i} + Q(x,y)\hat{j}$ (Where P and Q are functions of x and y)

$$\text{and } d\vec{S} = dx\hat{i} + dy\hat{j}$$

$$\text{Then, } \vec{F} \cdot d\vec{S} = (P\hat{i} + Q\hat{j}) \cdot (dx\hat{i} + dy\hat{j})$$

$$\text{Then, } \vec{F} \cdot d\vec{S} = P dx + Q dy \quad [\because \hat{i} \cdot \hat{i} = 1; \hat{j} \cdot \hat{j} = 1] \text{ -----(ii)}$$

Since $P dx + Q dy$ is a local estimate (from the tail to the head $d\vec{S}$) of the work, the total work is represented by a line integral:

$$\text{Work} = \int_{AB} \vec{F} \cdot d\vec{S} = \int_{AB} (P\hat{i} + Q\hat{j}) \cdot (dx\hat{i} + dy\hat{j}) = \int_{AB} (P dx + Q dy) \text{ -----(iii)}$$

In three dimension,

$$\text{Work} = \int_{AB} \vec{F} \cdot d\vec{S} = \int_{AB} (P\hat{i} + Q\hat{j} + R\hat{k}) \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k}) = \int_{AB} (P dx + Q dy + R dz) \text{ ----}$$

(iv)

Q # 75: How much work is accomplished by the force $\vec{F}(x,y) = xy\hat{i} + y\hat{j}$ in pushing a particle from $(0,0)$ to $(3,9)$ along the parabola $y = x^2$?

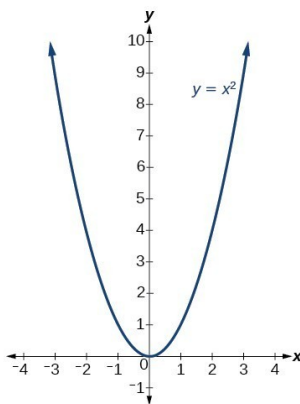


Figure # 86

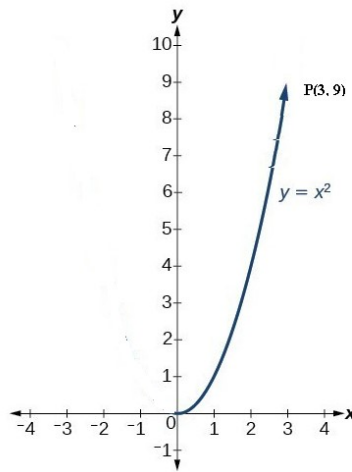


Figure # 87

Answer:

Figure shows the path of the particle, Call this path OP. Then

$$\text{Here, } \vec{F}(x, y) = xy \hat{i} + y \hat{j}, \text{-----(i)}$$

We know,

$$\vec{F}(x, y) = P \hat{i} + Q \hat{j} \text{-----(ii)}$$

Comparing (i) and (ii),

$$P = xy, Q = y$$

We know,

$$\text{Work} = \int_{OP} \vec{F} \cdot d\vec{S} = \int_{OP} (Pdx + Qdy)$$

$$\text{Work} = \int_{OP} (xydx + ydy) \quad [\because P = xy, Q = y]$$

To evaluate this integral, let us use x as the parameter,

Then, Given $y = x^2$

$$\frac{dy}{dx} = \frac{d}{dx}(x^2)$$

$$\frac{dy}{dx} = 2x$$

$$\therefore dy = 2x dx$$

$$\text{Work} = \int_0^3 (xydx + ydy)$$

$$\text{Work} = \int_0^3 [(x \cdot x^2 dx + x^2 \cdot 2x dx)]$$

$$\text{Work} = \int_0^3 (x^3 dx + 2x^3 dx)$$

$$\text{Work} = \int_0^3 3x^3 dx = 3 \left[\frac{x^{3+1}}{3+1} \right]_0^3 = \frac{3}{4} [x^4]_0^3 = \frac{3}{4} \times 3^4 = \frac{243}{4} \text{ Answer.}$$

Q # 76: Evaluate $\int_C xy dx$ from **B(1,0)** to **C(0,1)** along the curve C that is the portion of $x^2 + y^2 = 1$ in the first quadrant.

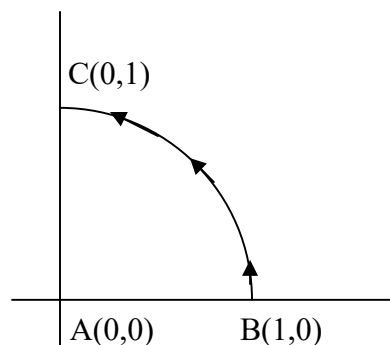


Figure # 88

Answer:

Given $x^2 + y^2 = 1$

$\Rightarrow y = \sqrt{1 - x^2}$

The curve BC is the first quadrant of the unit circle as shown in figure. On the

curve $y = \sqrt{1 - x^2}$, so that,

$$\int_C xy \, dx = \int_1^0 x \sqrt{1 - x^2} \, dx$$

Let $1 - x^2 = z$

$-2x \, dx = dz$

$x \, dx = -\frac{dz}{2}$

x	1	0
$1 - x^2 = z$	$z = 1 - x^2$	$z = 1 - x^2$
$\therefore z = 1 - x^2$	$z = 1 - 1$	$z = 1 - 0$
	$z = 0$	$z = 1$

$$\int_C xy \, dx = \int_1^0 x \sqrt{1 - x^2} \, dx =$$

$$-\int_0^1 \sqrt{z} \frac{dz}{2} = -\frac{1}{2} \int_0^1 \sqrt{z} \, dz = -\frac{1}{2} \left[\frac{z^{\frac{3}{2}}}{\frac{3}{2}} \right]_0^1 = -\frac{1}{2} \left[\frac{1^{\frac{3}{2}}}{\frac{3}{2}} - \frac{0^{\frac{3}{2}}}{\frac{3}{2}} \right] = -\frac{1}{2} \left[\frac{1}{\frac{3}{2}} - \frac{0}{\frac{3}{2}} \right]$$

$$= -\frac{1}{2} \times \frac{2}{3} (1 - 0) = -\frac{1}{3} \text{ Answer}$$

Q # 77: Find the value of the line integral when $\vec{F}(\mathbf{r}) = -y \hat{i} - xy \hat{j}$, where $\vec{r}$ is a function of t and C is the circular arc in Figure from A to B.

Or

Find the work done in moving a particle once around a quarter circle C in the xy plane, if the circle has center at the origin and radius 1 and if the force field is given by

$\vec{F}(\mathbf{r}) = -y \hat{i} - xy \hat{j}$

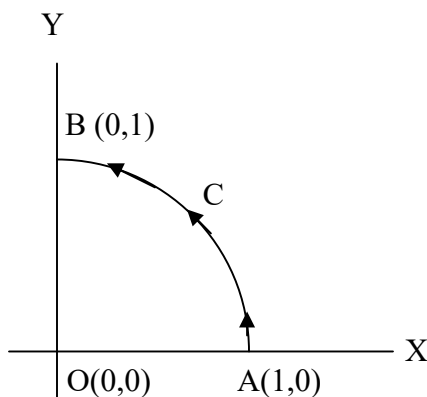


Figure # 89

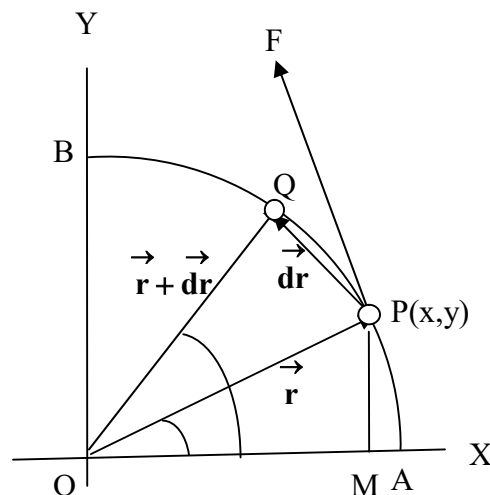


Figure # 90

$$\mathbf{W} = \left[\frac{-63}{6} \right] + \left[\frac{18}{3} \right] + \left[\frac{30}{4} \right]$$

$$\mathbf{W} = \frac{-126 + 72 + 90}{12}$$

$$\mathbf{W} = \frac{-126 + 162}{12}$$

$$\mathbf{W} = \frac{36}{12} = 3; \text{ where the units for } \mathbf{W} \text{ depend on the units chosen for force and distance}$$

Q # 80: Find the value of $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = (y - 2x)\hat{i} + (3x + 2y)\hat{j}$ and C is a circle in the xy- plane with center the origin and radius 2.

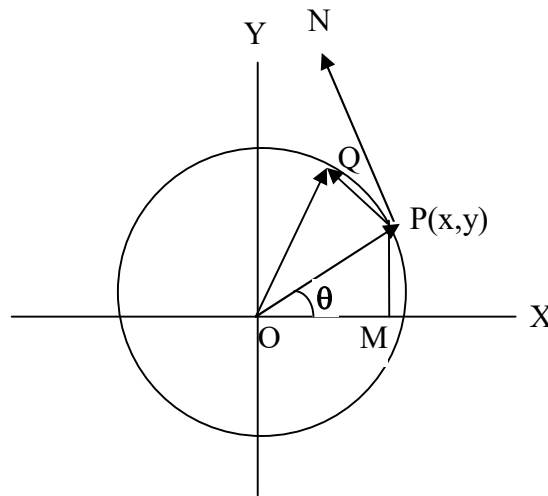


Figure # 94

The position Vector is

$$\vec{r} = x\hat{i} + y\hat{j} \text{-----(i)}$$

[Page no 48, Figure no 57, Equation no (i)]

Let,

$$\angle POM = \theta, \quad OP = 2$$

$$\frac{PM}{OP} = \sin \theta$$

$$\frac{y}{2} = \sin \theta \Rightarrow y = 2 \sin \theta$$

Similarly,

$$\frac{OM}{OP} = \cos \theta \Rightarrow \frac{x}{2} = \cos \theta \Rightarrow x = 2 \cos \theta$$

From (i),

$$\vec{r} = x\hat{i} + y\hat{j}$$

$$\vec{r}(\theta) = 2 \cos \theta \hat{i} + 2 \sin \theta \hat{j}$$

$$\frac{d\vec{r}}{d\theta} = -2 \sin \theta \hat{i} + 2 \cos \theta \hat{j}$$

$$\Rightarrow d\vec{r} = (-2 \sin \theta \hat{i} + 2 \cos \theta \hat{j}) d\theta$$

Given,

$$\vec{F} = (y - 2x) \hat{i} + (3x + 2y) \hat{j}$$

$$\therefore \int_C \vec{F} \cdot d\vec{r} = \int \{(y - 2x) \hat{i} + (3x + 2y) \hat{j}\} \cdot (-2 \sin \theta \hat{i} + 2 \cos \theta \hat{j}) d\theta$$

$$[\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0]$$

$$\therefore \int_C \vec{F} \cdot d\vec{r} = \int \{(y - 2x)(-2 \sin \theta) d\theta + (3x + 2y)(2 \cos \theta) d\theta\}.$$

$$\therefore \int_C \vec{F} \cdot d\vec{r} = \int \{(2 \sin \theta - 2 \times 2 \cos \theta)(-2 \sin \theta) d\theta + (3 \times 2 \cos \theta + 2 \times 2 \sin \theta)(2 \cos \theta) d\theta\}.$$

$$\therefore \int_C \vec{F} \cdot d\vec{r} = \int \{(2 \sin \theta - 4 \cos \theta)(-2 \sin \theta) d\theta + (6 \cos \theta + 4 \sin \theta)(2 \cos \theta) d\theta\}.$$

$$\therefore \int_C \vec{F} \cdot d\vec{r} = \int \{(-4 \sin^2 \theta + 8 \sin \theta \cos \theta) d\theta + (12 \cos^2 \theta + 8 \sin \theta \cos \theta) d\theta\}.$$

$$\therefore \int_C \vec{F} \cdot d\vec{r} = \int (-4 \sin^2 \theta + 16 \sin \theta \cos \theta + 12 \cos^2 \theta) d\theta.$$

$$\therefore \int_C \vec{F} \cdot d\vec{r} = \int_0^{2\pi} (-4 \sin^2 \theta + 16 \sin \theta \cos \theta + 12 \cos^2 \theta) d\theta.$$

$$\therefore \int_C \vec{F} \cdot d\vec{r} = \int_0^{2\pi} (-2 \times 2 \sin^2 \theta + 8 \times 2 \sin \theta \cos \theta + 6 \times 2 \cos^2 \theta) d\theta.$$

$$\therefore \int_C \vec{F} \cdot d\vec{r} = \int_0^{2\pi} \{-2(1 - \cos 2\theta) + 8 \times \sin 2\theta + 6(1 + \cos 2\theta)\} d\theta.$$

$$\therefore \int_C \vec{F} \cdot d\vec{r} = -2 \int_0^{2\pi} d\theta + 2 \int_0^{2\pi} \cos 2\theta d\theta + 8 \times \int_0^{2\pi} \sin 2\theta d\theta + 6 \int_0^{2\pi} d\theta + 6 \int_0^{2\pi} \cos 2\theta d\theta.$$

$$\therefore \int_C \vec{F} \cdot d\vec{r} = \left[-2\theta + 2 \frac{\sin 2\theta}{2} - 8 \times \frac{\cos 2\theta}{2} + 6\theta + 6 \frac{\sin 2\theta}{2} \right]_0^{2\pi}$$

$$\therefore \int_C \vec{F} \cdot d\vec{r} = \left[-2\left(\theta - \frac{\sin 2\theta}{2}\right) - 8 \times \frac{\cos 2\theta}{2} + 6\left(\theta + \frac{\sin 2\theta}{2}\right) \right]_0^{2\pi}$$

$$\therefore \int_C \vec{F} \cdot d\vec{r}$$

$$= -2 \left[\left(2\pi - \frac{\sin 2 \times 2\pi}{2} \right) - \left(0 - \frac{\sin 2 \times 0}{2} \right) \right] - 8 \left[\left(\frac{\cos 2 \times 2\pi}{2} - \frac{\cos 2 \times 0}{2} \right) \right] + 6 \left[\left(2\pi + \frac{\sin 2 \times 2\pi}{2} \right) - \left(0 + \frac{\sin 2 \times 0}{2} \right) \right]$$

$$\begin{aligned}
&= -2 \left[\left(2\pi - \frac{\sin 4\pi}{2} \right) - \left(0 - \frac{\sin 0}{2} \right) \right] - 8 \left[\left(\frac{\cos 4\pi}{2} - \frac{\cos 0}{2} \right) \right] + 6 \left[\left(2\pi + \frac{\sin 4\pi}{2} \right) - \left(0 + \frac{\sin 0}{2} \right) \right] \\
&= -2 \left[\left(2\pi - \frac{0}{2} \right) - \left(0 - \frac{0}{2} \right) \right] - 8 \left[\left(\frac{1}{2} - \frac{1}{2} \right) \right] + 6 \left[\left(2\pi + \frac{0}{2} \right) - \left(0 + \frac{0}{2} \right) \right] \\
&= -2 \left[2\pi - \left(-\frac{0}{2} \right) \right] - 8[0] + 6[2\pi - 0] \\
&= -2[2\pi + 0] - 0 + 6[2\pi] \\
&= -2[2\pi] + 6[2\pi] \\
\therefore \int_C \vec{F} \cdot d\vec{r} &= -4\pi + 12\pi = 8\pi \text{ Answer}
\end{aligned}$$

Q # 81: Find the value of $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = x^2 \hat{i} + 3xy \hat{j}$ if

- C is the straight line path from (0,0) to (1,2)
- C is the parabolic path $y = 2x^2$ from (0,0) to (1,2)
- C is composed of two straight-line paths the x axis from (0,0) to (1,0) and then a line parallel to the y-axis from (1,0) to (1,2)

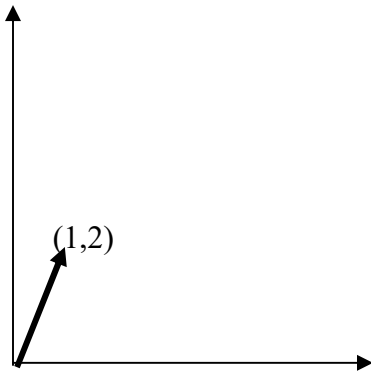


Figure # 95

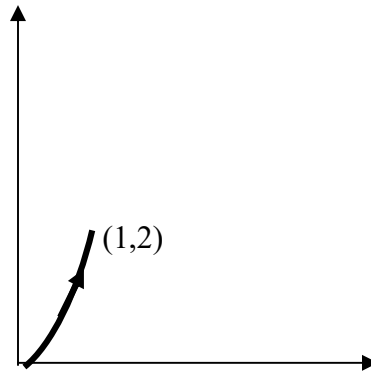


Figure # 96

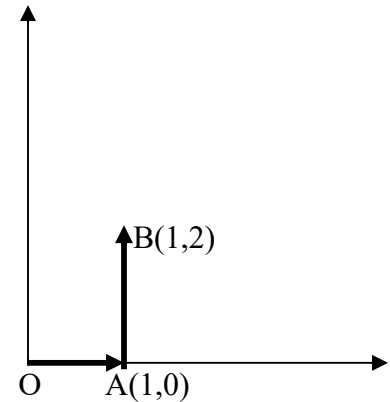


Figure # 97

Let, the position vector is: $\vec{r} = x \hat{i} + y \hat{j}$ [Page no 48, Figure no 57, Equation no (i)]

Since the line element $d\vec{r}$ lies in the xy-plane, we can express it as $d\vec{r} = dx \hat{i} + dy \hat{j}$, so that

$$\begin{aligned}
\int_C \vec{F} \cdot d\vec{r} &= \int_C (x^2 \hat{i} + 3xy \hat{j}) \cdot (dx \hat{i} + dy \hat{j}) \\
&= \int_C (x^2 dx + 3xy dy) \text{ -----(i)}
\end{aligned}$$

- On this path $y = 2x$, so we can convert (i) to a definite integral with respect to x, this is effectively a simple parameterization, $x = t, y = 2t, 0 \leq t \leq 1$
Hence,

$$\oint_C [y^2 dx - x^2 dy] = \int_2^0 [x^4 dx - x^2 (2x dx)] = \int_2^0 (x^4 dx - 2x^3) dx$$

$$= \left[\frac{1}{5} x^5 - \frac{1}{2} x^4 \right]_2^0 = \frac{8}{5}$$

Therefore, $\oint_C y^2 dx - x^2 dy = 0 - 16 + \frac{8}{5} = -\frac{72}{5}$

Q # 84: Evaluate the line integral, $\int \vec{F} \cdot d\vec{r}$ where the force field is given by

$\vec{F}(x, y) = 3xy \hat{i} - 5z \hat{j} + 10x \hat{k}$ along the curve, $x = t^2 + 1$, $y = 2t^2$, $z = t^3$ from $t = 1$ to $t = 2$

Answer:

We have the position vector

$$\therefore \vec{r} = x \hat{i} + y \hat{j} + z \hat{k} \quad [\text{Page no 48, Figure no 57, Equation no (i)}]$$

$$\therefore d\vec{r} = dx \hat{i} + dy \hat{j} + dz \hat{k}$$

$$\vec{F} \cdot d\vec{r} = [3xy \hat{i} - 5z \hat{j} + 10x \hat{k}] \cdot [dx \hat{i} + dy \hat{j} + dz \hat{k}]$$

$$\vec{F} \cdot d\vec{r} = 3xy dx - 5z dy + 10x dz \text{-----(i)}$$

Given

$$x = t^2 + 1$$

$$\therefore \frac{dx}{dt} = \frac{d}{dt}(t^2 + 1)$$

$$\therefore \frac{dx}{dt} = 2t$$

$$\therefore dx = 2t dt$$

$$y = 2t^2$$

$$\therefore \frac{dy}{dt} = \frac{d}{dt}(2t^2)$$

$$\therefore \frac{dy}{dt} = 4t$$

$$\therefore dy = 4t dt$$

$$z = t^3$$

$$\therefore \frac{dz}{dt} = \frac{d}{dt}(t^3)$$

$$\therefore \frac{dz}{dt} = 3t^2$$

$$\therefore dz = 3t^2 dt$$

From (i),

$$\vec{F} \cdot d\vec{r} = 3xydx - 5zdy + 10xdz$$

$$\vec{F} \cdot d\vec{r} = 3(t^2 + 1)(2t^2)2tdt - 5t^3 4tdt + 10(t^2 + 1)3t^2 dt$$

$$\vec{F} \cdot d\vec{r} = 3(t^2 + 1)(4t^3)dt - 20t^4 dt + 10(t^2 + 1)3t^2 dt$$

$$\vec{F} \cdot d\vec{r} = 3(4t^5 + 4t^3)dt - 20t^4 dt + 10(3t^4 + 3t^2)dt$$

$$\vec{F} \cdot d\vec{r} = (12t^5 + 12t^3)dt - 20t^4 dt + (30t^4 + 30t^2)dt$$

$$\vec{F} \cdot d\vec{r} = 12t^5 dt + 12t^3 dt - 20t^4 dt + 30t^4 dt + 30t^2 dt$$

$$\vec{F} \cdot d\vec{r} = 12t^5 dt + 10t^4 dt + 12t^3 dt + 30t^2 dt$$

$$\begin{aligned} \text{Total work done} &= \int_1^2 \vec{F} \cdot d\vec{r} = \int_1^2 (12t^5 dt + 10t^4 dt + 12t^3 dt + 30t^2 dt) \\ &= \int_1^2 12t^5 dt + \int_1^2 10t^4 dt + \int_1^2 12t^3 dt + \int_1^2 30t^2 dt \\ &= \left[12 \frac{t^6}{6} + 10 \frac{t^5}{5} + 12 \frac{t^4}{4} + 30 \frac{t^3}{3} \right]_1^2 \\ &= [2t^6 + 2t^5 + 3t^4 + 10t^3]_1^2 \\ &= (2 \times 2^6 + 2 \times 2^5 + 3 \times 2^4 + 10 \times 2^3) - (2 \times 1^6 + 2 \times 1^5 + 3 \times 1^4 + 10 \times 1^3) \\ &= (2 \times 64 + 2 \times 32 + 3 \times 16 + 10 \times 8) - (2 + 2 + 3 + 10) \\ &= (128 + 64 + 48 + 80) - (2 + 2 + 3 + 10) \\ &= (320) - (17) \\ &= 303 \end{aligned}$$

Q # 85: If $\vec{F} = xy\hat{i} - z\hat{j} + x^2\hat{k}$ and C is the curve curve, $x = t^2$, $y = 2t$, $z = t^3$ from $t = 0$ to $t = 1$, then evaluate the line integral, $\int_C \vec{F} \times d\vec{r}$

Answer: We have the position vector

$$\therefore \vec{r} = x\hat{i} + y\hat{j} + z\hat{k} \quad [\text{Page no 48, Figure no 57, Equation no (i)}]$$

$$\begin{aligned}
&= \left[9z\hat{i} + 6\frac{z^2}{2}\hat{j} - 8z\hat{k} \right]_0^4 \\
&= \left[9z\hat{i} + 3z^2\hat{j} - 8z\hat{k} \right]_0^4 \\
&= \left[(9 \times 4\hat{i} + 3 \times 4^2\hat{j} - 8 \times 4\hat{k}) - (9 \times 0\hat{i} + 3 \times 0^2\hat{j} - 8 \times 0\hat{k}) \right] \\
&= \left[36\hat{i} + 3 \times 16\hat{j} - 32\hat{k} \right] \\
&= 4 \left[9\hat{i} + 12\hat{j} - 8\hat{k} \right]
\end{aligned}
\quad \left[\because \int x^n dx = \frac{x^{n+1}}{n+1}; \int dz = z \right]$$

Q # 90: Show that $\iint_S \vec{F} \cdot \hat{\eta} ds = \frac{3}{2}$; Where $\vec{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$ and S is the surface of the cube bounded by the planes $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$

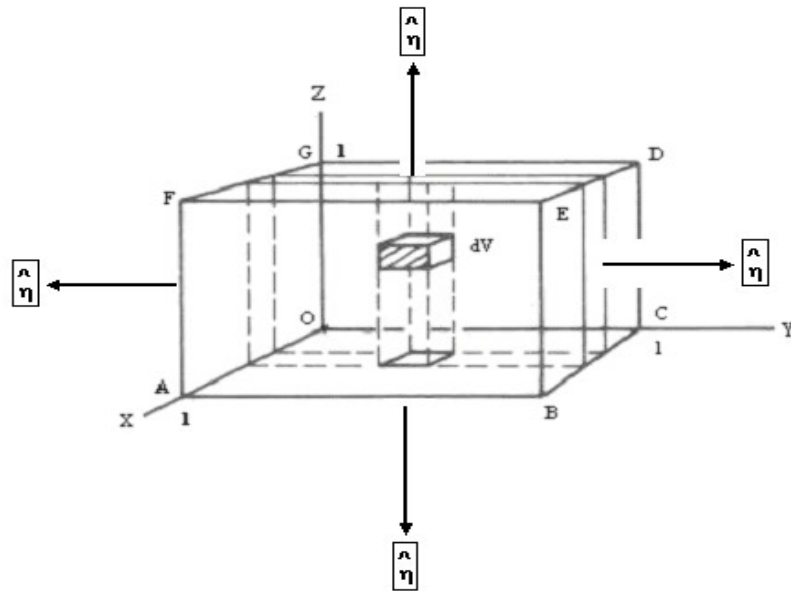


Figure # 113

Serial no	Surface	$\hat{\eta}=?$	ds	Plane
1	OABC	$\hat{\eta} = -\hat{k}$	$dx dy$	$z = 0$
2	DEFG	$\hat{\eta} = \hat{k}$	$dx dy$	$z = 1$
3	OAFG	$\hat{\eta} = -\hat{j}$	$dx dz$	$y = 0$
4	BCDE	$\hat{\eta} = \hat{j}$	$dx dz$	$y = 1$
5	OCDG	$\hat{\eta} = -\hat{i}$	$dy dz$	$x = 0$
6	ABEF	$\hat{\eta} = \hat{i}$	$dy dz$	$x = 1$

Now,

$$\iint_S \vec{F} \cdot \hat{n} ds = \iint_{OABC} \vec{F} \cdot \hat{n} ds + \iint_{DEFG} \vec{F} \cdot \hat{n} ds + \iint_{OAFG} \vec{F} \cdot \hat{n} ds + \iint_{BCDE} \vec{F} \cdot \hat{n} ds + \iint_{OCDG} \vec{F} \cdot \hat{n} ds + \iint_{ABEF} \vec{F} \cdot \hat{n} ds \quad (1)$$

$$1. \iint_{OABC} \vec{F} \cdot \hat{n} ds = \iint_S (4xz \hat{i} - y^2 \hat{j} + yz \hat{k}) \cdot \hat{n} ds = \iint_S (4xz \hat{i} - y^2 \hat{j} + yz \hat{k}) \cdot (-\hat{k}) dxdy$$

$$[\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0]$$

$$= \int_0^1 \int_0^1 -yz dxdy$$

$$= - \int_0^1 \int_0^1 yz dxdy = - \int_0^1 [yzx]_0^1 dy \quad [\because \int dx = x]$$

$$= - \int_0^1 [yz \times 1 - yz \times 0] dy = - \int_0^1 [yz(1 - 0)] dy$$

$$= - \left[z \frac{y^2}{2} \right]_0^1 = - \left[z \frac{1^2}{2} - z \frac{0^2}{2} \right] = - \frac{z}{2} (1 - 0) = - \frac{z}{2} = 0 [\because z = 0]$$

2.

$$\iint_{DEFG} \vec{F} \cdot \hat{n} ds = \iint_S (4xz \hat{i} - y^2 \hat{j} + yz \hat{k}) \cdot \hat{n} ds = \iint_S (4xz \hat{i} - y^2 \hat{j} + yz \hat{k}) \cdot (\hat{k}) dxdy = \int_0^1 \int_0^1 yz dxdy$$

$$[\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0]$$

$$= \int_0^1 \int_0^1 yz dxdy = \int_0^1 [yzx]_0^1 dy = \int_0^1 [yz \times 1 - yz \times 0] dy \quad [\because \int dx = x]$$

$$= \int_0^1 [yz] dy = \left[z \frac{y^2}{2} \right]_0^1 = \left[z \frac{1^2}{2} - z \frac{0^2}{2} \right] = \frac{z}{2} (1 - 0) = \frac{z}{2} = \frac{1}{2} [\because z = 1]$$

3.

$$\iint_{OAFG} \vec{F} \cdot \hat{n} ds = \iint_S (4xz \hat{i} - y^2 \hat{j} + yz \hat{k}) \cdot \hat{n} ds = \iint_S (4xz \hat{i} - y^2 \hat{j} + yz \hat{k}) \cdot (-\hat{j}) dxdz = \int_0^1 \int_0^1 y^2 dxdz$$

$$[\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0]$$

$$= \int_0^1 \int_0^1 y^2 dxdz = \int_0^1 [xy^2]_0^1 dz \quad [\because \int dx = x]$$

$$= \int_0^1 [1 \times y^2 - 0 \times y^2] dz$$

$$= \int_0^1 [y^2 - 0] dz = [y^2 z]_0^1 = [y^2 \times 1 - y^2 \times 0] \quad [\because \int dz = z]$$

$$= y^2 (1 - 0) = y^2 = 0 \quad [\because y = 0]$$

4.

$$\begin{aligned}
 \iint_{BCDE} \vec{F} \cdot \hat{n} ds &= \iint_S (4xz \hat{i} - y^2 \hat{j} + yz \hat{k}) \cdot \hat{n} ds = \iint_S (4xz \hat{i} - y^2 \hat{j} + yz \hat{k}) \cdot (\hat{j}) dx dz = \int_0^1 \int_0^1 -y^2 dx dz \\
 [\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0] \\
 &= -\int_0^1 \int_0^1 y^2 dx dz = -\int_0^1 [xy^2]_0^1 dz = -\int_0^1 [1 \times y^2 - 0 \times y^2] dz \quad [\because \int dx = x] \\
 &= -\int_0^1 [y^2 - 0] dz = -[y^2 z]_0^1 \quad [\because \int dz = z] \\
 &= -[y^2 \times 1 - y^2 \times 0] \\
 &= -y^2 (1 - 0) = -y^2 = -1 \quad [\because y = 1]
 \end{aligned}$$

$$\begin{aligned}
 5. \iint_{OCDE} \vec{F} \cdot \hat{n} ds &= \iint_S (4xz \hat{i} - y^2 \hat{j} + yz \hat{k}) \cdot \hat{n} ds = \iint_S (4xz \hat{i} - y^2 \hat{j} + yz \hat{k}) \cdot (-\hat{i}) dy dz \\
 [\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0] \\
 &= \int_0^1 \int_0^1 -4xz dy dz \\
 &= -\int_0^1 \int_0^1 4xz dy dz = -\int_0^1 [4xyz]_0^1 dz = -\int_0^1 [4x \times 1 \times z - 4x \times 0 \times z] dz \quad [\because \int dy = y] \\
 &= -\int_0^1 [4xz(1 - 0)] dz = -\left[4x \frac{z^2}{2}\right]_0^1 = -\left[4x \frac{1^2}{2} - 4x \frac{0^2}{2}\right] \\
 &= -4x \left(\frac{1}{2} - 0\right) = -2x = 0 \quad [\because x = 0]
 \end{aligned}$$

$$\begin{aligned}
 6. \iint_{ABEF} \vec{F} \cdot \hat{n} ds &= \iint_S (4xz \hat{i} - y^2 \hat{j} + yz \hat{k}) \cdot \hat{n} ds = \iint_S (4xz \hat{i} - y^2 \hat{j} + yz \hat{k}) \cdot (\hat{i}) dy dz \\
 [\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0] \\
 &= \int_0^1 \int_0^1 4xz dy dz \\
 &= \int_0^1 \int_0^1 4xz dy dz = \int_0^1 [4xyz]_0^1 dz = \int_0^1 [4x \times 1 \times z - 4x \times 0 \times z] dz \quad [\because \int dy = y] \\
 &= \int_0^1 [4xz(1 - 0)] dz = \left[4x \frac{z^2}{2}\right]_0^1 = \left[4x \frac{1^2}{2} - 4x \frac{0^2}{2}\right] \\
 &= 4x \left(\frac{1}{2} - 0\right) = 2x = 2 \quad [\because x = 1]
 \end{aligned}$$

Putting the values in (1),

Now,

$$\iint_S \vec{F} \cdot \hat{n} ds = \iint_{OABC} \vec{F} \cdot \hat{n} ds + \iint_{DEFG} \vec{F} \cdot \hat{n} ds + \iint_{OAFG} \vec{F} \cdot \hat{n} ds + \iint_{BCDE} \vec{F} \cdot \hat{n} ds + \iint_{OCDG} \vec{F} \cdot \hat{n} ds + \iint_{ABEF} \vec{F} \cdot \hat{n} ds$$

$$\iint_S \vec{F} \cdot \hat{n} ds = 0 + \frac{1}{2} + 0 - 1 + 0 + 2 = \frac{1}{2} + 1 = \frac{3}{2} \text{ (Proved)}$$

Area Enclosed by a closed Curve:

One of the earliest applications of integration is finding the area of a plane figure bounded by the x-axis, the curve $y = f(x)$ and ordinates at $x = x_1$ and $x = x_2$.

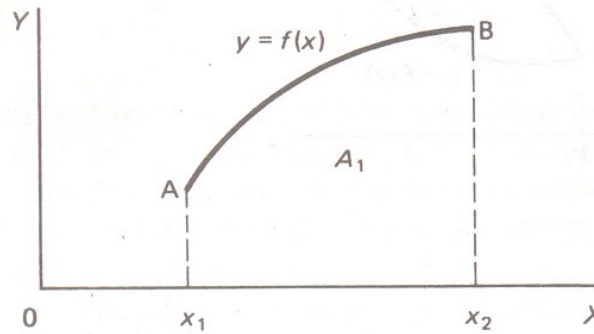


Figure # 114

$$A_1 = \int_{x_1}^{x_2} y dx = \int_{x_1}^{x_2} f(x) dx \text{ -----(i)}$$

If points A and B are joined by another curve $y = F(x)$

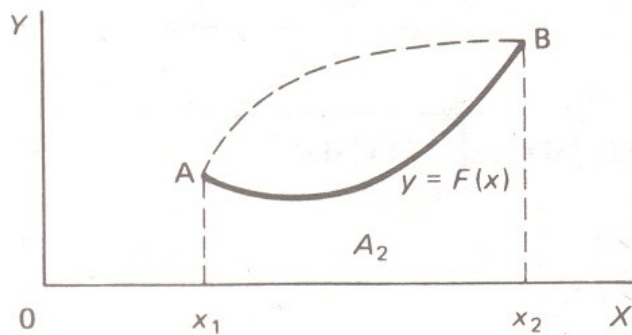


Figure # 115

$$A_2 = \int_{x_1}^{x_2} y dx = \int_{x_1}^{x_2} F(x) dx \text{ -----(ii)}$$

Combining the two figures, we have

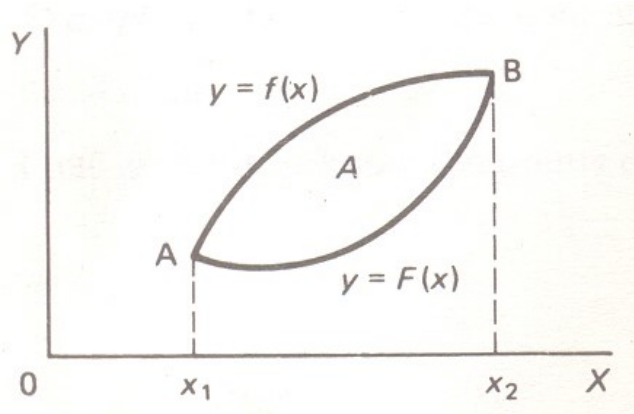


Figure # 116

$$A = A_1 - A_2$$

$$\therefore A = \int_{x_1}^{x_2} f(x)dx - \int_{x_1}^{x_2} F(x)dx \text{ -----(iii)}$$

It is convenient on occasions to arrange the limits so that the integration follows the path round the enclosed area in a regular order.

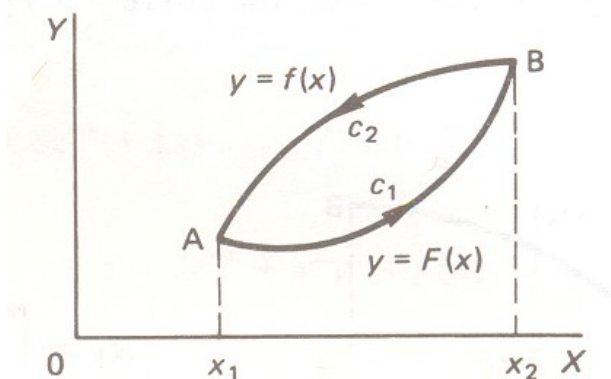


Figure # 117

For example

$\int_{x_1}^{x_2} F(x)dx$ gives A_2 as before, but integrating from B to A along c_2 with $y = f(x)$, i.e.

$\int_{x_1}^{x_2} f(x)dx$, is the integral for A_1 with the sign changed, i.e.

We can write, For Upper Curve $y = f(x)$;

$$\int_{x_2}^{x_1} f(x)dx = - \int_{x_1}^{x_2} f(x)dx \text{ -----(iv)}$$

$$4-2 = 2$$

$$-(2-4) = -(-2) = 2$$

$$\Rightarrow -\int_{x_2}^{x_1} f(x)dx = -\left[-\int_{x_1}^{x_2} f(x)dx\right] \quad [\text{Multiplying by negative sign on both sides}]$$

$$\Rightarrow -\int_{x_2}^{x_1} f(x)dx = \int_{x_1}^{x_2} f(x)dx$$

$$\therefore \int_{x_1}^{x_2} f(x)dx = -\int_{x_2}^{x_1} f(x)dx \quad [\text{Side change}]-----(\text{v})$$

From (iii),

$$\therefore \text{The result } A = A_1 - A_2 = \int_{x_1}^{x_2} f(x)dx - \int_{x_1}^{x_2} F(x)dx \quad [\text{From (iii)}]$$

$$= -\int_{x_2}^{x_1} f(x)dx - \int_{x_1}^{x_2} F(x)dx \quad [\text{From (v)}]$$

$$= -\left\{\int_{x_2}^{x_1} f(x)dx + \int_{x_1}^{x_2} F(x)dx\right\}$$

$$= -\left\{\int_{x_1}^{x_2} F(x)dx + \int_{x_2}^{x_1} f(x)dx\right\} -----(\text{vi})$$

If we proceed round the boundary in an anticlockwise manner, the enclosed area is kept on the left-hand side and the resulting area is considered positive. If we proceed round the boundary in a clockwise manner, the enclosed area remains on the right-hand side and the resulting area is negative.

The final result above can be written in the form

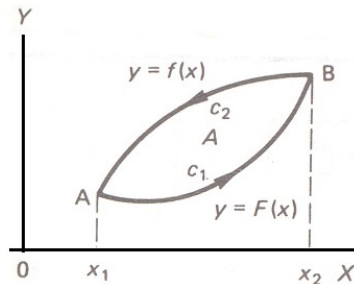


Figure # 118

$A = -\oint ydx$; Where the symbol $\oint$ indicates that the integral is to be evaluated round the closed boundary in the positive (i.e. anticlockwise) direction.

$$\therefore A = -\oint ydx = -\left\{\int_{x_1}^{x_2} F(x)dx + \int_{x_2}^{x_1} f(x)dx\right\} -----(\text{vii})$$

Q # 91: Determine the area enclosed by the graphs of $y = x^3$ and $y = 4x$ for $x \geq 0$

Answer: First we need to know the points of intersection.

Given, $y = x^3$ and $y = 4x$

Then, we can write, $x^3 = 4x$

$$\Rightarrow x^3 - 4x = 0$$

$$\Rightarrow x(x^2 - 4) = 0$$

$$\Rightarrow x(x+2)(x-2) = 0$$

$$\Rightarrow x = 0 \text{ and } x = -2, x = 2$$

But $x \geq 0$

$$\therefore x = 0 \text{ and } x = 2$$

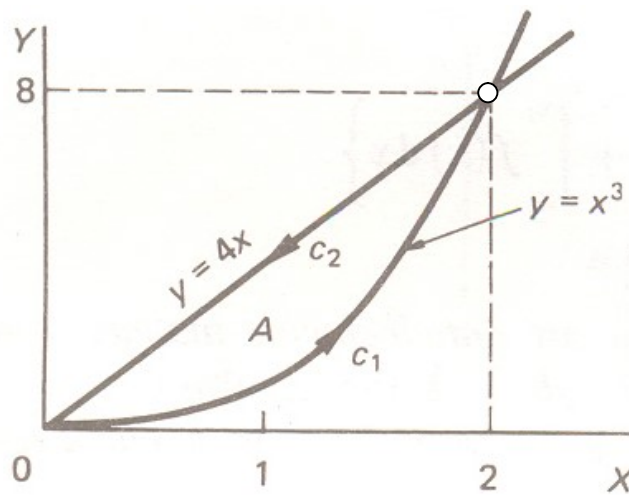


Figure # 119

We integrate in an anticlockwise manner

$c_1 : y = x^3$, Limits $x = 0$ to $x = 2$

$c_2 : y = 4x$, Limits $x = 2$ to $x = 0$

We have,

$$\begin{aligned} \therefore A &= -\oint y dx = -\left\{ \int_{x_1}^{x_2} F(x) dx + \int_{x_2}^{x_1} f(x) dx \right\} \\ &= -\left\{ \int_0^2 F(x) dx + \int_2^0 f(x) dx \right\} \\ &= -\left\{ \int_0^2 x^3 dx + \int_2^0 4x dx \right\} \\ &= -\left\{ \left[\frac{x^4}{4} \right]_0^2 + 4 \left[\frac{x^2}{2} \right]_2^0 \right\} \end{aligned}$$

$$\begin{aligned}
&= -\left\{\left[\frac{2^4}{4} - 0\right] + 4\left[\frac{0^2}{2} - \frac{2^2}{2}\right]\right\} \\
&= -\left\{\left[\frac{16}{4}\right] + 4\left[-\frac{4}{2}\right]\right\} \\
&= 4 \text{ Answer}
\end{aligned}$$

Q # 92: Find the area of the triangle with vertices **(0,0)** ,**(5,3)** and **(2,6)**

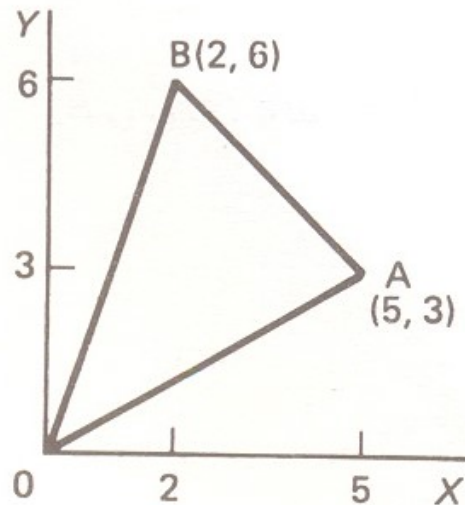


Figure # 120

Answer: We have,

$$\frac{y - y_1}{y_1 - y_2} = \frac{x - x_1}{x_1 - x_2}$$

∴ The equation of **OA** or **AO** is

$$\frac{y - 0}{0 - 3} = \frac{x - 0}{0 - 5}$$

$$\Rightarrow \frac{y}{-3} = \frac{x}{-5}$$

$$\Rightarrow \frac{y}{3} = \frac{x}{5}$$

$$\Rightarrow y = \frac{3x}{5}$$

The equation of **BA** or **AB** is

$$\frac{y - 6}{6 - 3} = \frac{x - 2}{2 - 5}$$

$$\Rightarrow \frac{y - 6}{3} = \frac{x - 2}{-3}$$

$$\Rightarrow \frac{y - 6}{1} = \frac{x - 2}{-1}$$

$$\Rightarrow \frac{y - 6}{1} = -x + 2$$

OA : c_1 is the line $y = 0 \therefore dy = 0$, Then

$I_1 = \oint (x^2 dx - 2xy dy)$ for this part becomes

$$I_1 = \int_0^1 (x^2 dx - 2x \cdot 0 \cdot 0) = \int_0^1 x^2 dx = \left[\frac{x^3}{3} \right]_0^1 = \frac{1}{3}$$

b) AB : c_2 is the line: $\frac{y-0}{0-1} = \frac{x-1}{1-0} \Rightarrow \frac{y}{-1} = \frac{x-1}{1} \Rightarrow y = -x + 1$

$$\therefore \frac{dy}{dx} = -1 + 0 \Rightarrow dy = -dx$$

Then

$I_2 = \oint (x^2 dx - 2xy dy)$ for this part becomes

$$I_2 = \int_1^0 \{x^2 dx - 2x(-x+1)(-dx)\} = \int_1^0 (x^2 - 2x^2 + 2x)(dx) = \int_1^0 (2x - x^2) dx = \left[\frac{2x^2}{2} - \frac{x^3}{3} \right]_1^0 = -\frac{2}{3}$$

c) BO : c_3 is the line $x = 0 \therefore dx = 0$, Then

$I_3 = \oint (x^2 dx - 2xy dy)$ for this part becomes

$$I_3 = \oint (0^2 \cdot 0 - 2 \cdot 0 \cdot y dy) = 0$$

$$\text{Finally } I = I_1 + I_2 + I_3 = \frac{1}{3} - \frac{2}{3} + 0 = -\frac{1}{3}$$

Q # 95: Evaluate the area of a circle $x^2 + y^2 = 4$.

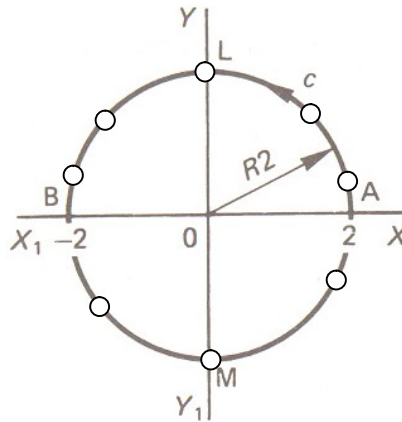


Figure # 127

Answer: Given $x^2 + y^2 = 4$

$$\Rightarrow y^2 = 4 - x^2$$

$$\Rightarrow y = \pm \sqrt{4 - x^2}$$

The Equation of the upper curve ALB: $y = +\sqrt{4 - x^2}$ between $x = 2$ and $x = -2$

And The equation of the lower curve BMA : $y = -\sqrt{4-x^2}$ between $x = -2$ and $x = 2$

We have, Area: $A = -\oint y dx = -\left\{ \int_{x_1}^{x_2} F(x) dx + \int_{x_2}^{x_1} f(x) dx \right\}$

$$\therefore A = -\left[\int_{-2}^2 \{-(\sqrt{4-x^2})\} dx + \int_2^{-2} \sqrt{4-x^2} dx \right]$$

$$\therefore A = -\left[\int_{-2}^2 \sqrt{4-x^2} dx + \int_{-2}^2 \{-(\sqrt{4-x^2})\} dx \right]$$

$$\therefore A = -\left[\int_{-2}^2 \sqrt{4-x^2} dx - \int_{-2}^2 \sqrt{4-x^2} dx \right]$$

$$\therefore A = -\left[\int_{-2}^2 \sqrt{4-x^2} dx + \int_{-2}^2 \sqrt{4-x^2} dx \right]$$

$$\therefore A = -\left[2 \int_{-2}^2 \sqrt{4-x^2} dx \right]$$

$$\therefore A = (-2)(-1) \int_{-2}^2 \sqrt{4-x^2} dx$$

$$\therefore A = 2 \int_{-2}^2 \sqrt{4-x^2} dx \text{ -----(i)}$$

$$\therefore A = 2 \times 2 \int_0^2 \sqrt{4-x^2} dx \quad \left[\because \int_{-a}^{+a} f(x) dx = 2 \int_0^a f(x) dx \right]$$

$$\therefore A = 4 \int_0^2 \sqrt{4-x^2} dx \text{ -----(ii)}$$

Let, $x = 2 \sin \theta$

$$\Rightarrow \frac{dx}{d\theta} = \frac{d}{d\theta} (2 \sin \theta)$$

$$\Rightarrow \frac{dx}{d\theta} = 2 \cos \theta$$

$$\Rightarrow dx = 2 \cos \theta d\theta$$

$$\text{Now, } \sqrt{4-x^2} = \sqrt{4-(2 \sin \theta)^2} = \sqrt{4-4 \sin^2 \theta} = \sqrt{4(1-\sin^2 \theta)} = \sqrt{4 \cos^2 \theta} = 2 \cos \theta$$

Given, $x = 2 \sin \theta$

$$\Rightarrow \theta = \sin^{-1}(x/2)$$

x	0	2
$\theta = \sin^{-1}(x/2)$	$\theta = \sin^{-1}(x/2)$ $\Rightarrow \theta = \sin^{-1}(0/2)$ $\Rightarrow \theta = \sin^{-1}(0)$	$\theta = \sin^{-1}(x/2)$ $\Rightarrow \theta = \sin^{-1}(2/2)$ $\Rightarrow \theta = \sin^{-1}(1)$

	$\Rightarrow \theta = \sin^{-1} \sin \theta$ $\Rightarrow \theta = 0$	$\Rightarrow \theta = \sin^{-1}(\sin \frac{\pi}{2})$ $\Rightarrow \theta = \frac{\pi}{2}$
--	---	---

From (ii),

$$\therefore A = 4 \int_0^2 \sqrt{4 - x^2} dx$$

$$\therefore A = 4 \int_0^{\pi/2} \sqrt{4 - (2 \sin \theta)^2} 2 \cos \theta d\theta$$

$$\therefore A = 4 \int_0^{\pi/2} \sqrt{4 - 4 \sin^2 \theta} 2 \cos \theta d\theta$$

$$\therefore A = 4 \int_0^{\pi/2} \sqrt{4(1 - \sin^2 \theta)} 2 \cos \theta d\theta$$

$$\therefore A = 4 \int_0^{\pi/2} \sqrt{4 \cos^2 \theta} 2 \cos \theta d\theta$$

$$\therefore A = 4 \int_0^{\pi/2} 2 \cos \theta \times 2 \cos \theta d\theta$$

$$\therefore A = 4 \int_0^{\pi/2} 4 \cos^2 \theta d\theta$$

$$\therefore A = 4 \times 2 \int_0^{\pi/2} 2 \cos^2 \theta d\theta$$

$$\therefore A = 8 \int_0^{\pi/2} 2 \cos^2 \theta d\theta$$

$$\therefore A = 8 \int_0^{\pi/2} (1 + \cos 2\theta) d\theta$$

$$[[2 \cos^2 \theta = 1 + \cos 2\theta]]$$

$$\therefore A = 8 \left[\theta + \frac{\sin 2\theta}{2} \right]_0^{\pi/2}$$

$$\therefore A = 8 \left[\left(\frac{\pi}{2} - 0 \right) + \left(\frac{\sin 2 \times \frac{\pi}{2}}{2} - 0 \right) \right]$$

$$\therefore A = 8 \left[\frac{\pi}{2} + \frac{\sin \pi}{2} \right]$$

$$\therefore A = 8 \left[\frac{\pi}{2} + \frac{0}{2} \right]$$

$$\therefore A = 8 \left[\frac{\pi}{2} \right]$$

$$\therefore A = 4\pi \quad \text{Answer}$$

Or from (i),

$$\therefore A = 2 \int_{-2}^2 \sqrt{4 - x^2} dx \text{ -----(i)}$$

$$\text{Let, } x = 2 \sin \theta$$

$$\Rightarrow \frac{dx}{d\theta} = \frac{d}{d\theta} (2 \sin \theta)$$

$$\Rightarrow \frac{dx}{d\theta} = 2 \cos \theta$$

$$\Rightarrow dx = 2 \cos \theta d\theta$$

$$\text{Now, } \sqrt{4 - x^2} = \sqrt{4 - (2 \sin \theta)^2} = \sqrt{4 - 4 \sin^2 \theta} = \sqrt{4(1 - \sin^2 \theta)} = \sqrt{4 \cos^2 \theta} = 2 \cos \theta$$

$$\text{Given, } x = 2 \sin \theta$$

$$\frac{x}{2} = \sin \theta$$

$$\sin \theta = \frac{x}{2}$$

$$\Rightarrow \theta = \sin^{-1} \left(\frac{x}{2} \right)$$

x	- 2	2
$\theta = \sin^{-1} \left(\frac{x}{2} \right)$	$\theta = \sin^{-1} \left(\frac{x}{2} \right)$ $\Rightarrow \theta = \sin^{-1} \left(-\frac{2}{2} \right)$ $\Rightarrow \theta = \sin^{-1} (-1)$ $\Rightarrow \theta = -\sin^{-1} (1)$ $\Rightarrow \theta = -\sin^{-1} \left(\sin \frac{\pi}{2} \right)$ $\Rightarrow \theta = -\frac{\pi}{2}$	$\theta = \sin^{-1} \left(\frac{x}{2} \right)$ $\Rightarrow \theta = \sin^{-1} \left(\frac{2}{2} \right)$ $\Rightarrow \theta = \sin^{-1} (1)$ $\Rightarrow \theta = \sin^{-1} \left(\sin \frac{\pi}{2} \right)$ $\Rightarrow \theta = \frac{\pi}{2}$

From (i),

$$\therefore A = 2 \int_{-2}^2 \sqrt{4 - x^2} dx$$

$$\therefore A = 2 \int_{-\pi/2}^{\pi/2} \sqrt{4 - (2 \sin \theta)^2} 2 \cos \theta d\theta$$

$$\therefore A = 2 \int_{-\pi/2}^{\pi/2} \sqrt{4 - 4 \sin^2 \theta} 2 \cos \theta d\theta$$

$$\therefore A = 2 \int_{-\pi/2}^{\pi/2} \sqrt{4(1 - \sin^2 \theta)} 2 \cos \theta d\theta$$

$$\therefore A = 2 \int_{-\pi/2}^{\pi/2} \sqrt{4 \cos^2 \theta} 2 \cos \theta d\theta$$

$$\therefore A = 2 \int_{-\pi/2}^{\pi/2} 2 \cos \theta \times 2 \cos \theta d\theta$$

$$\therefore A = 2 \int_{-\pi/2}^{\pi/2} 4 \cos^2 \theta d\theta$$

$$\therefore A = 2 \times 2 \int_{-\pi/2}^{\pi/2} 2 \cos^2 \theta d\theta$$

$$\therefore A = 4 \int_{-\pi/2}^{\pi/2} 2 \cos^2 \theta d\theta$$

$$\therefore A = 4 \int_{-\pi/2}^{\pi/2} (1 + \cos 2\theta) d\theta \quad [2 \cos^2 \theta = 1 + \cos 2\theta]$$

$$\therefore A = 4 \int_{-\pi/2}^{\pi/2} 1 d\theta + 4 \int_{-\pi/2}^{\pi/2} \cos 2\theta d\theta$$

$$\therefore A = 4 \left[\theta \right]_{-\pi/2}^{\pi/2} + 4 \left[\frac{\sin 2\theta}{2} \right]_{-\pi/2}^{\pi/2}$$

$$\therefore A = 4 \left[\left(\frac{\pi}{2} - \left(-\frac{\pi}{2} \right) \right) \right] + 4 \left[\left(\frac{\sin 2 \times \frac{\pi}{2}}{2} - \frac{\sin 2 \times \left(-\frac{\pi}{2} \right)}{2} \right) \right]$$

$$\therefore A = 4 \left[\left(\frac{\pi}{2} + \frac{\pi}{2} \right) \right] + 4 \left[\left(\frac{\sin \pi}{2} - \frac{\sin(-\pi)}{2} \right) \right]$$

$$\therefore A = 4 \left[\left(\frac{2\pi}{2} \right) \right] + 4 \left[\left(\frac{\sin \pi}{2} + \frac{\sin \pi}{2} \right) \right]$$

$$[\because \sin(-\theta) = -\sin \theta]$$

$$\therefore A = 4[\pi] + 4 \left[\left(\frac{0}{2} + \frac{0}{2} \right) \right]$$

$$[\because \sin \pi = 0]$$

$$\therefore A = 4[\pi] + 4.0$$

$$\therefore A = 4\pi \quad \text{Answer}$$

Justification

We know Area of a circle is πr^2

Here radius = $r = 2$

$$\therefore \text{Area of a circle is } \pi r^2 = \pi \times 2^2 = \pi \times 4 = 4\pi$$

Q # 96: If $\vec{F} = 2y\hat{i} - z\hat{j} + x^2\hat{k}$ and S is the surface of the parabolic cylinder $y^2 = 8x$ in the first octant bounded by the planes $y = 4$ and $z = 6$. Evaluate $\iint_S \vec{F} \cdot \hat{\eta} \, ds$

Answer: Given, $\vec{F} = 2y\hat{i} - z\hat{j} + x^2\hat{k}$ and the scalar function $y^2 = 8x$ i.e.
 $y^2 - 8x = 0$ implies $8x - y^2 = 0$

Let the scalar function $\phi(x, y) = 8x - y^2$ of the given parabolic surface.

We have, $\vec{\nabla} \phi$ is normal (perpendicular) vector to the surface.

Given, $\phi(x, y) = 8x - y^2$

$$\begin{aligned} \therefore \vec{\nabla} \phi &= \left(\frac{\delta}{\delta x} \hat{i} + \frac{\delta}{\delta y} \hat{j} + \frac{\delta}{\delta z} \hat{k} \right) \phi = \left(\frac{\delta \phi}{\delta x} \hat{i} + \frac{\delta \phi}{\delta y} \hat{j} + \frac{\delta \phi}{\delta z} \hat{k} \right) = \hat{i} \frac{\delta}{\delta x} \phi + \hat{j} \frac{\delta}{\delta y} \phi + \hat{k} \frac{\delta}{\delta z} \phi \\ &= \hat{i} \frac{\delta}{\delta x} (8x - y^2) + \hat{j} \frac{\delta}{\delta y} (8x - y^2) + \hat{k} \frac{\delta}{\delta z} (8x - y^2) \\ &= 8\hat{i} - 2y\hat{j} \end{aligned}$$

Let, $\hat{\eta}$ is the unit vector of $\vec{\nabla} \phi$

We can write,

$$\hat{\eta} = \frac{\vec{\nabla} \phi}{\left| \vec{\nabla} \phi \right|}$$

Greens Theorem

Q# 110: Verify Green's theorem for the integral $\oint_C \{(x^2 + y^2)dx + (x + 2y)dy\}$ taken

round the boundary curve c defined by

$$y = 0; 0 \leq x \leq 2$$

$$x^2 + y^2 = 4; 0 \leq x \leq 2$$

$$x = 0; 0 \leq y \leq 2$$

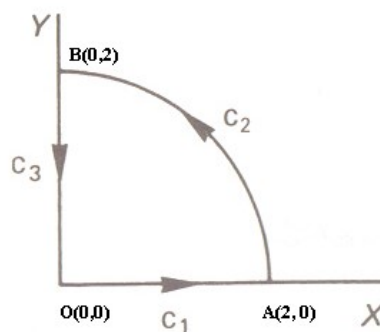


Figure # 161

Answer:

We have Green's theorem $\iint_s \left(\frac{\delta Q}{\delta x} - \frac{\delta P}{\delta y} \right) dx dy = \oint_c (P dx + Q dy)$

L.H.S.

First we have to evaluate $\iint_s \left(\frac{\delta Q}{\delta x} - \frac{\delta P}{\delta y} \right) dx dy$ -----(i)

Given, $\oint_c \{(x^2 + y^2) dx + (x + 2y) dy\}$ -----(ii)

Since, R.H.S $\oint_c (P dx + Q dy)$ -----(iii)

Comparing (ii) & (iii),

Here, $P = x^2 + y^2$

$$\therefore \frac{\delta P}{\delta y} = 0 + 2y = 2y$$

And $Q = x + 2y$

$$\therefore \frac{\delta Q}{\delta x} = 1 + 0 = 1$$

So, from (i), we get, [Putting the values of $\frac{\delta P}{\delta y}$ & $\frac{\delta Q}{\delta x}$ in (i)]

$$\iint_s \left(\frac{\delta Q}{\delta x} - \frac{\delta P}{\delta y} \right) dx dy = \iint_s (1 - 2y) dx dy$$
 -----(iv)

It will be more convenient to work in polar coordinates, so we make the substitutions

Let $x = r \cos \theta$ $y = r \sin \theta$

Then $\iint_s f(x, y) dx dy = \iint_s f(r, \theta) |J(r, \theta)| dr d\theta$ -----(v)

Here $f(x, y) = 1 - 2y$

From **Jacobian Determinant**,

$$\therefore |J(r, \theta)| = \frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix}$$
 -----(vi)

We have, $x = r \cos \theta$

$$\frac{\partial x}{\partial r} = \frac{\partial}{\partial r} (r \cos \theta)$$

$$\frac{\partial x}{\partial r} = \cos \theta \frac{\partial}{\partial r} (r)$$

$$\frac{\partial x}{\partial r} = \cos \theta \cdot 1$$

$$\frac{\partial x}{\partial r} = \cos \theta$$
 -----(vii)

Again

$$\frac{\partial x}{\partial \theta} = \frac{\partial}{\partial \theta}(r \cos \theta)$$

$$\frac{\partial x}{\partial \theta} = r \frac{\partial}{\partial \theta}(\cos \theta)$$

$$\frac{\partial x}{\partial \theta} = -r \sin \theta \quad \text{-----(viii)}$$

Again, $y = r \sin \theta$

$$\frac{\partial y}{\partial r} = \frac{\partial}{\partial r}(r \sin \theta)$$

$$\frac{\partial y}{\partial r} = \sin \theta \frac{\partial}{\partial r}(r)$$

$$\frac{\partial y}{\partial r} = \sin \theta \cdot 1$$

$$\frac{\partial y}{\partial r} = \sin \theta \quad \text{-----(ix)}$$

Again

$$\frac{\partial y}{\partial \theta} = \frac{\partial}{\partial \theta}(r \sin \theta)$$

$$\frac{\partial y}{\partial \theta} = r \frac{\partial}{\partial \theta}(\sin \theta)$$

$$\frac{\partial y}{\partial \theta} = r \cos \theta \quad \text{-----(x)}$$

From (vi),

$$\therefore |J(r, \theta)| = \frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix}$$

$$|J(r, \theta)| = \frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix}$$

$$|J(r, \theta)| = \frac{\partial(x, y)}{\partial(r, \theta)} = r \cos^2 \theta - (-r \sin^2 \theta)$$

$$|J(r, \theta)| = \frac{\partial(x, y)}{\partial(r, \theta)} = r \cos^2 \theta + r \sin^2 \theta$$

$$|J(r, \theta)| = \frac{\partial(x, y)}{\partial(r, \theta)} = r(\cos^2 \theta + \sin^2 \theta)$$

$$|J(r, \theta)| = \frac{\partial(x, y)}{\partial(r, \theta)} = r \cdot 1$$

$$|J(r, \theta)| = \frac{\partial(x, y)}{\partial(r, \theta)} = r \quad \text{-----(xi)}$$

From (v),

$$\iint_S f(x, y) dx dy = \iint_S f(r, \theta) |J(r, \theta)| dr d\theta \quad \text{-----(*)}$$

Hence, From (iv),

$$\iint_S (1 - 2y) dx dy = \iint_S (1 - 2r \sin \theta) r dr \times d\theta \quad \text{-----(xii)}$$

[Comparing * & xii,

[Here, $f(x, y) = 1 - 2y$, $y = r \sin \theta$, $f(r, \theta) = 1 - 2r \sin \theta$, $|J(r, \theta)| = r$]

Now,

When $x = 2$, $y = 0$ and $r = 2$ then

$$\begin{aligned} x &= r \cos \theta \\ \Rightarrow 2 &= 2 \cos \theta \\ \Rightarrow 1 &= \cos \theta \\ \Rightarrow \cos 0 &= \cos \theta \\ \Rightarrow \theta &= 0 \end{aligned}$$

Again

When $x = 0$, $y = 2$ and $r = 2$ then

$$\begin{aligned} y &= r \sin \theta \\ \Rightarrow 2 &= 2 \sin \theta \\ \Rightarrow 1 &= \sin \theta \\ \Rightarrow \sin \frac{\pi}{2} &= \sin \theta \\ \Rightarrow \theta &= \frac{\pi}{2} \end{aligned}$$

Hence the limit of θ is $0 \leq \theta \leq \frac{\pi}{2}$

Again

When $x = 2$ and $\theta = 0$ then

$$\begin{aligned} x &= r \cos \theta \\ 2 &= r \cos 0 \\ \Rightarrow 2 &= r \times 1 \\ \Rightarrow 2 &= r \\ \Rightarrow r &= 2 \end{aligned}$$

When $y = 2$ and $\theta = \frac{\pi}{2}$ then

$$\begin{aligned} y &= r \sin \theta \\ 2 &= r \sin \frac{\pi}{2} \\ \Rightarrow 2 &= r \times 1 \\ \Rightarrow 2 &= r \\ \Rightarrow r &= 2 \end{aligned}$$

Hence the limit of r is $0 \leq r \leq 2$

From (xii),

$$\begin{aligned}
 \iint_S (1-2y) dx dy &= \iint_S (1-2r \sin \theta) r dr d\theta - \\
 &= \int_0^{\pi/2} \int_0^2 (1-2r \sin \theta) r dr d\theta \\
 &= \int_0^{\pi/2} \int_0^2 (1-2r \sin \theta) r dr d\theta \\
 &= \int_0^{\pi/2} \int_0^2 (r-2r^2 \sin \theta) dr d\theta \\
 &= \int_0^{\pi/2} \left[\frac{r^2}{2} - 2 \frac{r^3}{3} \sin \theta \right]_0^2 d\theta \\
 &= \int_0^{\pi/2} \left[\frac{2^2}{2} - 2 \frac{2^3}{3} \sin \theta - 0 \right] d\theta \\
 &= \int_0^{\pi/2} \left[2 - \frac{16}{3} \sin \theta \right] d\theta \\
 &= \left[2\theta + \frac{16}{3} \cos \theta \right]_0^{\pi/2} \\
 &= \left[2 \times \frac{\pi}{2} + \frac{16}{3} \cos \frac{\pi}{2} - 0 - \frac{16}{3} \cos 0 \right] \\
 &= \left[\pi + \frac{16}{3} \times 0 - 0 - \frac{16}{3} \times 1 \right] \\
 &= \left[\pi - \frac{16}{3} \right] \\
 \therefore \iint_S \left(\frac{\delta Q}{\delta x} - \frac{\delta P}{\delta y} \right) dx dy &= \iint_S (1-2y) dx dy = \left[\pi - \frac{16}{3} \right] \text{-----(xiii)}
 \end{aligned}$$

Now we have to evaluate R.H.S of Green's Theorem $\oint_C (Pdx + Qdy)$

Given $\oint_C \{(x^2 + y^2)dx + (x + 2y)dy\}$

$\therefore P = x^2 + y^2$ and $Q = x + 2y$

We now take C_1, C_2, C_3 in turn

i) $C_1 : y = 0; \therefore dy = 0$

$$\begin{aligned}\therefore \oint_{c_1} (Pdx + Qdy) &= \int_0^2 (x^2 + y^2)dx + (x + 2y)dy = \int_0^2 (x^2 + 0^2)dx + (x + 2.0).0 \\ &= \int_0^2 x^2 dx = \frac{8}{3}\end{aligned}$$

$$\begin{aligned}\text{ii) } c_2 : x^2 + y^2 &= 4 \therefore y^2 = 4 - x^2 \therefore y = \sqrt{4 - x^2} = (4 - x^2)^{1/2} \\ \therefore \frac{dy}{dx} &= \frac{1}{2}(4 - x^2)^{-1/2} \cdot \frac{d}{dx}(4 - x^2) \\ \therefore \frac{dy}{dx} &= \frac{1}{2}(4 - x^2)^{-1/2} \cdot (0 - 2x) \\ \therefore dy &= -(4 - x^2)^{-1/2} \cdot x dx = \frac{-x dx}{\sqrt{4 - x^2}}\end{aligned}$$

$$\begin{aligned}\therefore \oint_{c_2} (Pdx + Qdy) &= \oint_{c_2} (x^2 + y^2)dx + (x + 2y)dy \\ &= \oint_{c_2} (x^2 + 4 - x^2)dx + (x + 2\sqrt{4 - x^2}) \cdot \frac{-x dx}{\sqrt{4 - x^2}} \\ &= \oint_{c_2} 4dx - (x + 2\sqrt{4 - x^2}) \frac{x dx}{\sqrt{4 - x^2}} \\ &= \oint_{c_2} 4dx - \frac{x^2 dx}{\sqrt{4 - x^2}} - 2\sqrt{4 - x^2} \frac{x dx}{\sqrt{4 - x^2}} \\ &= \oint_{c_2} 4dx - \frac{x^2 dx}{\sqrt{4 - x^2}} - 2x dx \\ &= \oint_{c_2} (4 - 2x)dx - \frac{x^2 dx}{\sqrt{4 - x^2}} \\ &= \oint_{c_2} (4 - 2x - \frac{x^2}{\sqrt{4 - x^2}})dx \\ &= \int_2^0 (4 - 2x - \frac{x^2}{\sqrt{4 - x^2}})dx \text{-----(xiv)}\end{aligned}$$

Let $x = 2 \sin \theta$

$$\Rightarrow dx = 2 \cos \theta d\theta$$

$$\therefore \sqrt{4 - x^2} = \sqrt{4 - (2 \sin \theta)^2} = \sqrt{4 - 4 \sin^2 \theta} = \sqrt{4(1 - \sin^2 \theta)} = \sqrt{4 \cos^2 \theta} = 2 \cos \theta$$

Given, $x = 2 \sin \theta$

$$\Rightarrow \theta = \sin^{-1}(x/2)$$

x	0	2
θ	$\theta = \sin^{-1}(x/2)$ $\Rightarrow \theta = \sin^{-1}(0/2)$ $\Rightarrow \theta = \sin^{-1}(0)$ $\Rightarrow \theta = \sin^{-1} \sin 0$ $\Rightarrow \theta = 0$	$\theta = \sin^{-1}(x/2)$ $\Rightarrow \theta = \sin^{-1}(2/2)$ $\Rightarrow \theta = \sin^{-1}(1)$ $\Rightarrow \theta = \sin^{-1}(\sin \frac{\pi}{2})$ $\Rightarrow \theta = \frac{\pi}{2}$

$$\begin{aligned}
 \therefore \oint_{c_2} (Pdx + Qdy) &= \int_2^0 (4 - 2x - \frac{x^2}{\sqrt{4-x^2}}) dx \\
 &= \int_{\pi/2}^0 \{ (4 - 2 \times 2 \sin \theta - \frac{(2 \sin \theta)^2}{2 \cos \theta}) \} 2 \cos \theta d\theta \\
 &= \int_{\pi/2}^0 \{ (8 \cos \theta - 4 \times 2 \sin \theta \cos \theta - 4 \sin^2 \theta) \} d\theta \\
 \therefore \oint_{c_2} (Pdx + Qdy) &= \int_{\pi/2}^0 8 \cos \theta d\theta - \int_{\pi/2}^0 4 \times 2 \sin \theta \cos \theta d\theta - \int_{\pi/2}^0 4 \sin^2 \theta d\theta \text{ -----(xv)}
 \end{aligned}$$

Now,

$$a) \int_{\pi/2}^0 8 \cos \theta d\theta = [8 \sin \theta]_{\pi/2}^0 = 8 \sin 0 - 8 \sin \frac{\pi}{2} = 0 - 8 \times 1 = -8$$

$$b) \int_{\pi/2}^0 4 \times 2 \sin \theta \cos \theta d\theta$$

Let $z = \sin \theta$

$$\Rightarrow dz = \cos \theta d\theta$$

Since $z = \sin \theta$

$$\Rightarrow \theta = \sin^{-1}(z/2)$$

θ	0	$\frac{\pi}{2}$
z	$z = \sin \theta$ $z = \sin 0$ $z = 0$	$z = \sin \theta$ $z = \sin \pi/2$ $z = 1$

$$\int_{\pi/2}^0 4 \times 2 \sin \theta \cos \theta d\theta = \int_1^0 4 \times 2 z dz = \left[8 \times \frac{z^2}{2} \right]_1^0 = 4(0 - 1) = -4$$

$$\begin{aligned}
\text{c) } \int_{\pi/2}^0 4 \sin^2 \theta d\theta &= 2 \int_{\pi/2}^0 2 \sin^2 \theta d\theta = 2 \int_{\pi/2}^0 (1 + \cos 2\theta) d\theta \\
&= 2 \left[\theta - \frac{\sin 2\theta}{2} \right]_{\pi/2}^0 = 2 \left(0 - \frac{1}{2} \sin 2 \cdot 0 - \frac{\pi}{2} + \frac{1}{2} \sin 2 \cdot \frac{\pi}{2} \right) = 2 \left(0 - 0 - \frac{\pi}{2} + \frac{1}{2} \cdot \sin \pi \right) = 2 \left(-\frac{\pi}{2} + \frac{1}{2} \cdot 0 \right) \\
&= -\pi
\end{aligned}$$

From (xv)

$$\begin{aligned}
\therefore \oint_{c_2} (Pdx + Qdy) &= \int_{\pi/2}^0 8 \cos \theta d\theta - \int_{\pi/2}^0 4 \times 2 \sin \theta \cos \theta d\theta - \int_{\pi/2}^0 4 \sin^2 \theta d\theta \\
&= -8 - (-4) - (-\pi) \\
&= -8 + 4 + \pi \\
&= \pi - 4
\end{aligned}$$

iii) $c_3 : x = 0; \therefore dx = 0$

$$\begin{aligned}
\therefore \oint_{c_3} (Pdx + Qdy) \\
&= \oint_{c_2} (x^2 + y^2) dx + (x + 2y) dy \\
&= \oint_{c_3} (0^2 + y^2) \cdot 0 + (0 + 2y) dy = \int_2^0 2y dy = \left[2 \frac{y^2}{2} \right]_2^0 = -4
\end{aligned}$$

Collecting our three partial results

$$\oint_c (Pdx + Qdy) = \frac{8}{3} + \pi - 4 - 4 = \pi - \frac{16}{3} \text{ -----(xvi)}$$

From (xiii) and (xvi), we can write

L.H.S. = R.H.S.

$$\therefore \iint_s \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy = \oint_c (Pdx + Qdy) \text{ (Proved)}$$

Q# 111: Use Green's theorem to evaluate $\int_C (x^2 + xy)dx + (x^2 + y^2)dy$ where C is the square formed ABCD by the lines $y = \pm 1, x = \pm 1$

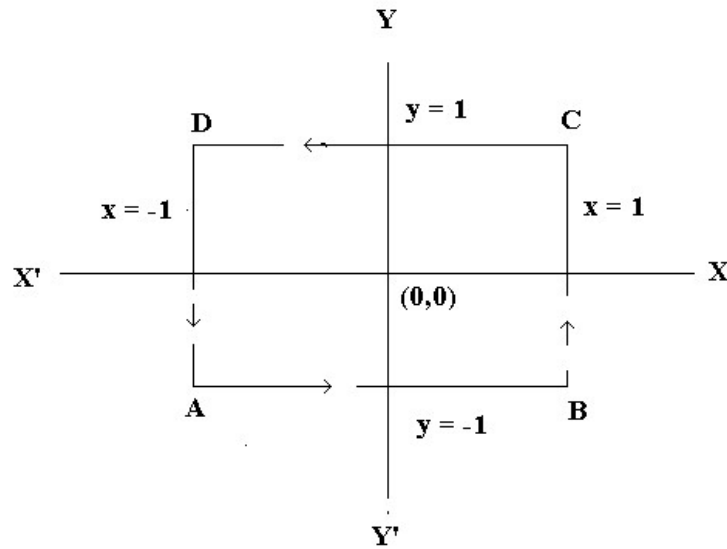


Figure # 162

Answer: From Green's theorem, we have,

$$\int_C Pdx + Qdy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dxdy$$

Here, $P = x^2 + xy$ and $Q = x^2 + y^2$

$$\int_C Pdx + Qdy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dxdy$$

L.H.S.

$$\int_C Pdx + Qdy = \text{TRY YOURSELF}$$

$$\text{Here, R.H.S} = \int_{-1}^1 \int_{-1}^1 \left\{ \left(\frac{\partial}{\partial x} (x^2 + y^2) \right) - \frac{\partial}{\partial y} (x^2 + xy) \right\} dxdy$$

$$= \int_{-1}^1 \int_{-1}^1 (2x - x) dxdy$$

$$= \int_{-1}^1 \int_{-1}^1 x dxdy$$

$$= \int_{-1}^1 \left[\frac{x^2}{2} \right]_{-1}^1 dy$$

$$\begin{aligned}
&= \int_{-1}^1 \left[\frac{1^2}{2} - \frac{(-1)^2}{2} \right] dy \\
&= \int_{-1}^1 \left[\frac{1^2}{2} - \frac{1^2}{2} \right] dy \\
&= 0 \text{ Answer}
\end{aligned}$$

Home Task:

State Green's theorem. Verify Green's theorem in the plane for

$\int_C (2xy - x^2)dx + (x + y^2)dy$ where the close curve of the region bounded by

$y = x, y^2 = x$

Chapter 4: General Theorem

Divergence theorem (Gauss' theorem)

The relation between surface and volume integrals (Gauss divergence theorem)

For a closed surface S , enclosing a region V in a vector field $\vec{F}$.

$$\text{Then, } \int_V \text{div } \vec{F} dV = \int_S \vec{F} \cdot d\vec{S}$$

Q # 99: Verify the divergence theorem for the vector field $\vec{F} = x^2 \hat{i} + z \hat{j} + y \hat{k}$ taken over the region bounded by the planes $z = 0, z = 2, x = 0, x = 1, y = 0, y = 3$

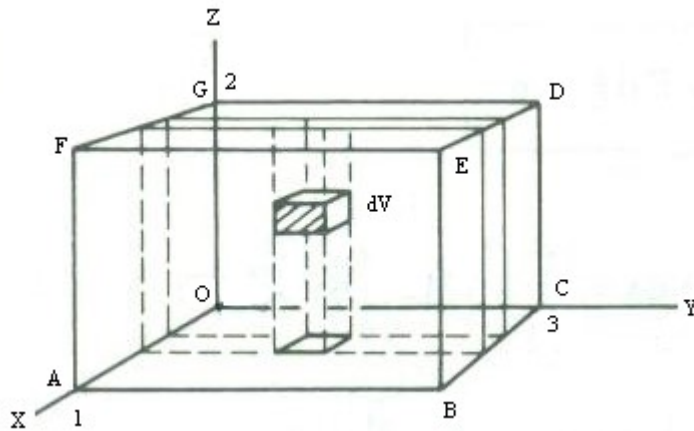


Figure # 132

Serial no	Surface	$\hat{\eta}=?$	ds	Plane
1	OABC	$\hat{\eta} = -\hat{k}$	$dx dy$	$z = 0$
2	DEFG	$\hat{\eta} = \hat{k}$	$dx dy$	$z = 2$
3	OAFG	$\hat{\eta} = -\hat{j}$	$dx dz$	$y = 0$
4	BCDE	$\hat{\eta} = \hat{j}$	$dx dz$	$y = 3$
5	OCDG	$\hat{\eta} = -\hat{i}$	$dy dz$	$x = 0$
6	ABEF	$\hat{\eta} = \hat{i}$	$dy dz$	$x = 1$

Answer:

$$dV = dx dy dz$$

$$\text{We have to show that } \int_V \text{div } \vec{F} dV = \int_S \vec{F} \cdot d\vec{S}$$

L.H.S.

Given, $\vec{F} = x^2 \hat{i} + z \hat{j} + y \hat{k}$

$$\therefore \text{div } \vec{F} = \nabla \cdot \vec{F} = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (x^2 \hat{i} + z \hat{j} + y \hat{k})$$

$$[\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0]$$

$$= \left(\frac{\partial}{\partial x} (x^2) + \frac{\partial}{\partial y} (z) + \frac{\partial}{\partial z} (y) \right)$$

$$= 2x + 0 + 0 = 2x$$

L.H.S.

$$\therefore \int_V \text{div } \vec{F} dV = \int_V 2x dV = \iiint_V 2x dz dy dx = \int_0^1 \int_0^3 \int_0^2 2x dz dy dx$$

$$= \int_0^1 \int_0^3 [2xz]_0^2 dy dx \quad \left[\int dz = z \right]$$

$$= \int_0^1 \int_0^3 [(2x \times 2) - (2x \times 0)] dy dx$$

$$= \int_0^1 \int_0^3 [4x] dy dx$$

$$= \int_0^1 [4xy]_0^3 dx \quad \left[\int dy = y \right]$$

$$= \int_0^1 [4x \times 3 - 4x \times 0] dx$$

$$= \int_0^1 [12x] dx$$

$$= \left[\frac{12x^2}{2} \right]_0^1 \quad \left[\because \int x^n dx = \frac{x^{n+1}}{n+1} \right]$$

$$= [6x^2]_0^1 = [6 \times 1^2 - 6 \times 0^2] = 6$$

$$\therefore \int_V \text{div } \vec{F} dV = 6$$

R.H.S.

We have,

Any Vector = Length of this Vector $\times$ Unit Vector

$$\therefore d\vec{S} = \left| \frac{\vec{r}}{|\vec{r}|} \right| \hat{n}$$

or

$$\therefore d\vec{S} = dS \hat{n}$$

So, we can write

$$\int_S \vec{F} \cdot d\vec{S} = \int_S \vec{F} \cdot dS \hat{n}$$

The enclosing surface S consists of six separate plane faces denoted as

$S_1, S_2, S_3, S_4, S_5, S_6$ as shown. We consider each face in turn $\vec{F} = x^2 \hat{i} + z \hat{j} + y \hat{k}$

(i) S_1 (OABC Base): $z = 0$; $\hat{n} = -\hat{k}$ (clockwise)

$$\vec{F} = x^2 \hat{i} + z \hat{j} + y \hat{k}$$

$$\therefore \vec{F} = x^2 \hat{i} + 0 \hat{j} + y \hat{k} = x^2 \hat{i} + y \hat{k} \text{ and } dS_1 = dxdy \text{ [} dS = dxdy \text{]}$$

$$\int_{S_1} \vec{F} \cdot \hat{n} dS_1 = \iint_{S_1} (x^2 \hat{i} + y \hat{k}) \cdot (-\hat{k}) dy dx = \int_0^1 \int_0^3 (-y) dy dx$$

$$[\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0]$$

$$= \int_0^1 \left[-\frac{y^2}{2} \right]_0^3 dx = \int_0^1 \left[-\frac{3^2}{2} + -\frac{0^2}{2} \right] dx = \int_0^1 \left[-\frac{9}{2} \right] dx = \left[-\frac{9}{2} x \right]_0^1 = -\frac{9}{2}(1-0) = -\frac{9}{2}$$

$$[\because \int dx = x]$$

ii) S_2 (DEFG Top): $z = 2$; $\hat{n} = \hat{k}$

$$\vec{F} = x^2 \hat{i} + z \hat{j} + y \hat{k}$$

$$\therefore \vec{F} = x^2 \hat{i} + 2 \hat{j} + y \hat{k} \text{ and } dS_2 = dxdy$$

$$\int_{S_2} \vec{F} \cdot \hat{n} dS_2 = \iint_{S_2} (x^2 \hat{i} + 2 \hat{j} + y \hat{k}) \cdot (\hat{k}) dy dx = \int_0^1 \int_0^3 (y) dy dx$$

$$[\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0]$$

$$= \int_0^1 \left[\frac{y^2}{2} \right]_0^3 dx = \int_0^1 \left[\frac{3^2}{2} - \frac{0^2}{2} \right] dx = \int_0^1 \left[\frac{9}{2} \right] dx = \left[\frac{9}{2} x \right]_0^1 = \frac{9}{2}(1-0) = \frac{9}{2} \quad [\because \int dx = x]$$

iii) S_3 (BCDE, Right-hand end): $y = 3$; $\hat{n} = \hat{j}$

$$\vec{F} = x^2 \hat{i} + z \hat{j} + y \hat{k}$$

$$\therefore \vec{F} = x^2 \hat{i} + z \hat{j} + 3 \hat{k} \text{ and } dS_3 = dx dz$$

$$\int_{S_3} \vec{F} \cdot \hat{n} dS_3 = \iint_{S_3} (x^2 \hat{i} + z \hat{j} + 3 \hat{k}) \cdot (\hat{j}) dz dx = \int_0^1 \int_0^2 (z) dz dx$$

$$[\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0]$$

$$= \int_0^1 \left[\frac{z^2}{2} \right]_0^2 dx = \int_0^1 \left[\frac{2^2}{2} - \frac{0^2}{2} \right] dx = \int_0^1 \left[\frac{4}{2} \right] dx = [2x]_0^1 = 2(1-0) = 2 \quad [\because \int dx = x]$$

$$\text{iv) } S_4 \text{ (OAFG, Left-hand end): } y = 0; \hat{n} = -\hat{j}$$

$$\vec{F} = x^2 \hat{i} + z \hat{j} + y \hat{k}$$

$$\therefore \vec{F} = x^2 \hat{i} + z \hat{j} + 0 \hat{k} \text{ and } dS_4 = dx dz$$

$$\int_{S_4} \vec{F} \cdot \hat{n} dS_4 = \iint_{S_4} (x^2 \hat{i} + z \hat{j} + 0 \hat{k}) \cdot (-\hat{j}) dz dx = \int_0^1 \int_0^2 (-z) dz dx$$

$$[\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0]$$

$$= \int_0^1 \left[-\frac{z^2}{2} \right]_0^2 dx = \int_0^1 \left[-\frac{2^2}{2} + -\frac{0^2}{2} \right] dx = \int_0^1 \left[-\frac{4}{2} \right] dx = [-2x]_0^1 = -2(1-0) = -2 \quad [\because \int dx = x]$$

$$\text{v) } S_5 \text{ (ABEF, Front): } x = 1; \hat{n} = \hat{i}$$

$$\vec{F} = x^2 \hat{i} + z \hat{j} + y \hat{k}$$

$$\therefore \vec{F} = 1^2 \hat{i} + z \hat{j} + y \hat{k} \text{ and } dS_5 = dy dz$$

$$\int_{S_5} \vec{F} \cdot \hat{n} dS_5 = \iint_{S_5} (1^2 \hat{i} + z \hat{j} + y \hat{k}) \cdot (\hat{i}) dy dz = \int_0^2 \int_0^3 dy dz$$

$$[\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0]$$

$$= \int_0^2 [y]_0^3 dz = \int_0^2 [3-0] dz = \int_0^2 [3] dz = [3z]_0^2 = 3(2-0) = 6 \quad [\because \int dy = y]$$

$$\text{vi) } S_6 \text{ (OCDG, Back): } x = 0; \hat{n} = -\hat{i}$$

$$\vec{F} = x^2 \hat{i} + z \hat{j} + y \hat{k}$$

$$\therefore \vec{F} = 0 \hat{i} + z \hat{j} + y \hat{k} \text{ and } dS_6 = dy dz$$

$$\int_{S_6} \vec{F} \cdot \hat{n} dS_6 = \iint_{S_6} (0 \hat{i} + z \hat{j} + y \hat{k}) \cdot (-\hat{i}) dy dz = 0$$

For the whole surface S we therefore have

$$\int_s \vec{F} \cdot d\vec{S} = -\frac{9}{2} + \frac{9}{2} + 2 - 2 + 6 + 0 = 6$$

Q # 100: Home task:

Verify the Divergence theorem for $\vec{F} = (2xy + z)\hat{i} + y^2\hat{j} - (x + 3y)\hat{k}$ taken over the region bounded by the planes, $2x + 2y + z = 6, x = 0, y = 0, z = 0$

Hints;

When, $x = 0, y = 0$ then $2x + 2y + z = 6$

$$2.0 + 2.0 + z = 6$$

$$z = 6$$

When, $y = 0, z = 0$ then $2x + 2y + z = 6$

$$2x + 2.0 + 0 = 6$$

$$2x = 6$$

$$x = 3$$

When, $x = 0, z = 0$ then $2x + 2y + z = 6$

$$2.0 + 2y + 0 = 6$$

$$2y = 6$$

$$y = 3$$

So, the above problem can be written as

Verify the Divergence theorem for $\vec{F} = (2xy + z)\hat{i} + y^2\hat{j} - (x + 3y)\hat{k}$ taken over the region bounded by the planes, $z = 0, z = 6, x = 0, x = 3, y = 0, y = 3$

Q # 101: Evaluate $I = \oint_C (3x^2y^2dx + 2x^3ydy)$ between $O(0,0)$ and $A(2,4)$

a) along c_1 i.e. $y = x^2$

b) along c_2 i.e. $y = 2x$

c) along c_3 i.e. $x = 0$ from $(0,0)$ to $(0,4)$ and $y = 4$ from $(0,4)$ to $(2,4)$

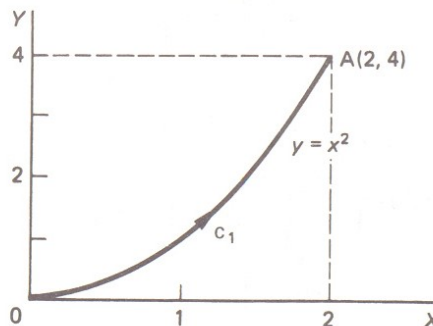


Figure # 133

$$I = \oint_C (3x^2y^2dx + 2x^3ydy)$$

The path c_1 is $y = x^2$